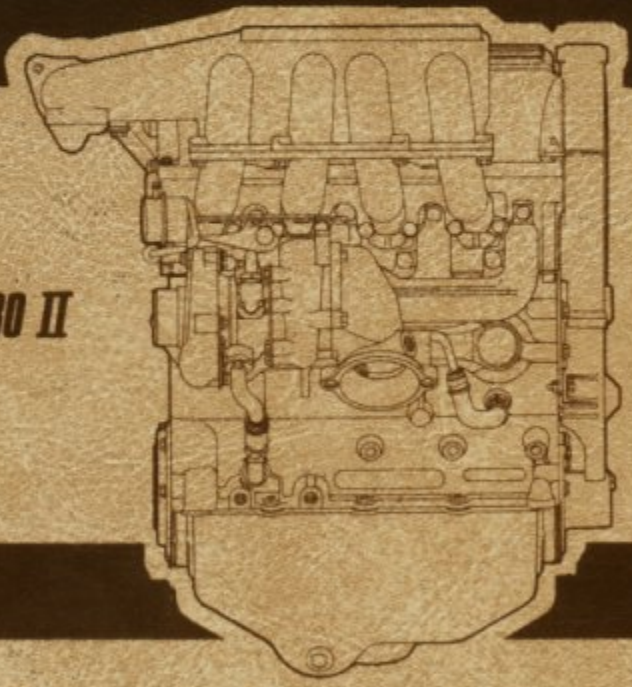
1984-1987 ADVANCED TURBO THEORY



TURBO II

Instructor's Guide

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How to use the Instructor's Guide

This instructor's guide is intended to be used to teach technicians who have repaired many turbo cars using the "Driveability Test Procedures". It is important that they know all the sensors and switches, their names and locations. The technicians also need a good understanding of the relationship between volts, amps and ohms. This instructor's guide is a sensor by sensor, switch by switch indepth analysis. It is intended to give the technician a more comprehensive understanding as to why they see the voltages they do. The instructor should start with the first sensor (MAP) and cover each component and circuit all the way through. The depth of what is taught is dependant on the caliber of the technician as determined by a pretest. It is imperative that the instructor **does not** teach everything that is in this instructor's guide. Teach only what can be understood and more importantly, teach only what can be used by the technician.

Hardware Suggested for Class

The following is a list of parts that will be helpful in teaching this class.

1. 1984 MAP Sensor, 1985 & '86 MAP Sensor, 1987 MAP Sensor
2. 1984 - '86 TPS Sensor, 1987 TPS Sensor
3. 1984 Coolant Sensor, 1985 - '87 Coolant Sensor
4. 1984 Charge Temperature Sensor, 1985 - '87 Charge Temperature Sensor
5. Oxygen (O_2) Sensor
6. Detonation Sensor
7. 1984 & '85 Turbo Distributor, 1986 & '87 Turbo Distributor
8. 1984 Speed Sensor, 1985 - '87 Speed Sensor
9. Ignition Coil
10. Fuel Rail and Injectors
11. Fuel Pump
12. 1984 - '86 AIS Motor, 1987 Turbo I AIS Motor, 1987 Turbo II AIS Motor
13. Radiator Fan Relay
14. A/C Cutout Relay
15. 1984 Purge and EGR Solenoid Bank, 1985 & '86 Purge/EGR/Wastegate/Baro Read Solenoid Bank, 1987 Combination Purge & EGR/Wastegate/Baro Read Solenoid Bank
16. 1984 Logic Module, 1985 & '86 Logic Module, 1987 Logic Module
17. 1984 Turbocharger, 1985 - '87 Turbocharger
18. 1984 & '85 EGR and Backpressure Valves, 1986 & '87 EGR and Backpressure Valves
19. Purge Canister
20. Throttle Body
21. Fuel Pressure Regulator

Manifold Pressure Pounds per square inch gauge and Inches of mercury Vacuum	Turbo MAP Sensor Signal Voltage
14.31 psig	4.90 V
11.41 psig	4.40 V
8.51 psig	3.90 V
5.61 psig	3.41 V
2.71 psig	2.91 V
.39 in Hg	2.41 V
6.30 in Hg	1.92 V
12.20 in Hg	1.42 V
18.11 in Hg	.93 V
24.01 in Hg	.43 V

Note: Due to differences in altitude from area to area, do not use these values to test a turbo MAP sensor. Instead, measure the voltage on the MAP sensor signal line at 20 in Hg and at 5 in Hg. Then subtract the voltages. The voltage differences should be 1.13 V - 1.39 V for a good turbo MAP sensor.

Fig. 1 MAP Sensor Voltages at Different Manifold Pressures

MAP (Manifold Absolute Pressure) Sensor

The MAP sensor is a 3 wire sensor with a nipple on it. The 1984 MAP sensor is mounted above the logic module. The 1985 & '86 MAP sensors are mounted on the logic module housing. The 1987 MAP sensor is mounted under the hood by the right strut tower. The MAP sensor "gives" the logic module (the microprocessor) 2 basic pieces of information:

1. Atmospheric pressure (altitude)
2. Manifold pressure (vacuum or boost level)

The MAP sensor "gives" the logic module this information by sending a voltage that is proportional to manifold pressure. As the manifold pressure increases, the MAP sensor signal voltage increases. As the manifold pressure decreases, the MAP sensor signal voltage decreases. See Fig. 1 for MAP sensor signal voltage values at different manifold pressures. There are 2 types of MAP sensors, the 1984, '85 & '86 style and the 1987 style. There are 2 types of MAP sensor circuits, the 1984 style and the 1985, '86 & '87 style. The MAP sensor will be discussed first.

1984, '85 & '86 MAP Sensor

It was stated earlier that the MAP sensor sends out a voltage that is proportional to manifold pressure. Internally, the MAP sensor has 6 major components. See Fig. 2. Note that the MAP sensor schematics have been simplified for explanation purposes. The 1st and 2nd major components are 2 temperature compensating resistors RT+ & RT−. These correct for any error in voltage due to a change in temperature. The 3rd major component is the piezoresistive chip. See Fig. 3. This chip is the pressure sensing element of the MAP sensor. The chip will output a pressure sensitive voltage at point "B" and "C".

The piezoresistive chip has 4 variable resistors on it: R_1 R_2 R_3 R_4. These resistors are hooked up in such a way called a "wheatstone bridge". See Fig. 3. The chip is made of silicon and measures $^1/_4$" x $^1/_4$". The thickness of the chip around the outer edges is .030" or .045". The center of the chip is .001" thick. This allows the center of the chip to move and act as a diaphragm. See Fig. 3. Below the center of the chip is 0 psia (perfect vacuum 29.92 in Hg). This puts the center of the chip in a concave position because the higher atmospheric pressure (14.69 psia) on the top of the diaphragm is pushing down toward the lower pressure (0 psia) below the diaphragm. By having 0 psia below the diaphragm this makes the chip pressure sensitive. See Fig. 8 for pressure relationship chart. The wheatstone bridge is fed voltage at point "A" and provided a ground at point "D". See Fig. 3. The wheatstone bridge is essentially 2 series circuits made up of R_1 & R_4 and R_2 & R_3. The outputs of the wheatstone bridge are at points "B" and "C". These outputs are just the voltage remaining after the first resistors R_1 & R_2. In short, the wheatstone bridge is 2 series circuits. Both circuits are fed voltage at point "A" and both circuits are provided a ground at point "D". The resistors change resistance with a change in pressure. The outputs of the wheatstone bridge at point "B" and "C" will change with pressure, so at any intake manifold pressure, the wheatstone bridge outputs 2 voltages.

If there were 0 psia applied to the nipple of the MAP sensor, this would make R_1 R_2 R_3 R_4 equal in resistance. This is called having the wheatstone bridge in balance. If there are 2 resistors in a series circuit of equal resistance, each resistor uses ½ the source voltage. So the outputs at point "B" and "C" are now equal. As the pressure on the nipple rises above 0 psia, R_1 and R_3 increase in resistance while R_2 and R_4 decrease in resistance by the same amount that R_1 and R_3 increased. For example, if R_1 and R_3 increase resistance by 1 ohm, R_2 and R_4 will decrease resistance by 1 ohm. This makes the voltage at the output point "C" go up with an increase in pressure and the voltage at output point "B" go down with an increase in pressure. See Fig. 3. Remember, the wheatstone bridge is 2 series circuits. In the R_1 & R_4 series circuit, R_1 is now the larger resistor and will use more than ½ the source voltage. So as R_1 increases in resistance, the voltage at point "B" decreases. The same is true of the other series circuit however; it reacts just the opposite with a increasing voltage output at point "C".

In summary, the piezoresistive chip has 4 pressure variable resistors. This resistance changes the voltage at the outputs of the wheatstone bridge.

The 4th major component is the unity gain amplifier. See Fig. 2. A unity gain amplifier is an amplifier that doesn't amplify (no gain). The output of the unity gain amplifier at pin "7" is equal to the voltage at input pin "6". The reason for a unity gain amplifier is to limit current flow so the voltage at the wheatstone bridge won't be lowered by excessive current flow. The unity gain amplifier calibrates the piezoresistive chip since each chip is a little different. It also offers some temperature compensation. The unity gain amplifier has 2 inputs and 1 output. It is fed 5 volts and provided a signal ground. See Fig. 2. The input at pin "5" will always be ½ source voltage because R_3 & R_4 are equal in resistance. The other input at pin "6" is fed voltage from the output of the wheatstone bridge at point "B". See Fig. 2.

In summary, the output of the unity gain amplifier is the same as the voltage output of the wheatstone bridge at point "B".

The 5th major component is the differential amplifier. See Fig. 2. The differential amplifier amplifies the difference of the voltage at its input pins by 60 times. The reason for the differential amplifier is because the voltage difference from the

1984 MAP SENSOR

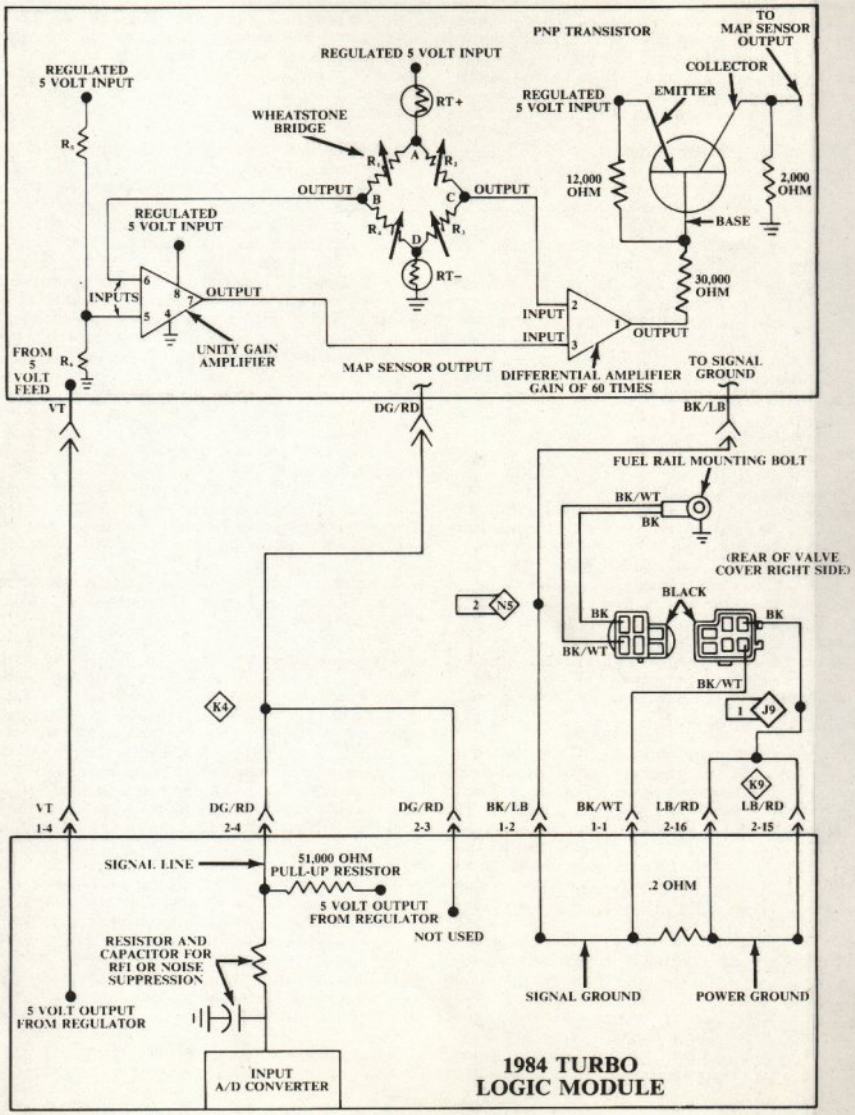


Fig. 2

wheatstone bridge is so small (.055 volts maximum difference) that it must be amplified to a usable range. The output is then inverted or opposite. For example, if the input voltages were equal, the differential amplifier will subtract the input voltages, this equals 0. Then this voltage is multiplied by 60. This still equals 0 volts. Next the output is inverted to 3.3 volts (maximum differential amplifier output). So if the inputs are equal the output is high (voltage) and if the inputs are at there maximum difference, the output is low (voltage).

In summary, the output of the differential amplifier decreases as intake manifold pressure increases.

The 6th major component is the PNP transistor. See Fig. 2 . This is needed because the maximum output of the differential amplifier is only 3.3 volts. The MAP sensor output has to go up to 4.94 volts. This is done by using the PNP transistor as an amplifier. We usually think of a transistor as a solid state relay. This is not the case in the MAP sensor. The emmiter is fed 5 volts and the collector is the output of the MAP sensor. The base circuit is connected to the output of the differential amplifier through a 30,000 ohm resistor. See Fig. 2. The output voltage is controlled by varying the base circuit voltage. In a silicon transistor, the base circuit only needs .7 volts to be completely "on". In other words, if the emmiter is fed 5 volts, the base circuit doesn't need a ground to be "on" only 4.3 volts or less. The base circuit voltage varies from 4.45 - 4.33 volts. This is only .120 volts variation on the base circuit and this causes the collector voltage to vary from .4 volts - 4.94 volts. So in this way a small voltage change on the base circuit causes a much larger change in the collector voltage. This makes the transistor an amplifier. If you're looking at the base circuit in Fig. 2 you're probably wondering how the voltage is varied when the PNP transistor is hooked up the way it is. We stated earlier that if 0 psia were applied to the nipple of the MAP sensor the wheatstone bridge would be in balance and its outputs would be equal. This puts equal voltages at the inputs of the differential amplifier, causing the output of the differential amplifier go up to 3.3 volts. So now there is 3.3 volts applied to the 30,000 ohm resistor. There is a 12,000 ohm resistor that is connected from the emmiter to the base circuit. The 12,000 ohm resistor is fed 5 volts on the other end. See Fig. 2. So what we have is opposing positive voltages at the base of the transistor. The current flow is from the emitter through the 12,000 ohm resistor on to the 30,000 ohm resistor. The circuit gets its ground through the output of the differential amplifier. What the base circuit "sees" is the voltage remaining after the 12,000 ohm resistor. Once the current flow is calculated, the voltage drop across the 12,000 ohm resistor will be known. This has to be subtracted from 5 (the feed voltage) and this will result in approximately 4.52 volts on the base circuit. This will completely shut the transistor "off" and the output of the MAP sensor will be 0 volts when 0 psia is applied to the nipple of the MAP sensor (theoretically). As the pressure on the nipple of the MAP sensor increases, the output of the differential amplifier goes down (lower voltage). This causes more current to flow resulting in more voltage drop across the 12,000 ohm resistor. This lowers the base voltage a little bit more which turns on the transistor more and increases the output voltage.

In summary, as the base circuit voltage is lowered, the output voltage of the MAP sensor increases. The output of the PNP transistor is connected to the DG/RD wire in 1984 and the WT wire in 1985 & '86.

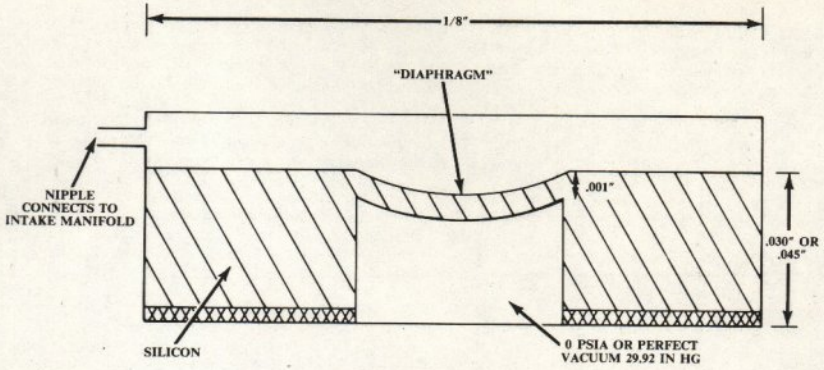
1987 MAP Sensor

This MAP sensor works the same as the previous years. Internally the only difference is that there is only 5 major components. See Fig. 6. The PNP transistor is now part of the differential amplifier, so the output of the differential amplifier is the output of the MAP sensor.

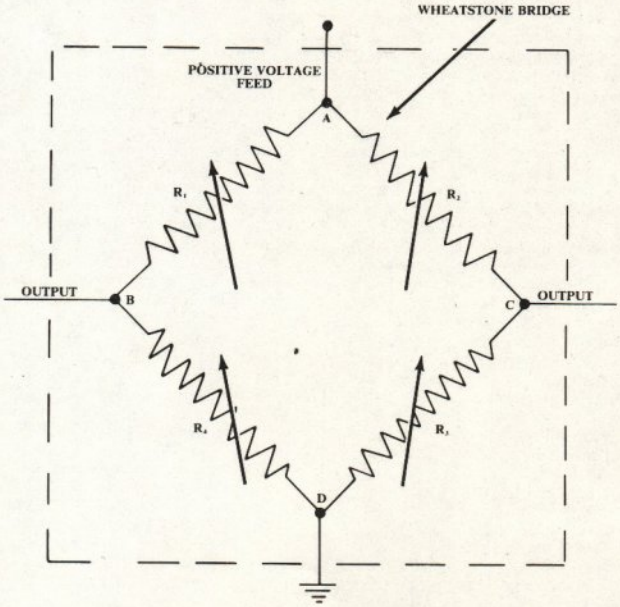
1984 MAP Sensor and Circuit

Anytime the ignition key is "on", the logic module sends a regulated 5 volts on the VT wire to the MAP sensor. See Fig. 2. The MAP sensor is provided a signal ground on the BK/LB wire. There are 2 types of grounds in this system, a signal ground and a power ground. The power ground is for all the relatively heavy current flow components (i.e., coil, injectors, etc.). Because of the heavy current flow through the power ground, it may drift (change voltage relative to engine ground). If the sensors were hooked up to a ground that drifts, this would change the sensor signal voltage when the sensor hasn't "seen" a change. For this reason the signal ground is separated from the power ground. The .2 ohm resistor that is between the signal and power ground (in the logic module) is to protect the electronics from high voltage surges which may happen if the BK/WT wire became open. See Fig. 2. The .2 ohm resistor allows the signal ground circuit to be completed through the power ground if the BK/WT wire is open. The sensor signal voltage may be inaccurate at this time, but high voltage surges have been avoided. The BK/LB wire from the MAP sensor goes into the logic module and comes out of the logic module on the BK/WT wire. The BK/WT wire is connected to the fuel rail mounting bolt where the signal and power grounds are connected to the engine. This is commonly referred to as the system ground in the driveability books or the reference ground in the service manual.

Now, the MAP sensor is powered up and will output a voltage on the DG/RD wire that is proportional to manifold pressure. This voltage goes into the logic on 2 pins. See Fig. 2. Pin 2-3 is not used. Pin 2-4 is connected to the signal line that goes to the input of the A/D (analog to digital)



SIDE VIEW OF PIEZORESISTIVE CHIP



TOP VIEW OF PIEZORESISTIVE CHIP

Fig. 3

converter. There is a resistor in series and a capacitor in parallel with the signal line to the A/D converter. These are for RFI or noise suppression. See Fig. 2. The A/D converter figuratively speaking is the logic module's "voltmeter". Just as we need a voltmeter to "see" or understand the voltage in a circuit, so does the logic module. The A/D converter has very high input resistance, so like a voltmeter, very little current flows through it. The A/D converter is a device that takes a analog voltage signal (constantly varying with no discrete states) and "converts" it or changes it into a digital voltage signal (binary number). This is needed because the logic module cannot "understand" an analog voltage signal. So any analog voltage signal that is an input to the logic module is converted to a digital voltage signal.

Hooked up to the signal line is a 51,000 ohm pull-up resistor. The reason for the pull-up resistor is to put 5 volts on the signal line in the case of an open circuit on the DG/RD or BK/LB wire. This allows the logic module to identify an open circuit by the signal line always being high. See Fig. 2. This happens because when there is no current flow, a resistor doesn't use any voltage. When the MAP sensor is hooked up and working, the 5 volts that is fed to the 51,000 ohm pull-up resistor is almost all used up. The pull-up resistor is in series with a 2,000 ohm resistor connected to the MAP sensor output. See Fig. 2. Since the pull-up resistor is much larger than the 2,000 ohm resistor, most of the voltage is used up by the pull-up resistor when the circuit is complete and current is flowing.

1985, '86 & '87 MAP Sensor Circuit

This circuit has a few changes from 1984 but basically works the same. These changes are noted on Figs. 4, 5 and 6. The wire colors changed along with the pull-up resistor value. There was a capacitor added along with the .2 ohm resistor that is in between the signal and power grounds in the logic module.

MAP Sensor Fault Codes

The logic module monitors the MAP sensor signal line during cranking at the start to run transfer. The MAP sensor signal voltage must decrease by at least .04 volts. This "tells" the logic module that the manifold pressure is decreasing due to engine start-up. If the voltage changes less than .04 volts, this will turn on the power loss/limited light. The system is now in "limp-in". Fault code 13 is stored in the logic module's memory. At this time the logic module will completely close the AIS motor and guess at what the manifold pressure is based on TPS and the reference pickup (engine RPM). Since the manifold pressure is not known, the engine must be protected from overboost. This is done by limiting engine RPM to 2,000 RPM when the throttle is opened more than 22 degrees.

Once the engine has started, the logic module monitors the MAP sensor signal line every time the reference pickup changes states (when the ignition coil is fired). If the voltage on the MAP sensor signal line goes below .02 volts or above 4.94 volts, the logic module will turn on the power loss/limited light. The system is now in "limp-in". Fault code 14 is stored in the logic module's memory. At this time the logic module will do the same thing as it did for fault code 13 to protect the engine from overboost. This system will remain in "limp-in" until 2 things happen:

1. The problem is fixed.
2. The ignition key is turned "off" then back "on" again.

Also, the logic module "knows" the boost level by the MAP sensor signal voltage. If the voltage goes too high, the logic module turns on the power loss/limited light. Fault code 45 is stored in the logic module's memory. At this time the logic module will cut all the fuel "off" until the MAP sensor signal voltage falls below a certain level. Then the fuel is turned back "on". This turning the fuel "on" and "off" will continue as long as the engine is overboosted. This will feel as a severe bucking. Depending on the model year and what transaxle is in the car, different MAP voltages are allowed to attain different boost levels. See Fig. 7. The power loss/limited light will only be on when the engine is in overboost, however; once the engine has been overboosted, fault code 45 will remain in memory.

1985 MAP SENSOR

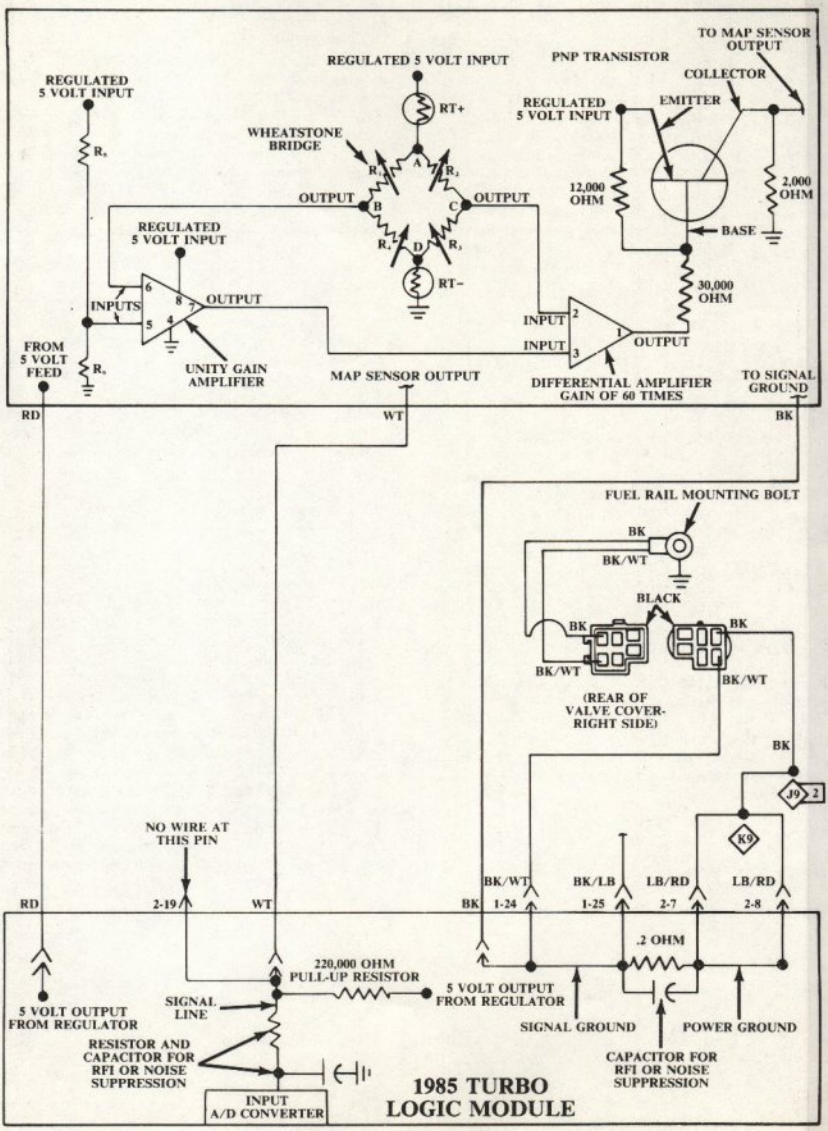


Fig. 4

1986 MAP SENSOR

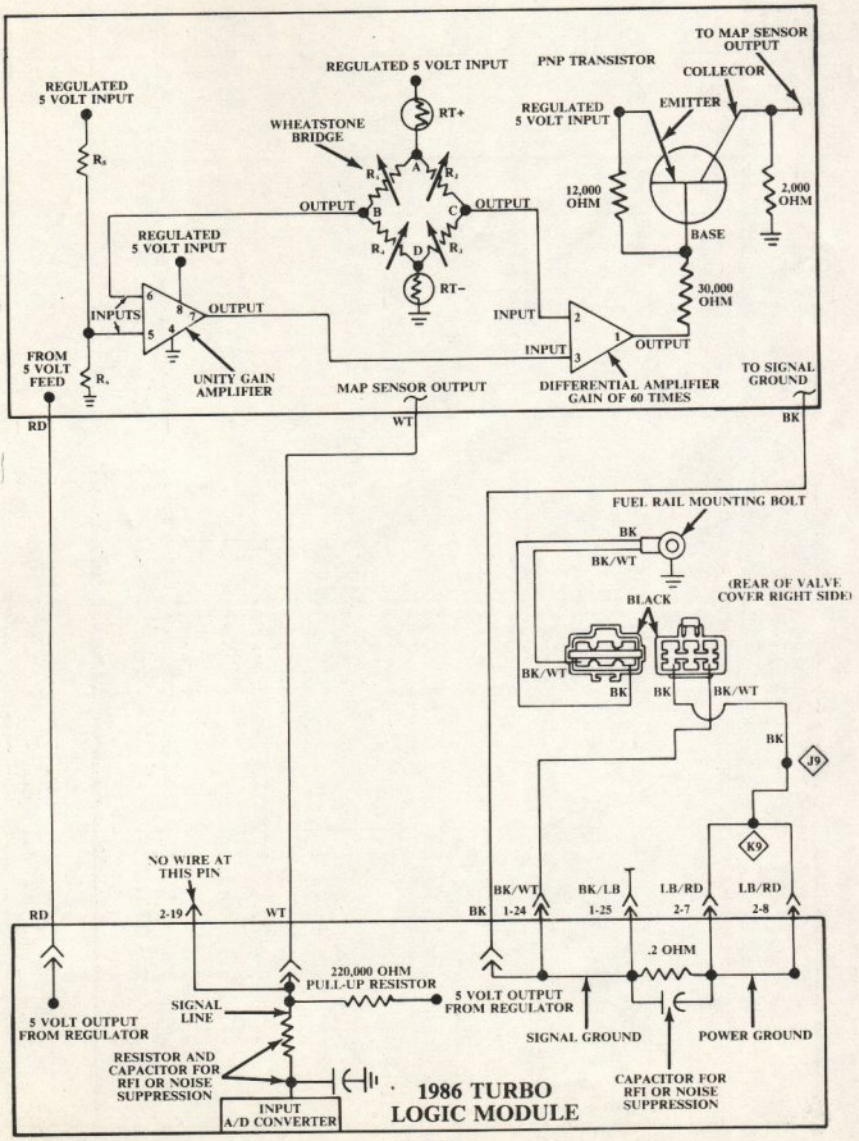


Fig. 5

1987 MAP SENSOR

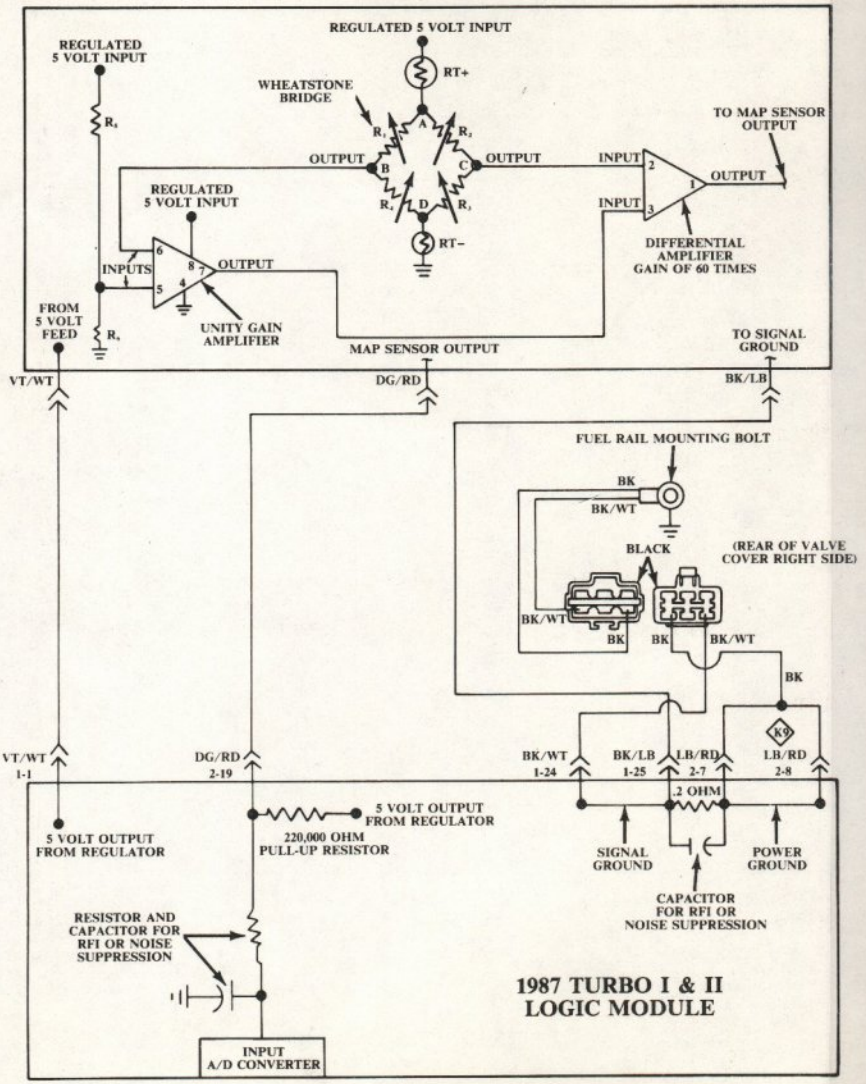


Fig. 6

Model Year	Auto Transaxle	Man Transaxle	Time For Fault Code To Appear
1984	4.30 Volts	4.30 Volts	Instant
1985	4.65 Volts	4.54 Volts	2 Seconds
1986	4.88 Volts	4.65 Volts	1 Second
1987 Turbo I	4.88 Volts	4.65 Volts	1 Second
1987 Turbo II	4.96 Volts	4.65 Volts	1 Second

Fig. 7 Maximum MAP Sensor Voltage Before Overboost

Pounds per square inch Absolute	Pounds per square inch Gauge and Inches of mercury vacuum	Kilopascals Gauge and centimeters of mercury vacuum	Kilopascals Absolute	Inches of Water
29.70 psia	15 psig	104.43 kPag	204.76 kPaa	—
26.70 psia	12 psig	82.74 kPag	184.07 kPaa	—
24.70 psia	10 psig	68.95 kPag	170.28 kPaa	—
21.70 psia	7 psig	48.27 kPag	149.60 kPaa	—
19.70 psia	5 psig	34.48 kPag	135.81 kPaa	—
16.70 psia	2 psig	13.79 kPag	115.12 kPaa	—
14.697 psia	0 psig/in Hg	0.00 kPag/cm Hg	101.33 kPaa	0.00 in H_2O
13.71 psia	2 in Hg	5.08 cm Hg	94.53 kPaa	27.24 in H_2O
12.24 psia	5 in Hg	12.70 cm Hg	84.39 kPaa	68.10 in H_2O
11.26 psia	7 in Hg	17.78 cm Hg	77.63 kPaa	95.34 in H_2O
9.78 psia	10 in Hg	25.40 cm Hg	67.43 kPaa	136.20 in H_2O
8.80 psia	12 in Hg	30.48 cm Hg	60.68 kPaa	163.44 in H_2O
7.33 psia	15 in Hg	38.10 cm Hg	50.54 kPaa	204.30 in H_2O
5.85 psia	18 in Hg	45.72 cm Hg	40.34 kPaa	245.16 in H_2O
4.87 psia	20 in Hg	50.80 cm Hg	33.58 kPaa	272.40 in H_2O
3.89 psia	22 in Hg	55.88 cm Hg	26.82 kPaa	299.64 in H_2O
2.42 psia	25 in Hg	63.50 cm Hg	16.68 kPaa	340.50 in H_2O
0.00 psia	29.92 in Hg	75.996 cm Hg	0.00 kPaa	407.52 in H_2O

Note: All values are taken at sea level

Fig. 8

	Altitude (Feet)	Inches of Mercury Vacuum
	−1,000 ft	31.0185 in Hg
	−900 ft	30.9073 in Hg
Seal Level	0 ft	29.9213 in Hg
	500 ft	29.3846 in Hg
	1,000 ft	28.8557 in Hg
	1,500 ft	28.3345 in Hg
	2,000 ft	27.8210 in Hg
	3,000 ft	26.8167 in Hg
	4,000 ft	25.8418 in Hg
	6,000 ft	23.9782 in Hg
	8,000 ft	22.2250 in Hg
	10,000 ft	20.5770 in Hg
	12,000 ft	19.0294 in Hg
	14,000 ft	17.5774 in Hg
	16,000 ft	16.2164 in Hg
	18,000 ft	14.9421 in Hg
	20,000 ft	13.7501 in Hg
	22,000 ft	12.6363 in Hg
	25,000 ft	11.1035 in Hg
	30,000 ft	8.88544 in Hg
	35,000 ft	7.04062 in Hg
	40,000 ft	5.53802 in Hg
	45,000 ft	4.35488 in Hg
	49,900 ft	3.44112 in Hg
	50,000 ft	3.42466 in Hg

Fig. 9 The Effects of Altitude on a Perfect Vacuum

1984 TPS (THROTTLE POSITION SENSOR)

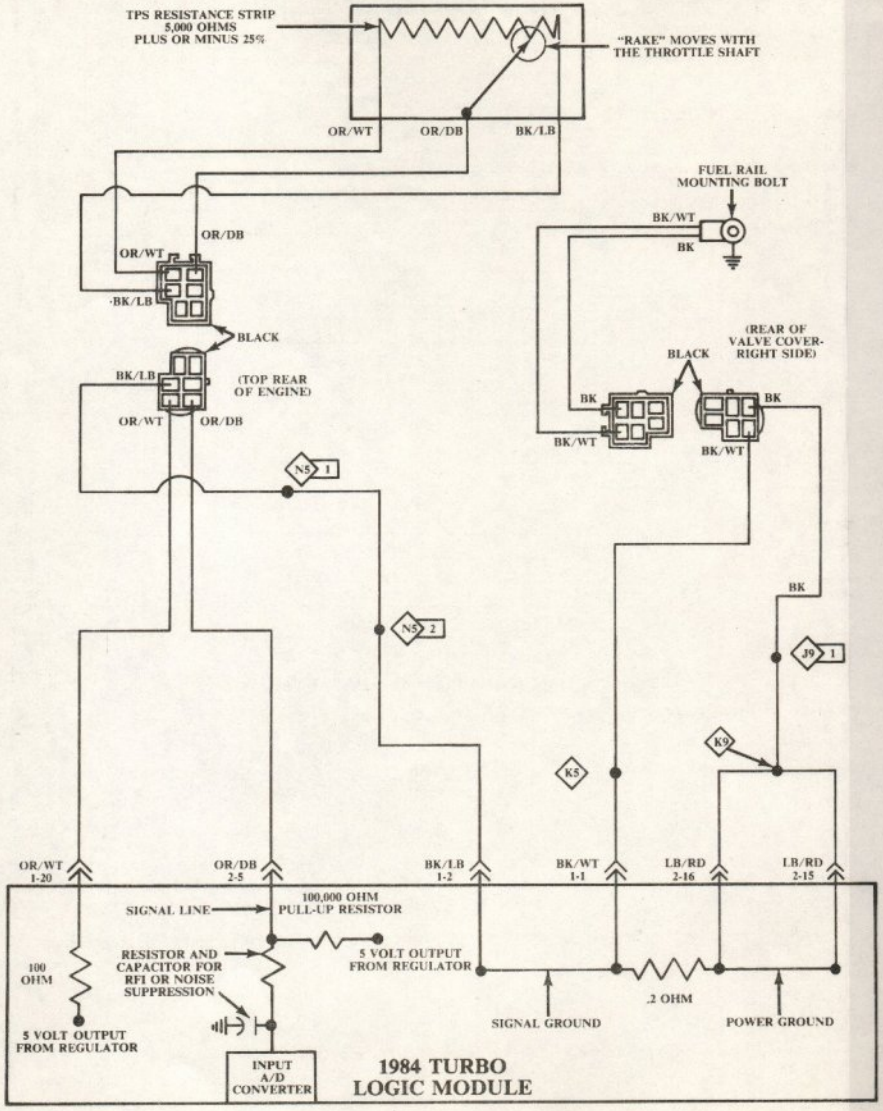


Fig. 10

TPS (Throttle Position Sensor)

The TPS is a 3 wire sensor that is mounted on the throttle body. The TPS "gives" the logic module (the microprocessor) 4 basic pieces of information:

1. Closed throttle (minimum TPS voltage)
2. Wide open throttle (2.6 volts above minimum TPS voltage)
3. Throttle position (where the throttle is between closed throttle and W.O.T.)
4. Rate of opening (how fast did the throttle get to its position)

The TPS "gives" the logic module this information by sending out a voltage that is proportional to throttle opening. As the throttle is opened, the TPS signal voltage increases. As the throttle is closed, the TPS signal voltage decreases.

Internally, the TPS has a resistance strip. This resistance strip has a fixed resistance of 5,000 ohms (plus or minus 25%). See Fig. 10. There is also a "rake" inside the TPS that moves with the throttle shaft. This "rake" is in contact with the resistance strip.

Anytime the ignition key is "on", the logic module sends a regulated 5 volts to a 100 ohm resistor. See Fig. 10. This resistor is to limit current flow in case of a grounded OR/WT wire. The other end of the 100 ohm resistor is connected to the OR/WT wire. The OR/WT wire is connected to the TPS resistance strip. The other end of the TPS resistance strip is connected to the BK/LB wire. This wire goes into the logic module and comes out of the logic module on the BK/WT wire. The BK/WT wire is connected to the fuel rail mounting bolt for signal ground. See Fig. 10.

So now the TPS is powered up and there is current flowing through the TPS resistance strip. Any resistor that has current passing through it, is using voltage. Inside the TPS, the "rake" that is in contact with resistance strip, is electrically connected to the OR/DB wire. So when the ignition key is "on" and the throttle is closed, the voltage on the OR/DB wire will be at its lowest (1.00 volt approximately). The reason for this is that the "rake" is near the signal ground end of the TPS resistance strip. See Fig. 10. The 5 volts (from the logic module) is almost all used up by the TPS resistance strip. As the throttle opens, the "rake" moves toward the voltage source side of the TPS resistance strip. This will increase the voltage on the OR/DB wire because less of the source voltage is being used by the TPS resistance strip. In effect, as the throttle is opened, the resistance between the OR/WT wire and the OR/DB wire is lowered. Lower resistance value will cause a lower voltage drop. A lower voltage drop will result in a higher remaining voltage (TPS signal voltage). The TPS signal voltage is calculated by subtracting the voltage drop across the TPS resistance strip (at the point where the "rake" contacts the TPS resistance strip) from the 5 volt source.

At 2.6 volts above closed throttle voltage, the logic module will "consider" this wide open throttle. 2.6 volts above closed throttle voltage happens at 79.3° of throttle opening. See Fig. 11 for throttle opening vs. increase in minimum TPS voltage. The voltage on the OR/DB wire will increase .0375 volts per degree of throttle opening.

Let's recap what was just said. The TPS is powered up and will output a voltage that is proportional to throttle opening on the OR/DB wire. This voltage goes into the logic module and is present on the signal line. See Fig. 10. The signal line is connected to the input of the A/D converter. There is a resistor in series and a capacitor in parallel with the signal line. These are for RFI or noise suppression. Connected to the signal line is a 100,000 ohm pull-up resistor. See Fig. 10. The reason for the pull-up resistor is to put 5 volts on the signal line if the OR/DB wire or BK/LB wire becomes open. This "tells" the logic module that there is an open circuit by the signal line always being high (5 volts). This happens because when there is no current flow (open circuit) a resistor doesn't use any voltage.

When the TPS is hooked up and working, the 5 volts that is fed to the 100,000 ohm pull-up resistor, is almost all used up by the pull-up resistor. The pull-up resistor is in series with the TPS resistance strip to ground. See Fig. 10. Since the pull-up resistor is much larger in resistance than the TPS resistance strip, it uses up most of the 5 volts.

TPS Fault Code

The logic module monitors the TPS signal line (OR/DB wire) every 11 milliseconds (91 times a second) anytime the ignition key is "on". The TPS signal voltage must not be below .16 volts or above 4.7 volts. If either of these limits are exceeded, the logic module will turn on the power loss/limited light. The system will now be in "limp-in". Fault code 24 will be stored in the logic module's memory.

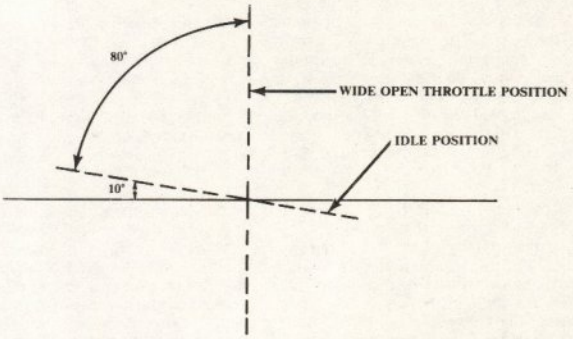
At this time, the logic module will use the MAP sensor signal voltage as an indication of throttle position. If the MAP sensor signal voltage is less than 2 volts, the logic module will "consider" the throttle closed. If the MAP signal sensor voltage is above 2 volts, the logic module will "consider" the throttle wide open. The system will remain in "limp-in" until 2 things happen:

1. The problem is fixed.
2. The ignition key is turned "off" then back "on" again.

THROTTLE POSITION vs. INCREASE IN MINIMUM TPS VOLTAGE

Throttle Position In Degrees Of Opening		Increase Of Minimum TPS Voltage On OR/DB Wire
*Closed Throttle	10°	0 Volts
	20°	.38 Volts
	30°	.75 Volts
Half Throttle	45°	1.31 Volts
	60°	1.86 Volts
	79.3°	**2.60 Volts
Wide Open Throttle	90°	3.00 Volts

THROTTLE BLADE MOVEMENT RANGE



* At idle the throttle blade is 10° open. When the throttle blade is opened from idle to wide open, it has only traveled 80°.

** The logic module will "consider" an increase of 2.60 volts or more over the minimum TPS signal voltage, to be wide open throttle.

Fig. 11

1985 TPS (THROTTLE POSITION SENSOR)

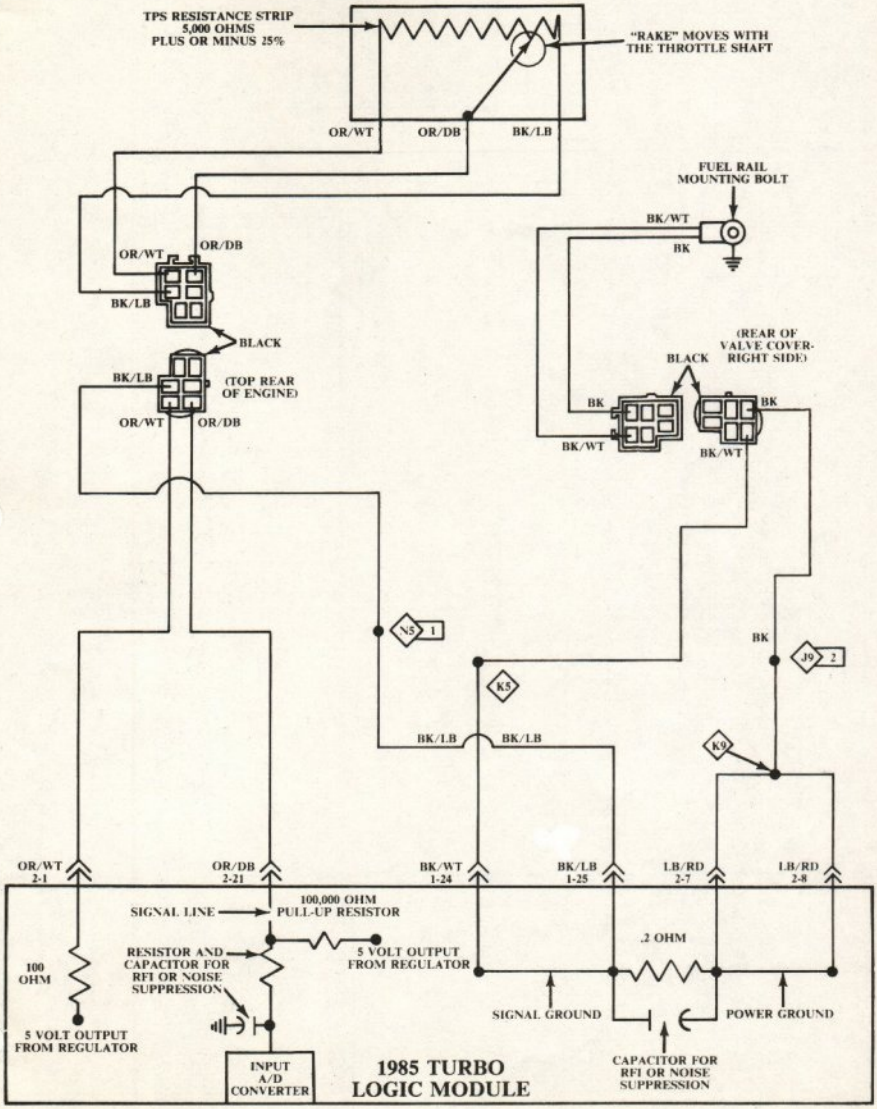


Fig. 12

1986 & '87 TPS (THROTTLE POSITION SENSOR)

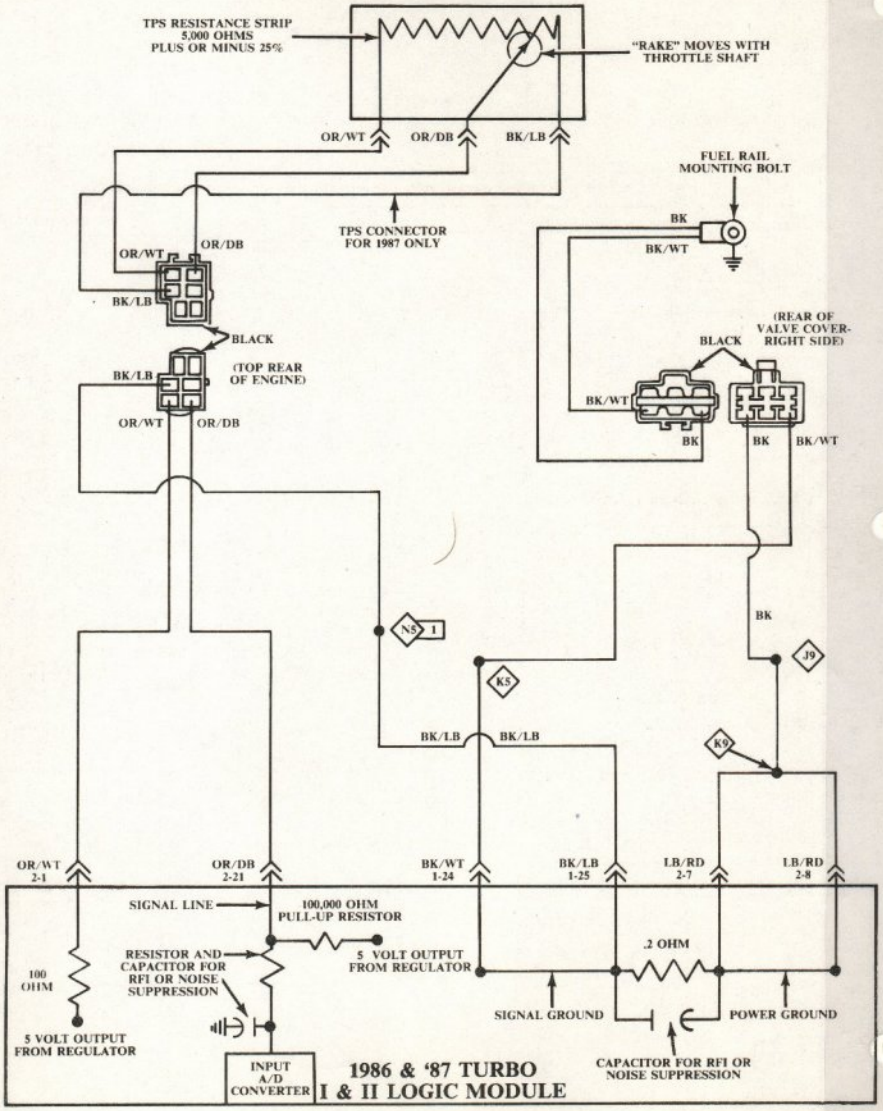


Fig. 13

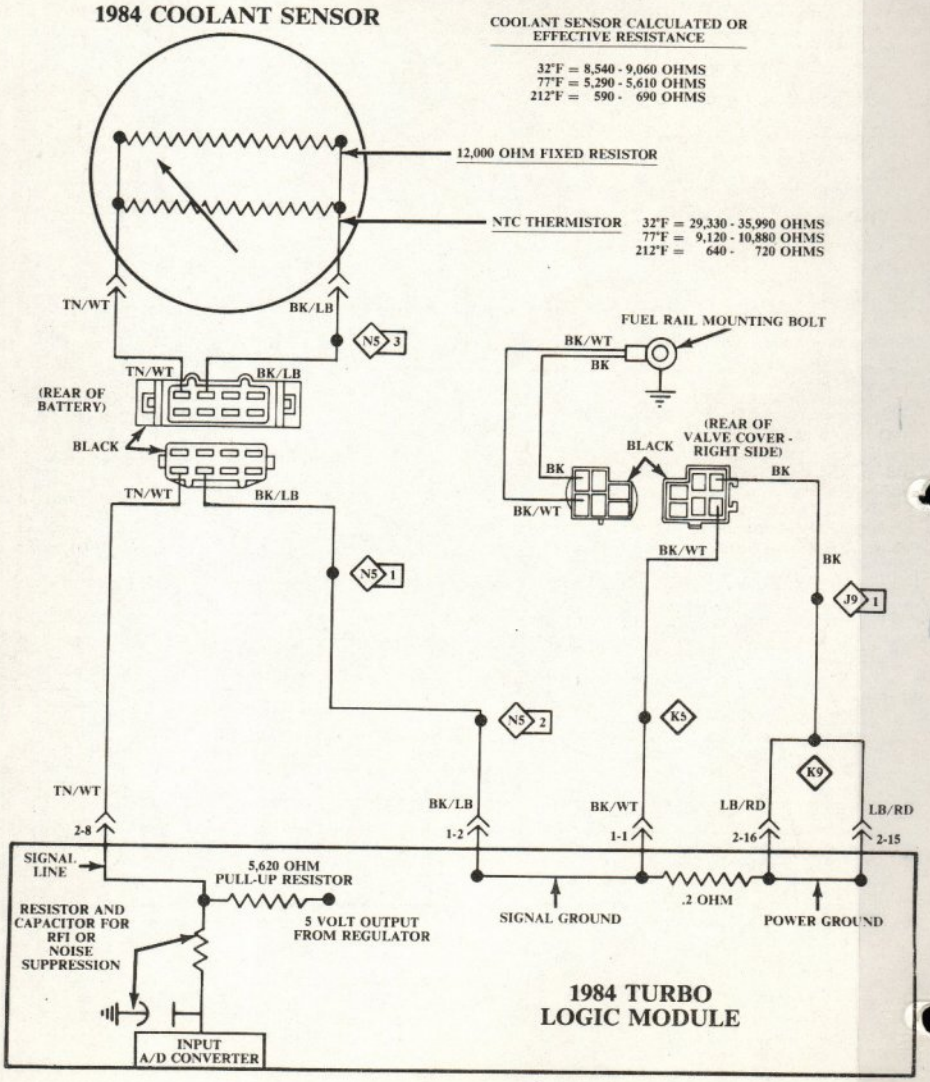

1984 COOLANT SENSOR
COOLANT SENSOR CALCULATED OR EFFECTIVE RESISTANCE
32°F = 8,540 - 9,060 OHMS
77°F = 5,290 - 5,610 OHMS
212°F = 590 - 690 OHMS
12,000 OHM FIXED RESISTOR
NTC THERMISTOR
32°F = 29,330 - 35,990 OHMS
77°F = 9,120 - 10,880 OHMS
212°F = 640 - 720 OHMS
TN/WT
BK/LB
N5 3
(REAR OF BATTERY)
BLACK
N5 1
N5 2
FUEL RAIL MOUNTING BOLT
BK/WT
BK
(REAR OF VALVE COVER - RIGHT SIDE)
K5
J9 1
K9
2-8
1-2
1-1
2-16
2-15
LB/RD
SIGNAL LINE
5,620 OHM PULL-UP RESISTOR
RESISTOR AND CAPACITOR FOR RFI OR NOISE SUPPRESSION
5 VOLT OUTPUT FROM REGULATOR
.2 OHM
SIGNAL GROUND
POWER GROUND
INPUT A/D CONVERTER
1984 TURBO LOGIC MODULE

Fig. 14

Coolant Sensor

The coolant sensor is a 2 wire sensor that is mounted in the thermostat housing. The coolant sensor "gives" the logic module 2 basic pieces of information:

1. Engine coolant temperature
2. Rate of engine coolant temperature change

The coolant sensor "gives" the logic module this information by changing the voltage on the coolant sensor signal line inversely proportional to engine temperature. As the engine temperature increases, the coolant sensor signal voltage decreases. As the engine temperature decreases, the coolant sensor signal voltage increases. There are 2 types of coolant sensors and coolant sensor circuits. The 1984 style and the 1985, '86 & '87 style. The 1984 style will be discussed first.

1984 Coolant Sensor and Circuit

Internally, the 1984 coolant sensor has a 12,000 ohm fixed resistor in parallel with a negative temperature coefficient (NTC) thermistor rated at 9,120 ohms - 10,880 ohms at 77°F. See Fig. 14. With these 2 resistors in parallel, this makes their calculated or effective resistance (what is measured across the terminals of the coolant sensor with an ohmmeter) 5,290 ohms - 5,610 ohms at 77°F. The reason for the 12,000 ohm fixed resistor is, to maintain coolant sensor accuracy at higher (hot) engine temperatures. A NTC thermistor is a resistor that will radically change its resistance with temperature. As the temperature goes up, the resistance goes down. As the temperature goes down, the resistance goes up, hence the name "negative" temperature coefficient thermistor.

Anytime the ignition key is "on", the logic module sends a regulated 5 volts to the 5,620 ohm pull-up resistor. The pull-up resistor is connected to the signal line. See Fig. 14. The reason for the pull-up resistor is two fold. First, the pull-up resistor uses some of the 5 volts it is fed with, when the circuit is complete and current is flowing. The voltage remaining after the pull-up resistor is the coolant sensor signal voltage. This voltage leaves the logic module on the TN/WT wire and goes to the coolant sensor and then to signal ground at the fuel rail mounting bolt. This voltage is also present at the input to the A/D converter. The resistor and capacitor at the input to the A/D converter are for RFI or noise suppression. See Fig. 14.

The second reason for the pull-up resistor is to put 5 volts at the input of the A/D converter in case of an open circuit on the TN/WT wire or BK/LB wire. This "tells" the logic module there is an open circuit by the signal line always being high (5 volts). This happens because, when there is no current flow (open circuit), a resistor doesn't use any voltage.

In summary, the coolant sensor signal voltage is taken after the 5,620 ohm pull-up resistor in the logic module. The coolant sensor signal voltage is actually the voltage remaining between the pull-up resistor and the coolant sensor. To calculate coolant sensor signal voltage, subtract the voltage drop across the pull-up resistor from the 5 volt source. For example, as the engine warms up, the coolant sensor resistance will decrease. This causes the voltage drop across the 5,620 ohm pull-up resistor to increase with engine temperature. As the voltage drop increases, the coolant sensor signal voltage will decrease. See Fig. 17.

1984 Coolant Sensor Fault Code

The logic module monitors the coolant sensor signal line (TN/WT wire) every 11 milliseconds (91 times a second), anytime the ignition key is "on". The coolant sensor signal voltage must not be below .157 volts or above 3.76 volts. Also, the coolant sensor signal voltage must not change too fast to be real (over .5 volts in less than 88 milliseconds). If any of these limits are exceeded, the logic module will turn on the power loss/limited light. The system will now be in "limp-in". Fault code 22 will be stored in the logic module's memory. Also, for 1984 only, the logic module will **not** turn on the cooling fan. At this time the logic module will use the charge temperature sensor signal as an indication of engine temperature. This system will remain in "limp-in" until 2 things happen:

1. The problem is fixed.
2. The ignition key is turned "off" then back "on" again.

1985, '86 & '87 Coolant Sensor and Circuit

Internally, the 1985, '86 & '87 coolant sensor also has NTC thermistor, however; there is no fixed resistor in parallel with the NTC thermistor. See Fig. 15. This makes the coolant sensor's resistance higher. It is rated at 9,120 ohms - 10,880 ohms at 77°F. Different coolant sensor circuits inside the logic module are used depending on coolant sensor temperature. The following sequence of events is what happens on a cold engine (32°F) that is warming up to normal operating temperature. Inside the logic module there are 2 pull-up resistors. When the ignition key is first turned "on", the microprocessor will remove the ground on the base circuit of the PNP transistor. See Fig. 15. This will turn "off" the PNP transistor. With the PNP transistor turned "off", the emitter is no longer connected to the collector. This shuts off the 5 volts to the 1,000 ohm pull-up resistor. This is called switching out the 1,000 ohm pull-up resistor. There is a 2,700 ohm resistor that is connected from the emitter to the base of the PNP transistor. The resistor is there to make sure the transistor turns "off" when the microprocessor removes the ground from the base circuit. This will put 5 volts on the

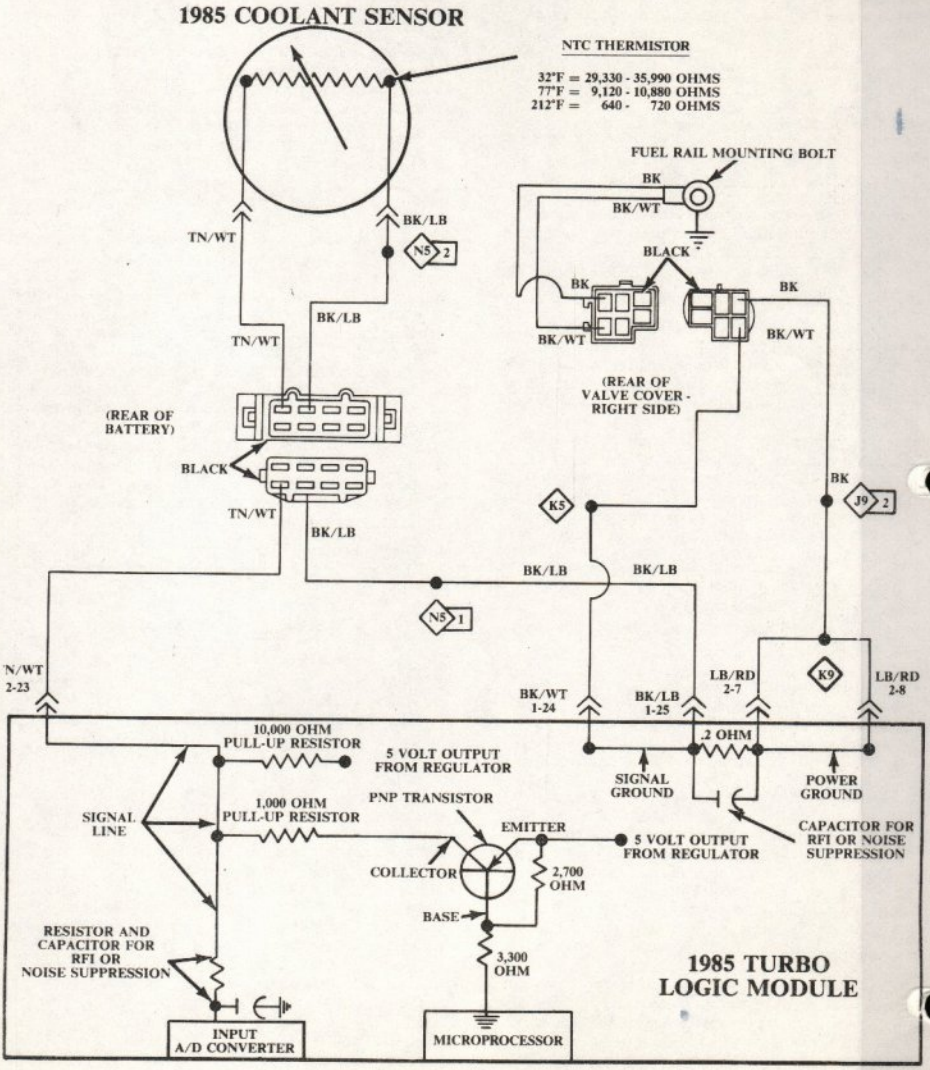

1985 COOLANT SENSOR
NTC THERMISTOR
32°F = 29,330 - 35,990 OHMS
77°F = 9,120 - 10,880 OHMS
212°F = 640 - 720 OHMS
FUEL RAIL MOUNTING BOLT
BK
BK/WT
BLACK
BK
BK
BK/WT
BK/WT
TN/WT
BK/LB
N5 2
TN/WT
BK/LB
(REAR OF BATTERY)
BLACK
TN/WT
BK/LB
(REAR OF VALVE COVER - RIGHT SIDE)
K5
BK
J9 2
BK/LB
BK/LB
N5 1
N/WT 2-23
BK/WT 1-24
BK/LB 1-25
LB/RD 2-7
K9
LB/RD 2-8
10,000 OHM PULL-UP RESISTOR
5 VOLT OUTPUT FROM REGULATOR
.2 OHM
SIGNAL GROUND
POWER GROUND
SIGNAL LINE
1,000 OHM PULL-UP RESISTOR
PNP TRANSISTOR
EMITTER
5 VOLT OUTPUT FROM REGULATOR
CAPACITOR FOR RFI OR NOISE SUPPRESSION
COLLECTOR
2,700 OHM
BASE
RESISTOR AND CAPACITOR FOR RFI OR NOISE SUPPRESSION
3,300 OHM
1985 TURBO LOGIC MODULE
INPUT A/D CONVERTER
MICROPROCESSOR

Fig. 15

base circuit and the transistor needs a grounded base circuit to be "on".

The 1,000 ohm pull-up resistor is switched out when the ignition key is first turned "on". Then the logic module sends a regulated 5 volts to the 10,000 ohm pull-up resistor. See Fig. 15. After the 10,000 ohm pull-up resistor, the 5 volts are dropped to some lower voltage (depending on the coolant sensor resistance). This voltage leaves the logic module on the TN/WT wire and goes to the coolant sensor and then to signal ground at the fuel rail mounting bolt. This voltage is also present at the input to the A/D converter. The resistor and capacitor at the input to the A/D converter are for RFI or noise suppression.

The coolant sensor signal voltage is taken after the pull-up resistors. The coolant sensor signal voltage is, with the 1,000 ohm pull-up resistor switched out, actually the voltage remaining between the 10,000 ohm pull-up resistor and the coolant sensor.

To calculate the coolant sensor signal voltage, subtract the voltage drop across the 10,000 ohm pull-up resistor from the 5 volt source. For example, a cold (32°F) engine may have 3.89 volts for a coolant sensor signal voltage. As the engine warms up, the coolant sensor resistance decreases. This causes the voltage drop across the 10,000 ohm pull-up resistor to increase with engine temperature. As the voltage drop increases, the coolant sensor signal voltage will decrease with engine temperature. For example, the cold (32°F) engine is now at 77°F. This will decrease the coolant sensor resistance to 10,000 ohms (approximately). Now there are 2 10,000 ohm resistors in series (the pull-up resistor in the logic module and the coolant sensor thermistor). They will each use 1/2 of the 5 volt source. The coolant sensor signal voltage has decreased from 3.89 volts at 32°F to 2.5 volts at 77°F. The coolant sensor signal voltage will continue to drop as the engine warms up. At approximately 125°F, the coolant sensor signal voltage is 1.25 volts. At this point the engine temperature may increase by a 100°F or more, but there is only 1.25 volts left after the 10,000 ohm pull-up resistor. This is true because the 10,000 ohm pull-up resistor is using up most of the 5 volt source. The coolant sensor resistance has decreased so much (from 35,000 ohms at 32°F to 3,400 ohms at 125°F) that it would take a large temperature change to affect the coolant sensor signal voltage. This would make the coolant sensor signal voltage inaccurate at higher engine temperatures. Hot engine temperature accuracy is needed to determine when to turn the cooling fan on and off.

At approximately 125°F, inside the logic module, the microprocessor will ground the base circuit of the PNP transistor. See Fig. 15. This will turn "on" the PNP transistor. With the PNP transistor turned "on", the 5 volts on the emitter will be connected across the collector to the 1,000 ohm pull-up resistor. This is called switching in the 1,000 ohm pull-up resistor. There is a 3,300 ohm resistor in the base circuit of the PNP transistor. This resistor is there to limit current flow and thus reduce heat. Now, in the logic module there is, the 10,000 ohm pull-up resistor in parallel with the 1,000 ohm pull-up resistor. The calculated or effective resistance of the pull-up resistors is 909 ohms. This will decrease the voltage drop because the coolant sensor is the larger resistor again. For example, with the 1,000 ohm pull-up resistor switched in, the calculated or effective resistance of the pull-up resistors is 909 ohms, this is in series with a 3,400 ohm resistor (the coolant sensor). The voltage drop will decrease dramatically (when the 1,000 ohm pull-up resistor is switched in) to 1.05 volts. This will increase the coolant sensor voltage to 3.95 volts. From this point, the coolant sensor signal voltage will continue to drop again as the engine temperature increases. The coolant sensor signal voltage is now very sensitive to engine temperatures that are on the hot end of the temperature scale.

In summary, the 10,000 ohm pull-up resistor is for cold temperature accuracy while the 1,000 ohm pull-up resistor is for hot temperature accuracy.

When checking the coolant sensor signal voltage, one must know if the 1,000 ohm pull-up resistor is switched in or out. Then it will be known which temperature vs. voltage curve to use to translate a voltage reading into a temperature. For example, a coolant sensor signal voltage may read 3.6 volts. This could mean the engine temperature is 40°F or 140°F. Use the cold curve when the engine temperature is below 100°F. See Fig. 18. Use the hot curve when the engine temperature is above 125°F. There is an overlap period of 25°F where the logic module could have the 1,000 ohm pull-up resistor switched in or out. The reason for the overlap period is to stabilize the coolant sensor signal voltage during the switch over point. If there were no overlap period and the engine temperature was right at the switch point, the logic module may switch the 1,000 ohm pull-up resistor in and out with only a slight change in engine temperature. This would change the coolant sensor signal voltage back and forth from cold to the hot curve with a large change in voltage. With the 25°F overlap period, once the logic module has switched in the 1,000 ohm pull-up resistor (hot curve), the engine would have to cool down 25°F before the logic module would switch out (cold curve) the 1,000 ohm pull-up resistor. The overlap period stabilizes the coolant sensor voltage during the switch, from cold to hot or hot to cold curve.

For the overlap period between 100°F and 125°F, if the engine is above 100°F and the temperature is rising, use the cold curve. If the engine was above 125°F and is cooling down into the overlap period, use the hot curve till the engine temperature is below 100°F.

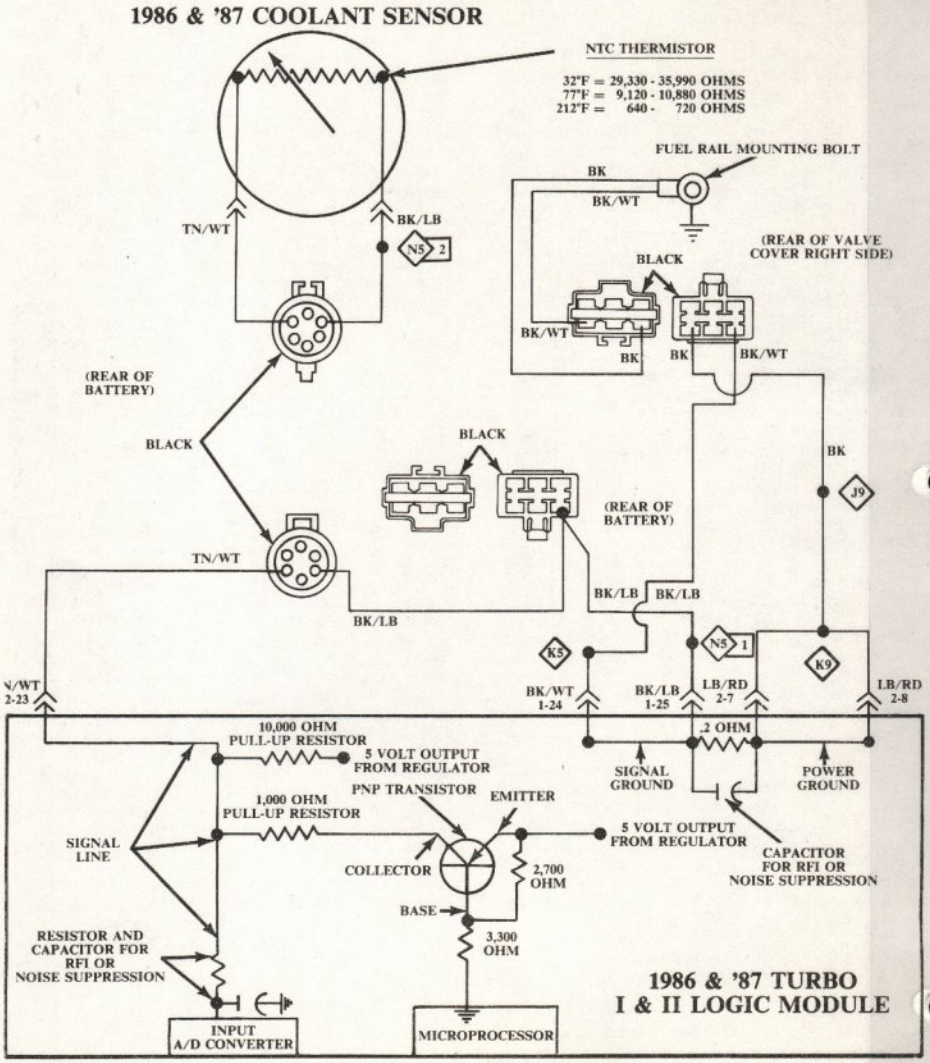

1986 & '87 COOLANT SENSOR
NTC THERMISTOR
32°F = 29,330 - 35,990 OHMS
77°F = 9,120 - 10,880 OHMS
212°F = 640 - 720 OHMS
FUEL RAIL MOUNTING BOLT
BK
BK/WT
(REAR OF VALVE COVER RIGHT SIDE)
BLACK
BK/WT
BK
BK
BK/WT
TN/WT
BK/LB
N5 2
(REAR OF BATTERY)
BLACK
BLACK
(REAR OF BATTERY)
BK
J9
TN/WT
BK/LB
BK/LB
BK/LB
K5
N5 1
K9
N/WT 2-23
BK/WT 1-24
BK/LB 1-25
LB/RD 2-7
LB/RD 2-8
10,000 OHM PULL-UP RESISTOR
5 VOLT OUTPUT FROM REGULATOR
.2 OHM
SIGNAL GROUND
POWER GROUND
PNP TRANSISTOR
EMITTER
1,000 OHM PULL-UP RESISTOR
SIGNAL LINE
5 VOLT OUTPUT FROM REGULATOR
CAPACITOR FOR RFI OR NOISE SUPPRESSION
COLLECTOR
2,700 OHM
BASE
3,300 OHM
RESISTOR AND CAPACITOR FOR RFI OR NOISE SUPPRESSION
1986 & '87 TURBO I & II LOGIC MODULE
INPUT A/D CONVERTER
MICROPROCESSOR

Fig. 16

1985, '86 & '87 Coolant Sensor Fault Code 22

The logic module will monitor the coolant sensor signal line (TN/WT wire) every 700 milliseconds (1.4 times a second), anytime the ignition key is "on". With the 1,000 ohm pull-up resistor switched in, the coolant sensor signal voltage must not be below .51 volts. With the 1,000 ohm pull-up resistor switched out, the coolant sensor signal voltage must not be above 4.96 volts. If either of these limits are exceeded, the logic module will turn on the power loss/limited light. The system will now be in "limp-in". Fault code 22 will be stored in the logic module's memory. The logic will turn on the cooling fan. At this time, the logic module will use the charge temperature sensor signal as an indication of engine temperature. This system will remain in "limp-in" until 2 things happen:

1. The problem is fixed.
2. The ignition key is turned "off" then back "on" again.

1986 & '87 Coolant Sensor Fault Code 17

For 1986, when the engine is cranked, if the coolant sensor indicates a temperature between −20°F and 212°F, a 20 minute timer is started. When the timer runs out, the coolant sensor must indicate at least 160°F. If not, fault code 17 is stored in the logic module's memory.

For 1987, 12 minutes after engine start-up, the logic module starts an 8 minute timer. After the timer runs out and the car speed is above 28 MPH and the logic module is not in the default mode for fault code 22; the logic module monitors the coolant sensor signal line for a too cool too long condition. If all these parameters are met and the coolant sensor does not indicate at least 174°F, fault code 17 is stored in the logic module's memory.

1984 COOLANT SENSOR TEMPERATURE vs. VOLTAGE CURVE

Temperature	Voltage
−4°F	3.28 V
14°F	3.19 V
32°F	3.05 V
50°F	2.86 V
68°F	2.61 V
77°F	2.46 V
86°F	2.31 V
104°F	1.98 V
122°F	1.65 V
140°F	1.34 V
158°F	1.07 V
176°F	.84 V
194°F	.66 V
212°F	.51 V
230°F	.40 V
248°F	.32 V

Fig. 17

1985, '86, & '87 COOLANT SENSOR COLD & HOT TEMPERATURE vs. VOLTAGE CURVES

Cold Curve 1,000 ohm pull-up resistor switched out	
−20°F	4.70 V
−10°F	4.57 V
0°F	4.45 V
10°F	4.30 V
20°F	4.10 V
30°F	3.90 V
40°F	3.60 V
50°F	3.30 V
60°F	3.00 V
70°F	2.75 V
80°F	2.44 V
90°F	2.15 V
100°F	1.83 V
110°F	1.57 V
120°F	1.25 V

Hot Curve 1,000 ohm pull-up resistor switched in	
110°F	4.20 V
120°F	4.00 V
130°F	3.77 V
140°F	3.60 V
150°F	3.40 V
160°F	3.20 V
170°F	3.02 V
180°F	2.80 V
190°F	2.60 V
200°F	2.40 V
210°F	2.20 V
220°F	2.00 V
230°F	1.80 V
240°F	1.62 V
250°F	1.45 V

Fig. 18

1984 CHARGE TEMPERATURE SENSOR

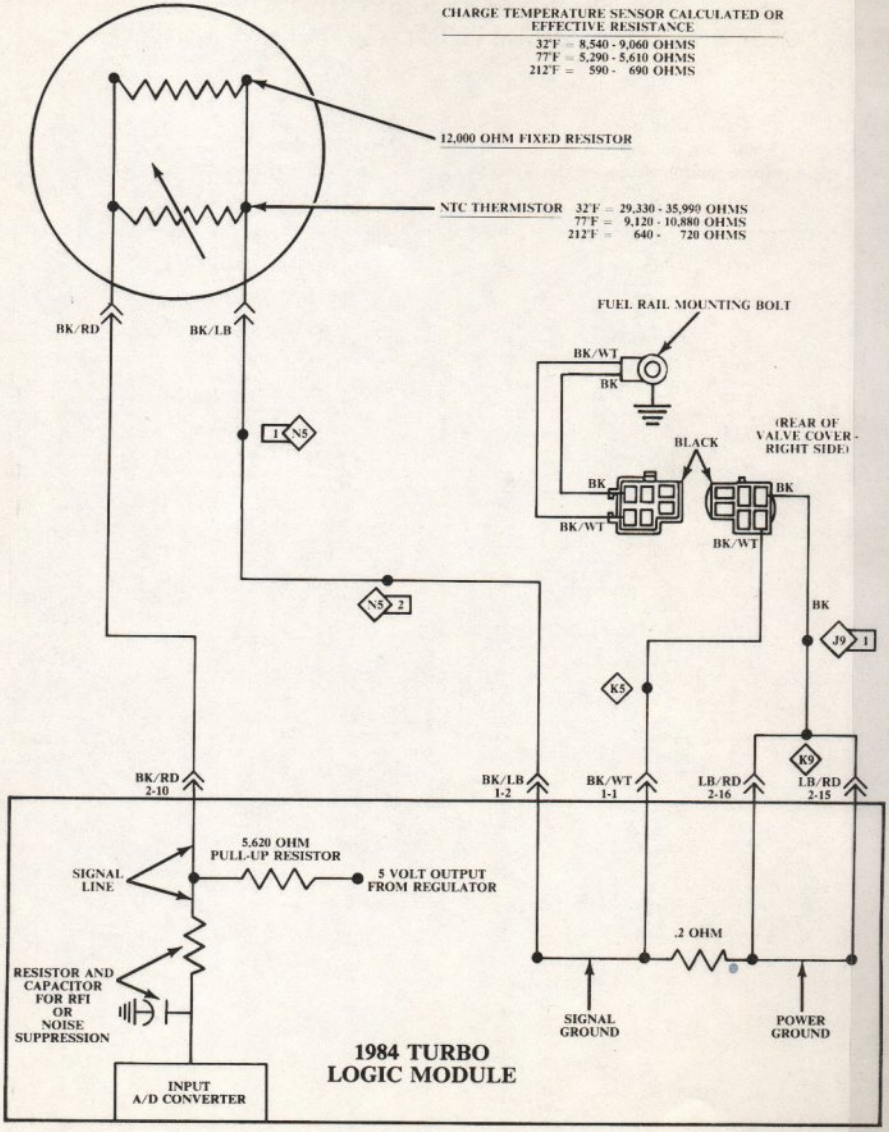


Fig. 19

Charge Temperature Sensor

The charge temperature sensor is a 2 wire sensor that is mounted in the intake manifold. The charge temperature sensor "gives" the logic module 1 basic piece of information:

1. Intake manifold air temperature

The charge temperature sensor "gives" the logic module this information by changing the voltage on the charge temperature sensor signal line inversely proportional to intake manifold air temperature. As the intake manifold air temperature increases, the charge temperature voltage decreases. As the intake manifold air temperature decreases, the charge temperature voltage increases. There are 2 types of charge temperature sensors, the 1984 style and the 1985, '86 & '87 style. The 1984 charge temperature sensor will be discussed first.

1984 Charge Temperature Sensor

Internally, the charge temperature sensor has a 12,000 ohm fixed resistor in parallel with a negative temperature coefficient (NTC) thermistor rated at 9,120 ohms - 10,880 ohms at 77°F. See Fig. 19. With these 2 resistors in parallel, that will make their calculated or effective resistance (what is measured across the terminals of the charge temperature sensor with an ohmmeter) 5,290 ohms -5,610 ohms at 77°F. See Fig. 22. This is the same as what's inside the 1984 coolant sensor.

1985, '86 & '87 Charge Temperature Sensor

This charge temperature sensor also has an NTC thermistor, however; there is no fixed resistor in parallel with the NTC thermistor. This is the same as what's inside the 1985, '86 & '87 coolant sensor. This makes the charge temperature sensor's resistance higher. It's rated at 9,120 ohms - 10,880 at 77°F. See Fig. 22.

1984, '85, '86 & '87 Charge Temperature Sensor and Circuit

Anytime the ignition key is "on", the logic module sends a regulated 5 volts to the 5,620 ohm pull-up resistor. The pull-up resistor is connected to the signal line. See Fig. 20. The reason for the pull-up resistor is two fold. First, the pull-up resistor uses some of the 5 volts it is fed with, when the circuit is complete and current is flowing. The voltage after the pull-up resistor is the charge temperature sensor signal voltage. This voltage leaves the logic module on the BK/RD wire and goes to the charge temperature sensor and then to signal ground at the fuel rail mounting bolt. This voltage is also present at the input to the A/D converter. The resistor and capacitor at the input to the A/D converter are for RFI or noise suppression.

The second reason for the pull-up resistor is to put 5 volts at the input of the A/D converter in case of an open circuit on the BK/RD wire or BK/LB wire. This "tells" the logic module there is an open circuit by the signal line always being high (5 volts). This happens because when there is no current flow a resistor doesn't use any voltage.

In summary, the charge temperature sensor signal voltage is taken after the 5,620 ohm pull-up resistor in the logic module. The charge temperature sensor signal voltage is actually the voltage remaining between the pull-up resistor and the charge temperature sensor. To calculate charge temperature sensor signal voltage, subtract the voltage drop across the pull-up resistor from the 5 volt source. For example, as the engine warms up, the charge temperature sensor resistance will decrease. This causes the voltage drop across the 5,620 ohm pull-up resistor to increase with engine temperature. As the voltage drop increases, the charge temperature sensor signal voltage will decrease.

Charge Temperature Sensor Fault Code

The logic module monitors the charge temperature sensor signal line (BK/RD wire) every 11 milliseconds (91 times a second), anytime the ignition key is "on". In 1984, the charge temperature sensor signal voltage must not be below .16 volts or above 3.76 volts. In 1985, '86 & '87, the charge temperature sensor signal voltage must not be below .06 volts or above 4.98 volts. If either of these limits are exceeded, the logic module will turn on the power loss/limited light. The system is now in "limp-in". Fault code 23 is stored in the logic module's memory. At this time the logic module will ignore the charge temperature sensor signal voltage and assume that the intake manifold air temperature is 70°F. This system remains in "limp-in" until 2 things happen:

1. The problem is fixed.
2. The ignition key is turned "off" then back "on" again.

1985 CHARGE TEMPERATURE SENSOR

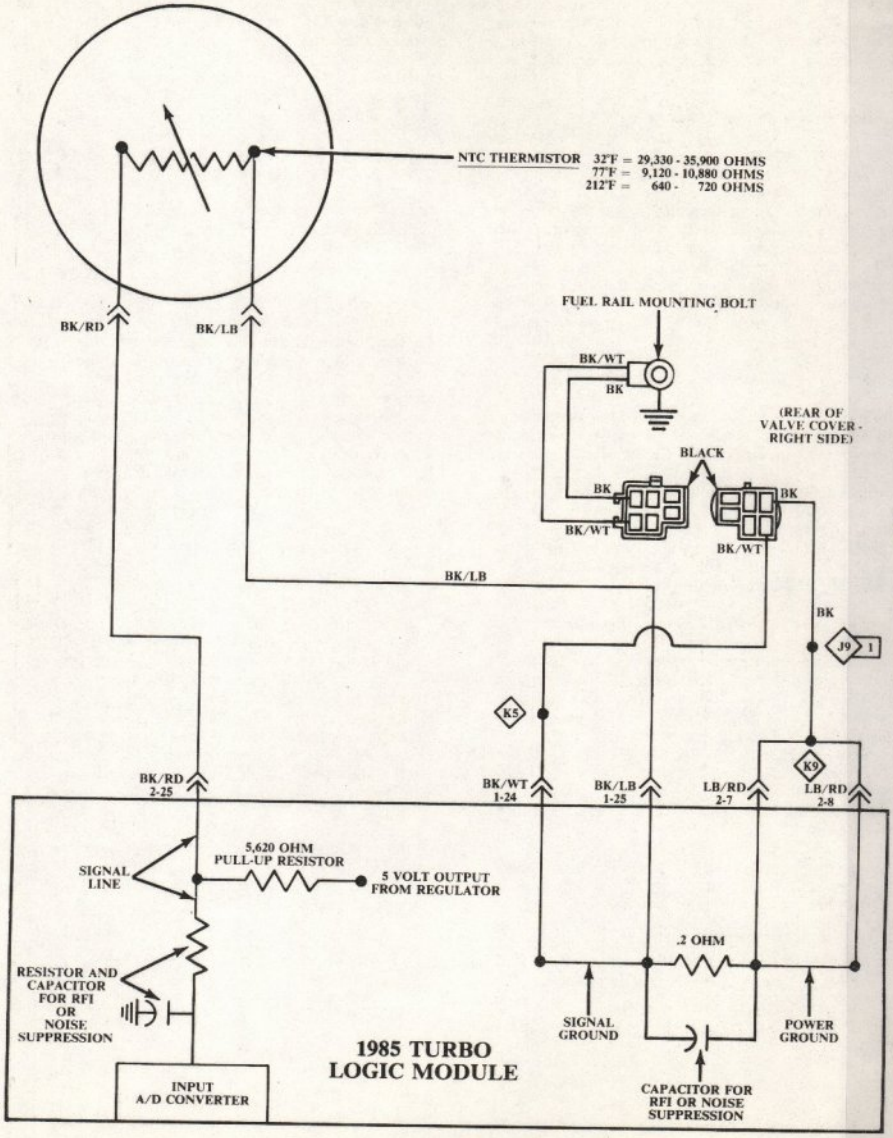


Fig. 20

1986 & '87 CHARGE TEMPERATURE SENSOR

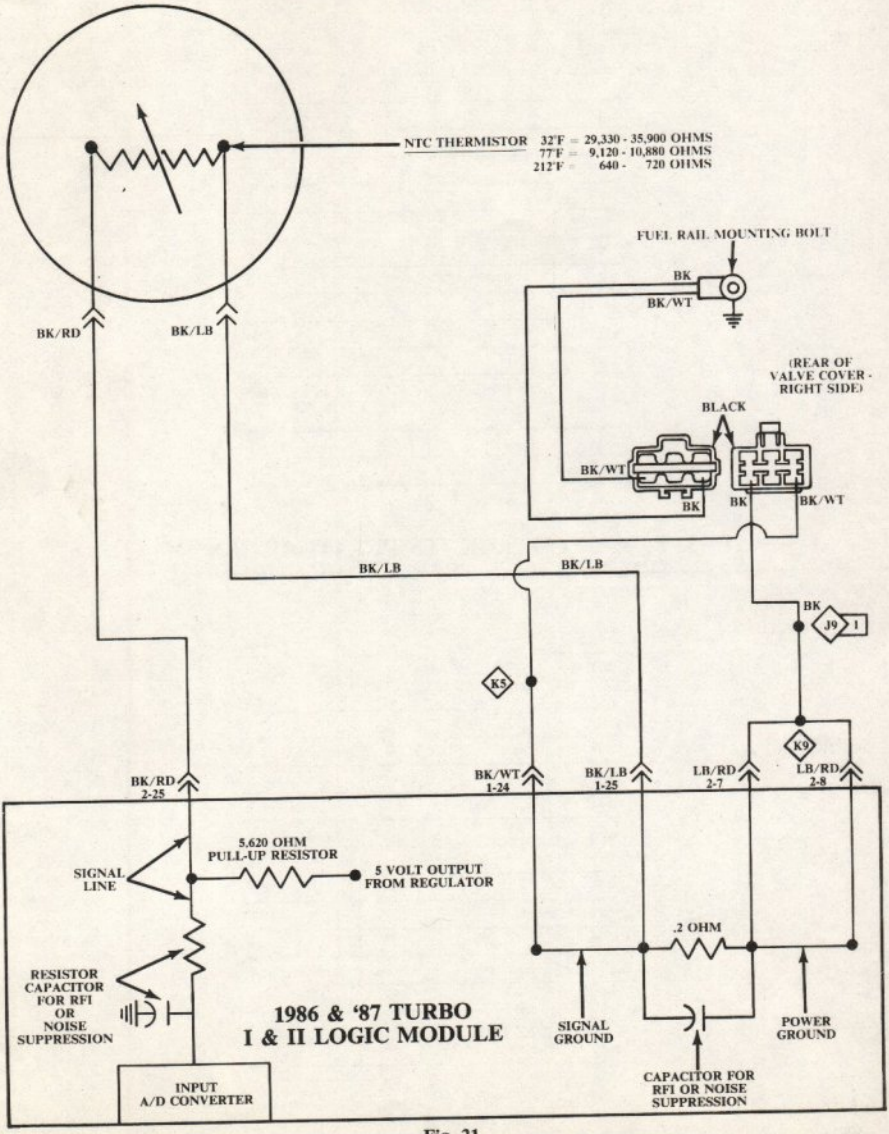


Fig. 21

1984 CHARGE TEMPERATURE SENSOR TEMPERATURE VS. VOLTAGE CURVE

Temperature	Voltage
−4°F	3.28 V
14°F	3.19 V
32°F	3.05 V
50°F	2.86 V
68°F	2.61 V
77°F	2.46 V
86°F	2.31 V
104°F	1.98 V
122°F	1.65 V
140°F	1.34 V
158°F	1.07 V
176°F	.84 V
194°F	.66 V
212°F	.51 V
230°F	.40 V
248°F	.32 V

1985, '86 & '87 CHARGE TEMPERATURE SENSOR TEMPERATURE VS. VOLTAGE CURVE

Temperature	Voltage
−20°F	4.81 V
0°F	4.70 V
20°F	4.47 V
40°F	4.11 V
60°F	3.67 V
80°F	3.08 V
100°F	2.51 V
120°F	1.97 V
140°F	1.52 V
160°F	1.15 V
180°F	.86 V
200−F	.65 V
220°F	.48 V
240°F	.35 V
260°F	.28 V

Fig. 22

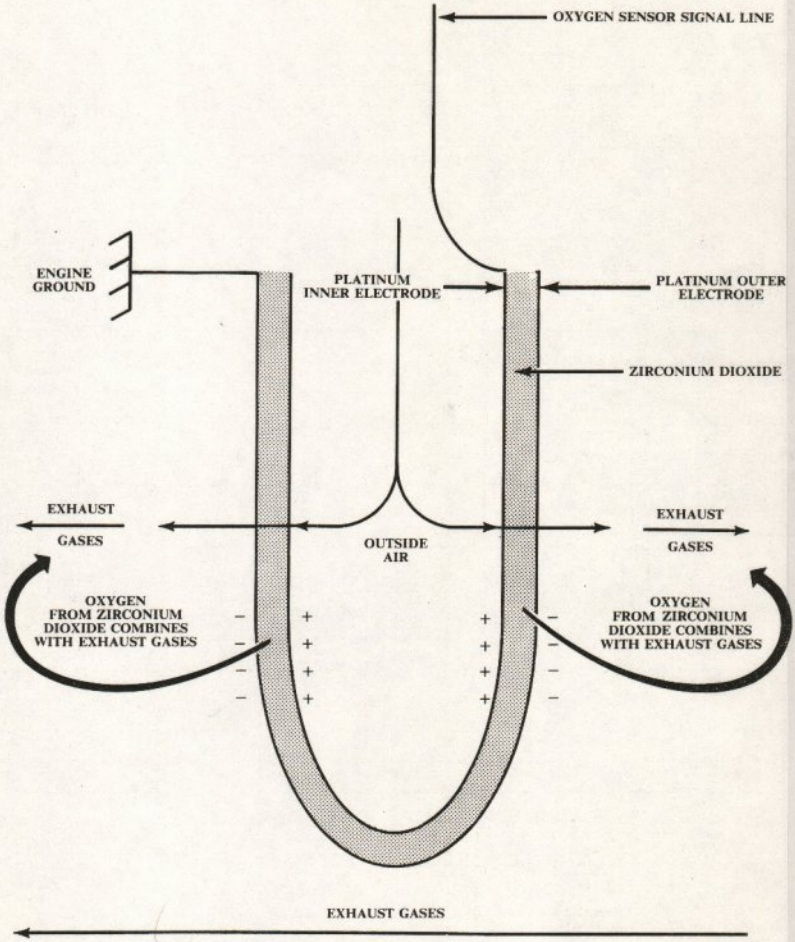


Fig. 23 Oxygen Sensor "Cone"

Oxygen (O_2) Sensor

The oxygen sensor is a 1 wire sensor in 1984, '85 & '86 and a 3 wire sensor in 1987. It is mounted in the exhaust side of the turbocharger housing. The oxygen sensor "gives" the logic module (the microprocessor) 1 basic piece of information:

1. Exhaust oxygen content (either a lean or a rich condition)

The oxygen sensor "gives" the logic module this information by sending a voltage that is dependant on the air/fuel ratio once it has warmed up to its operating temperature. There is 1 type of oxygen sensor. The 1987 oxygen sensor has a heater in it and this accounts for the other 2 wires. There are 2 types of oxygen sensor circuits, the 1984 style and the 1985, '86 & '87 style. The oxygen sensor will be discussed first.

1984, '85, '86 & '87 Oxygen Sensor

Internally, the oxygen sensor has zirconium dioxide that is shaped like a "cone". See Fig. 23. The zirconium dioxide is the electrolyte. The inside and outside of the zirconium dioxide "cone" is coated with platinum to form electrodes. The inner electrode is exposed to the air that is outside the exhaust manifold. The outer electrode is exposed to the exhaust gases. At temperatures below 662°F, the zirconium dioxide does not act as electrolyte and the oxygen sensor will not generate a voltage of its own. Once the oxygen sensor is heated up to 662°F, the zirconium dioxide becomes active. Instead of dissimilar metals in the presence of electrolyte (how a car battery generates voltage), the oxygen sensor uses dissimilar amounts of oxygen in the presence of electrolyte (the zirconium dioxide) to generate its voltage. The oxygen sensor's voltage will vary from a low of 0 volts (a lean condition) to a high of 1 volt (a rich condition) depending upon the difference between the oxygen levels present at the two electrodes.

Two conditions must be met before the oxygen sensor will generate a voltage of its own. First, there must be a rich condition. Second, the zirconium dioxide "cone" must be heated up to a minimum of 662°F. Under these conditions the zirconium dioxide "cone" will give up some of its oxygen atoms to the exhaust gases. See Fig. 23. This happens because there is not enough oxygen in the exhaust to combine with the other exhaust gases. As the oxygen migrates out of the zirconium dioxide "cone", the oxygen atoms deposit some of their electrons on the outer electrode leaving it with a negative charge. The outer electrode is connected to engine ground. The outside air that is around the inner electrode will now pass through the zirconium dioxide "cone" into the exhaust stream. See Fig. 23. The oxygen that is in the air will collect some electrons from the inner electrode as oxygen passes through the zirconium dioxide "cone". This will give the inner electrode a positive charge. The oxygen sensor signal wire is connected to the inner electrode.

The leaner the mixture, the more oxygen that is around the outer electrode. This causes less oxygen from the zirconium dioxide "cone" to migrate into the exhaust stream. That in turn causes less electrons to be deposited on the outer electrode. That results in less oxygen from the air passing through the zirconium dioxide "cone" and collecting less electrons from the inner electrode. The result is a lower charge at the inner and outer electrodes which will be seen as a lower oxygen sensor signal voltage.

For 1987 there is a positive temperature coefficient (PTC) heater inside the zirconium dioxide "cone". The reason for the heater is to keep exhaust contaminants off the zirconium dioxide "cone". This increases oxygen sensor life to 120,000 miles. The heater is fed 12 volts (WT wire) with the ignition key "on" and provided a ground (WT wire). Because the heater is a PTC heater, it will increase its resistance with an increase in temperature. This regulates the temperature of the heater. The tip of the zirconium dioxide "cone" will be heated up to 932°F - 1292°F without any heat from the exhaust gases. The heater is made up of platinum coated ceramic and its resistance should be 5 - 10 ohms at 77°F.

1984 Oxygen Sensor and Circuit

Anytime the ignition key is "on", the logic module sends a regulated 5 volts to the 10 million ohm pull-up resistor. See Fig. 24. The pull-up resistor is fed 5 volts on one end and the other end is connected to the signal line. Connected in series with the 10 million ohm pull-up resistor is a 1 million ohm pull-down resistor. A pull-down resistor is connected to signal ground on one end and the other end is connected to the signal line. With both these resistors hooked up this way they do one thing; act as an open circuit indicator. In the case of an open circuit the signal line will go up or be "pulled-up" to .45 volts. This "tells" the logic module that there is an open circuit. Both the pull-up and pull-down resistors are needed the oxygen sensor circuit because the maximum voltage on the oxygen sensor signal line should be 1 volt. The lowest voltage available is 5 volts. If just the pull-up resistor was used, open circuit voltage would be 5 volts. This voltage can be lowered by the use of a pull-down resistor. Using Ohm's Law, you can calculate how .45 volts is obtained from a 5 volt source. First the total circuit resistance must be calculated. This is done by adding the pull-up resistor (10 million ohms) with the pull-down resistor (1 million ohms). This equals 11 million ohms. Next the current flow must be calculated. This is done by dividing the voltage source (5 volts) by the total circuit resistance (11 million ohms). This equals .0000004 amps. Now,

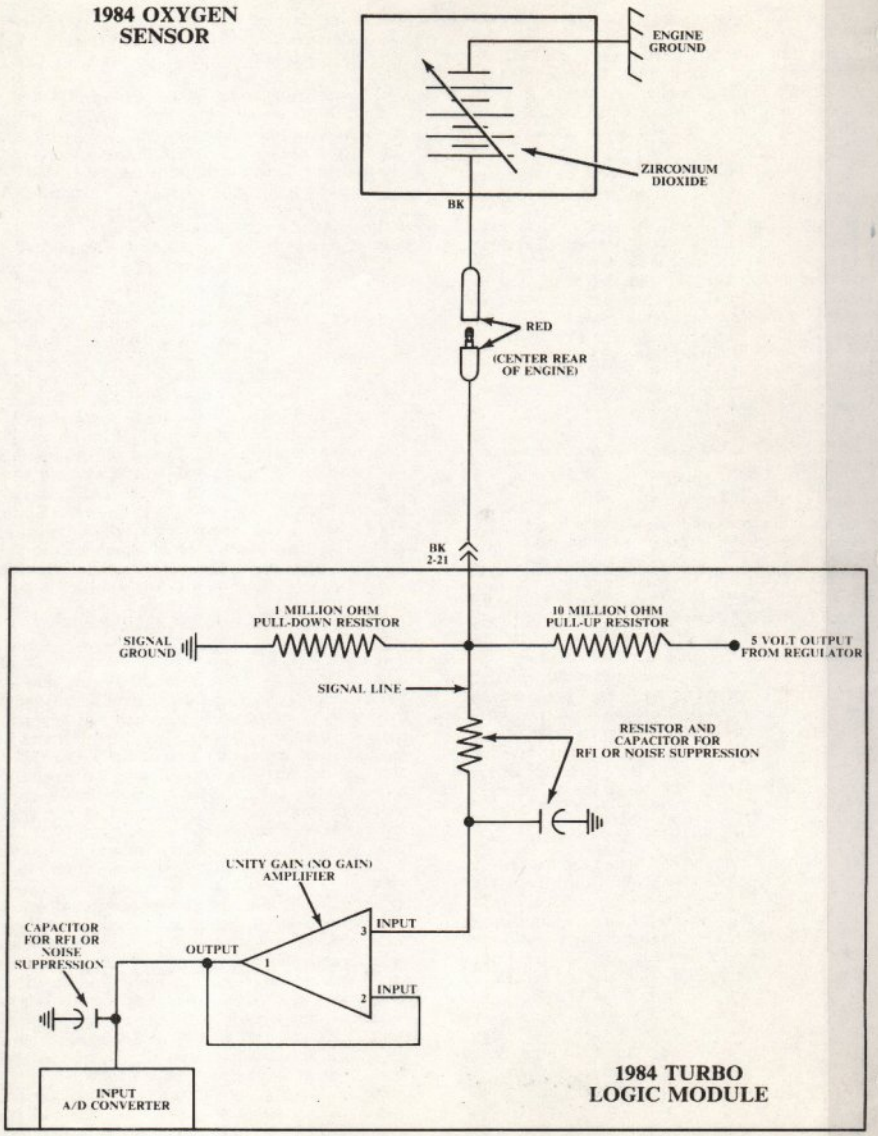

1984 OXYGEN SENSOR
ENGINE GROUND
ZIRCONIUM DIOXIDE
BK
RED
(CENTER REAR OF ENGINE)
BK 2-21
1 MILLION OHM PULL-DOWN RESISTOR
10 MILLION OHM PULL-UP RESISTOR
SIGNAL GROUND
5 VOLT OUTPUT FROM REGULATOR
SIGNAL LINE
RESISTOR AND CAPACITOR FOR RFI OR NOISE SUPPRESSION
UNITY GAIN (NO GAIN) AMPLIFIER
CAPACITOR FOR RFI OR NOISE SUPPRESSION
OUTPUT
INPUT
INPUT
INPUT A/D CONVERTER
1984 TURBO LOGIC MODULE

Fig. 24

multiply the current flow (.0000004 amps) by the pull-up resistor (10 million ohms). This equals the voltage drop across the pull-up resistor (4.5454 volts). Subtract the voltage drop (4.5454 volts) from the voltage source (5 volts). This equals the oxygen sensor open circuit voltage of .45 volts. This .45 volts goes 2 places. One place the .45 volts goes is on the signal line to input pin "3" of the unity gain amplifier. See Fig. 24. There is a resistor in series and a capacitor in parallel with the signal line to the unity gain amplifier. These are for RFI or noise suppression. A unity gain amplifier is an amplifier that doesn't amplify (no gain). The output of the unity gain amplifier at pin "1" is equal to the input voltage at pin "3". See Fig. 24. So what ever voltage that is on the signal line is also applied to the input of the A/D converter. The reason for the unity gain amplifier is that it has very high resistance at its input pins. This is needed because the oxygen sensor can only put out a limited amount of current (microamps). The oxygen sensor, like any battery, once its current capabilities have been exceeded, the voltage drops off. This would make the oxygen sensor signal voltage inaccurate. The very high resistance of the unity gain amplifier at its input pins limits the current flow out of the oxygen sensor. The unity gain amplifier has low resistance at its output pin "1". This is needed by the A/D converter. The output of the unity gain amplifier at pin "1" is connected to the input of the A/D converter. See Fig. 24. There is a capacitor connected to the input of the A/D converter. This is for RFI or noise suppression.

Let's review what was just covered. There is a pull-up and pull-down resistor that act as a open circuit indicator. The voltage that is on the signal line is applied to the input of the A/D converter. The second place the .45 volts goes is out of the logic module on the BK wire to the oxygen sensor. With a cold oxygen sensor (below 200°F), its internal resistance will be very high or effectively infinite (above 100 million ohms) resistance. This makes the voltage that is seen on the oxygen sensor signal line (BK wire) when the ignition key is first turned "on" .45 volts. This is true because the resistance of the cold oxygen sensor is the same as an open circuit. Once the engine is started and the oxygen sensor begins to warm up, its internal resistance begins to drop. This happens before the oxygen sensor starts generating a voltage of its own. What will be seen is the .45 volts slowly decreasing as the oxygen sensor first begins to warm up. This happens because the oxygen sensor's resistance has fallen low enough for it to become a path to ground. This makes the 10 million ohm pull-up resistor effectively a larger and larger resistor in the circuit. The result is a lower and lower oxygen sensor signal voltage as the oxygen sensor resistance decreases. Soon after the .45 volts starts to drop, at some point, the oxygen sensor is up to operating temperature (662°F) and starts generating a voltage of its own. Now the oxygen sensor signal voltage begins to rise to indicate the oxygen content of the exhaust.

As the oxygen sensor's voltage varies up and down, this voltage is sent to the logic module on the BK wire. This goes to input pin "3" of the unity gain amplifier. See Fig. 24. The output of the unity gain amplifier at pin "1" is the same as the oxygen sensor's voltage. This voltage is then sent to the input of the A/D converter and converted so the logic module can "understand" the voltage signal.

1985, '86 & '87 Oxygen Sensor And Circuit

Anytime the ignition key is "on", the logic module sends a regulated 5 volts to the 100,000 ohm pull-up resistor. See Fig. 25. Connected in series with the 100,000 ohm pull-up resistor is a 10,000 ohm pull-down resistor. These 2 resistors work the same as in the 1984 circuit and still generate the same .45 volts open circuit voltage. This is true because the pull-up resistor is still 10 times greater than the pull-down resistor. The .45 volts then goes to a 2 million ohm resistor. See Fig. 25. This resistor is there to prevent the oxygen sensor voltage from going to an "easy" ground through the 10,000 ohm pull-down resistor. This limits the oxygen sensor's current flow by adding high resistance to the oxygen sensor circuit. The 2 million ohm resistor is connected to the signal line. From this point the voltage goes 2 places. One place the .45 volts goes is to input pin "3" of the comparator. See Fig. 25. There is a resistor in series and a capacitor in parallel with the signal line. These are for RFI or noise suppression.

A comparator is a device that when hooked up as in Fig. 25 it will act as a unity gain amplifier. That is, input pin "3" of the comparator will be the same as the output pin "1". The reason for the comparator is that higher resistance was needed at the input pins which was not possible with a unity gain amplifier. So a comparator was used instead of a unity gain amplifier. The output pin "1" of the comparator is connected with the input of the A/D converter. The resistor and capacitor on the output of the comparator these are for RFI or noise suppression.

The second place the .45 volts goes is out of the logic module on the BK wire to the oxygen sensor. From this point on, the circuit acts the same as previously mentioned. The output of the oxygen sensor eventually goes to the input of the A/D converter and is converted so the logic module can "understand" the voltage signal.

Oxygen Sensor Fault Codes

The logic module monitors the oxygen sensor signal line when the system is in closed loop. Closed loop starts after a 7.39 second timer runs out. The 7.39 second timer is started at different coolant sensor

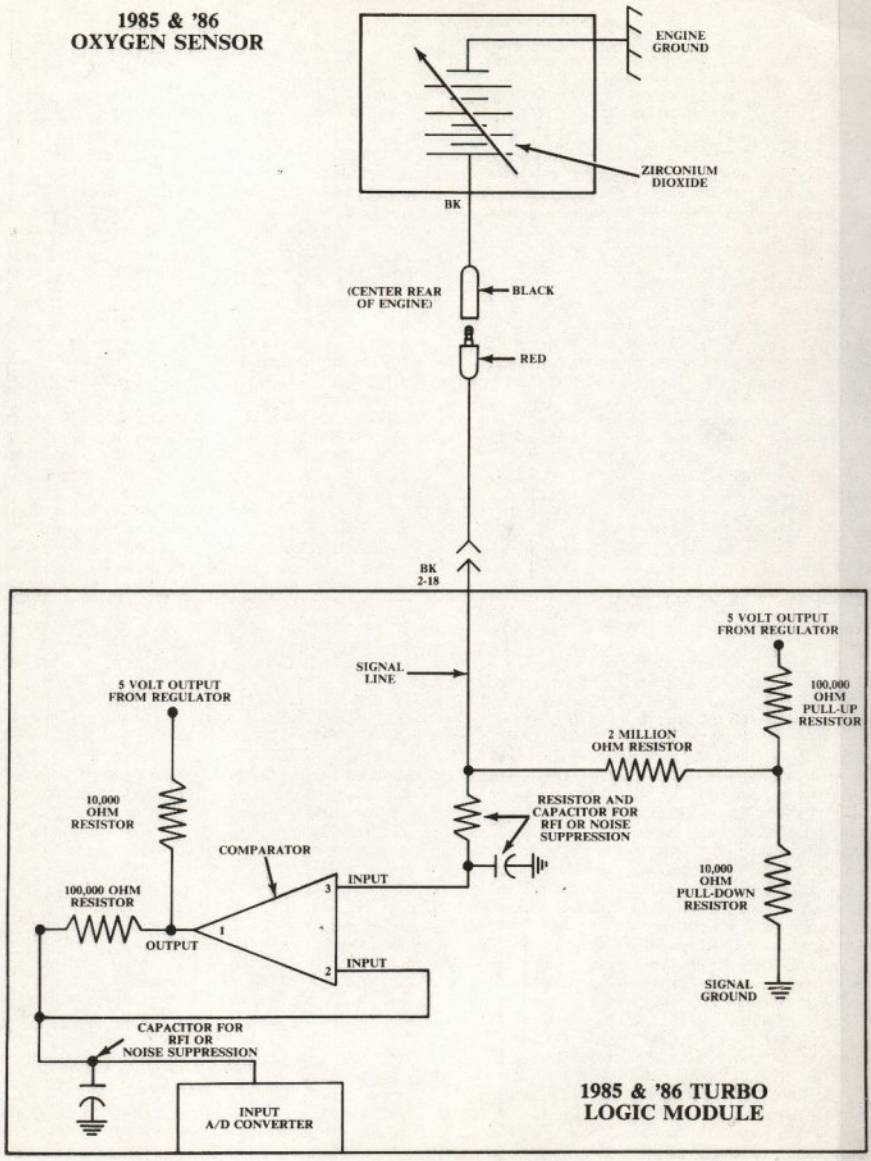

1985 & '86
OXYGEN SENSOR
ENGINE
GROUND
ZIRCONIUM
DIOXIDE
BK
(CENTER REAR
OF ENGINE)
BLACK
RED
BK
2-18
5 VOLT OUTPUT
FROM REGULATOR
SIGNAL
LINE
5 VOLT OUTPUT
FROM REGULATOR
100,000
OHM
PULL-UP
RESISTOR
2 MILLION
OHM RESISTOR
10,000
OHM
RESISTOR
RESISTOR AND
CAPACITOR FOR
RFI OR NOISE
SUPPRESSION
COMPARATOR
100,000 OHM
RESISTOR
INPUT
10,000
OHM
PULL-DOWN
RESISTOR
OUTPUT
INPUT
SIGNAL
GROUND
CAPACITOR FOR
RFI OR
NOISE SUPPRESSION
INPUT
A/D CONVERTER
1985 & '86 TURBO
LOGIC MODULE

Fig. 25

temperatures depending on the model year. See Fig. 27 for specific temperatures. So once the engine is started, a certain coolant sensor temperature must be reached. Once this temperature is reached, the 7.39 second timer starts. Once the timer runs out, the system goes into closed loop. 59 seconds after closed loop in 1984 or 718 seconds (11 minutes 58 seconds) after closed loop in 1985 - '87, the logic module begins monitoring the neutral region (neither lean nor rich) of the oxygen sensor signal voltage. There are different neutral regions depending on the model year. See Fig. 28 for specific neutral region voltages. If the oxygen sensor signal voltage remains in the neutral region for more than 21 seconds, after the previously mentioned timers run out, fault code 21 is stored in the logic module's memory. Once fault code 21 is stored in memory, the power loss/limited light does not come on, however; the logic module still monitors the oxygen sensor signal line. The system will be allowed to go rich but is prevented from going lean.

In 1984, once in closed loop, if the oxygen sensor signal voltage stays above or below the neutral region for more than 121 seconds (2 minutes 1 second), fault code 51 is stored in the logic module's memory.

In 1985, once in closed loop, if the oxygen sensor signal voltage stays below the neutral region for more than 121 seconds, fault code 51 is stored in the logic module's memory. If the oxygen sensor signal voltage stays above the neutral region for more than 121 seconds, fault code 52 is stored in the logic module's memory.

In 1986 & '87, once in closed loop, if the oxygen sensor signal voltage stays below the neutral region for more than 715 seconds (11 minutes 55 seconds), fault code 51 is stored in the logic module's memory. If the oxygen sensor signal voltage stays above the neutral region for more than 715 seconds, fault code 52 is stored in the logic module's memory.

In 1984 - '87, once fault code 51 or 52 is stored in memory, the logic module continues to monitor the oxygen sensor signal line. If either fault code persists for 4 times what it took to set the fault code originally, the logic module will allow the system to stay rich but will prevent the system from going lean.

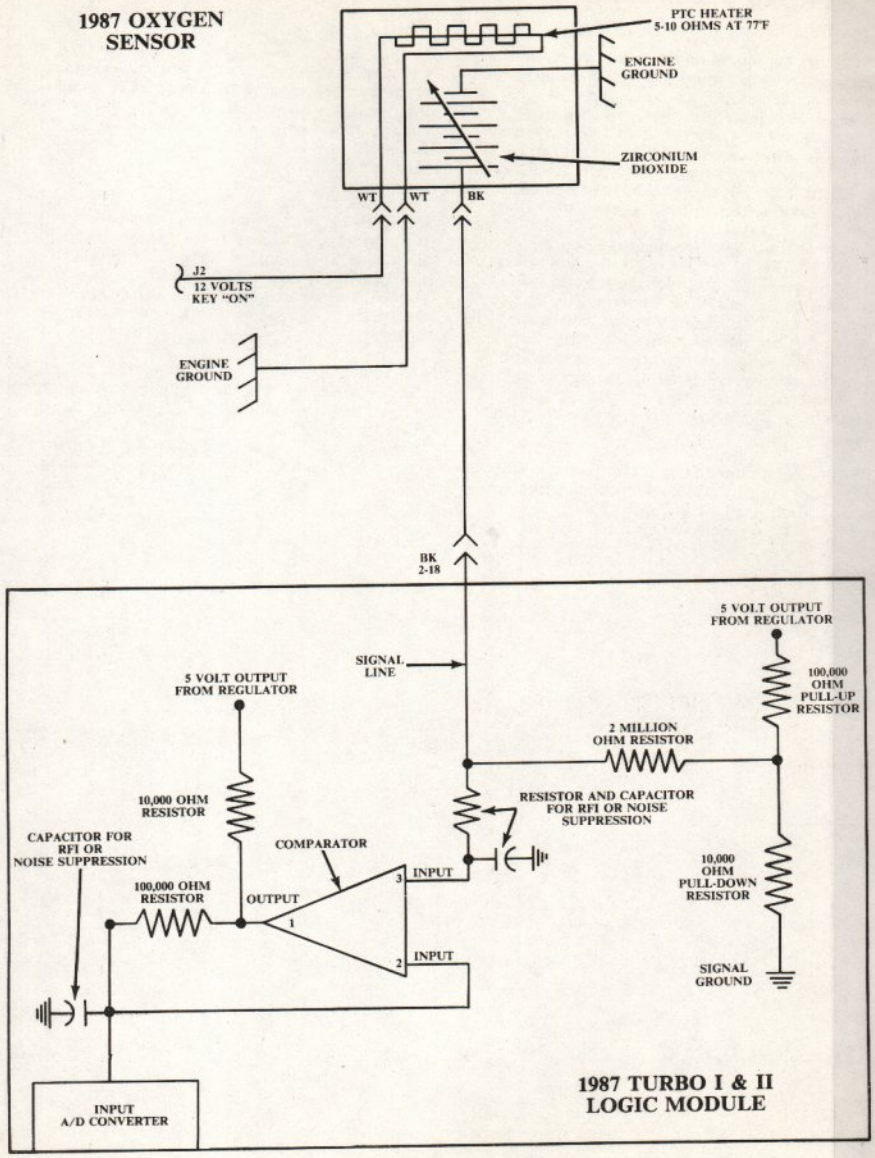

1987 OXYGEN SENSOR
PTC HEATER 5-10 OHMS AT 77°F
ENGINE GROUND
ZIRCONIUM DIOXIDE
WT
WT
BK
J2
12 VOLTS KEY "ON"
ENGINE GROUND
BK 2-18
5 VOLT OUTPUT FROM REGULATOR
SIGNAL LINE
5 VOLT OUTPUT FROM REGULATOR
100,000 OHM PULL-UP RESISTOR
2 MILLION OHM RESISTOR
10,000 OHM RESISTOR
RESISTOR AND CAPACITOR FOR RFI OR NOISE SUPPRESSION
CAPACITOR FOR RFI OR NOISE SUPPRESSION
COMPARATOR
INPUT
10,000 OHM PULL-DOWN RESISTOR
100,000 OHM RESISTOR
OUTPUT
INPUT
SIGNAL GROUND
INPUT A/D CONVERTER
1987 TURBO I & II LOGIC MODULE

Fig. 26

MODEL YEAR	COOLANT SENSOR TEMPERATURE
1984	89.6°F
1985	95.0°F
1986	95.0°F
1987 TURBO I	95.0°F
1987 TURBO II	80.6°F

Fig. 27 Closed Loop Timer Start Temperatures

MODEL YEAR	LEAN	NEUTRAL REGION	RICH
1984	←	.391-.547 VOLTS	→
1985	←	.352-.547 VOLTS	→
1986	←	.352-.547 VOLTS	→
1987	←	.352-.547 VOLTS	→

Fig. 28 Oxygen Sensor Neutral Regions

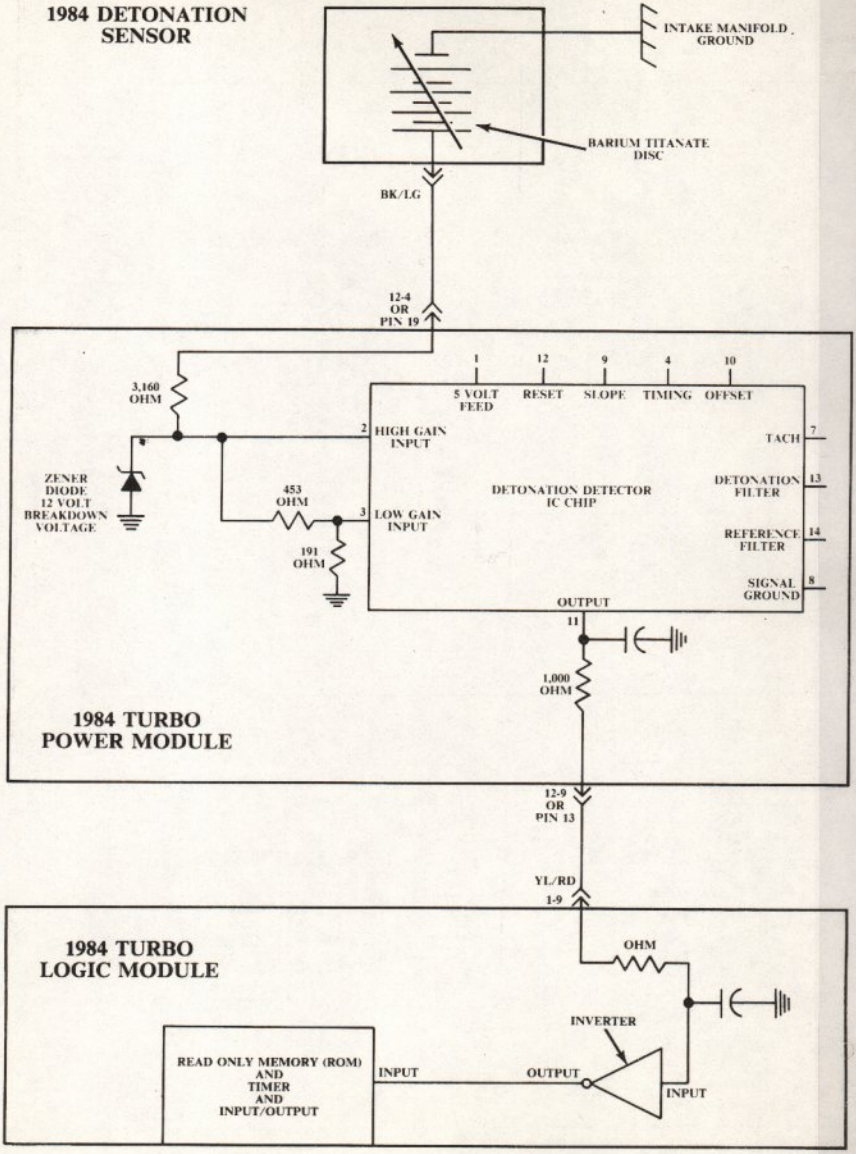

1984 DETONATION SENSOR
INTAKE MANIFOLD GROUND
BARIUM TITANATE DISC
BK/LG
12-4 OR PIN 19
1
12
9
4
10
5 VOLT FEED
RESET
SLOPE
TIMING
OFFSET
3,160 OHM
2
HIGH GAIN INPUT
TACH
7
ZENER DIODE 12 VOLT BREAKDOWN VOLTAGE
453 OHM
3
LOW GAIN INPUT
DETONATION DETECTOR IC CHIP
DETONATION FILTER
13
REFERENCE FILTER
14
191 OHM
SIGNAL GROUND
8
OUTPUT
11
1,000 OHM
1984 TURBO POWER MODULE
12-9 OR PIN 13
YL/RD
1-9
1984 TURBO LOGIC MODULE
OHM
INVERTER
READ ONLY MEMORY (ROM) AND TIMER AND INPUT/OUTPUT
INPUT
OUTPUT
INPUT

Fig. 29

Detonation Sensor

The detonation sensor is a 1 wire sensor that is mounted in the intake manifold. The detonation sensor "gives" the logic module (the microprocessor) 1 basic piece of information:

1. When the engine is detonating

The detonation sensor "gives" the logic module this information by sending out an a.c. voltage sine wave that is proportional to engine detonation intensity. There is 1 type of detonation sensor for 1984, '85, '86 & '87. There are 2 types of detonation sensor circuits, the 1984 style and the 1985, '86 & '87 style. The detonation sensor will be discussed first.

1984, '85, '86 & '87 Detonation Sensor

Internally the detonation sensor has a piezoelectric crystal that is shaped like a disc. The disc is made of a barium titanium compound called barium titanate. The disc measures approximately $\frac{1}{8}$" to $\frac{1}{2}$" in diameter and is .010" to .020" thick. The disc is coated with silver on both sides to form electrodes. The compound is called piezoelectric because it will output a voltage when squeezed or flexed. When the tone or pitch of the sound is in the 6,000 to 6,300 herz range, the detonation sensor outputs a voltage. The amount of the voltage is dependent on the volume or amplitude of the detonation in the engine. The louder the detonation, the larger the a.c. sine wave (higher voltage). The quieter the detonation, the smaller the a.c. sine wave (lower voltage).

When the engine is detonating, it is actually vibrating violently. This vibration flexes the barium titanate disc back and forth creating the a.c. voltage. The vibration can be measured in G forces of acceleration. At 1 G force of acceleration (32 feet per second second), the detonation sensor outputs .4 volts. The output of the detonation sensor may go as high as 6 or 8 volts under severe detonation. This voltage sine wave happens so fast that an oscilliscope is needed to see the actual output of the detonation sensor.

1984 Detonation Sensor and Circuit

Anytime the engine is detonating (above .8 volts at 2,000 RPM or above 4 volts at 4,000 RPM), the detonation sensor outputs an a.c. voltage on the BK/LG wire. See Fig. 29. This voltage goes into the power module to a detonation detector IC chip. The output of the detonation detector IC chip is now low (0 volts) when the engine is detonating. This low voltage leaves the power module on the YL/RD wire and goes into the logic module to an inverter See Fig. 29. The inverter takes the low voltage (0 volts) and inverts it into a high voltage (5 volts). The 5 volts from the inverter then goes to the I/O (Input/Output). See Fig. 29. The microprocessor uses the I/O to "read" digital inputs like the detonation detector IC chip. Now the logic module begins taking away the maximum allowed advance (retards timing) by 2° on all 4 cylinders. If the voltage remains high at the I/O in the logic module, the maximum allowed advance is lowered to 14° less than that of normal timing. If the engine stops detonating due to the maximum timing advance being reduced, the logic module will return the system to normal timing.

If poor engine performance is suspected due to the logic module taking away the maximum allowed advance, the diagnostic readout box (C-4805) can be used to confirm this. Plug in the diagnostic readout box and take the car for a test drive. If the logic module "hears" detonation and is reducing the maximum timing advance, the diagnostic readout box will display an "80" or "81".

1985, '86 & '87 Detonation Sensor and Circuit

Anytime the engine is detonating (above .8 volts at 2,000 RPM or above 4 volts at 4,000 RPM), the detonation sensor outputs an a.c. voltage on the BK/LG wire. This voltage goes into the logic module to the base circuit of Transistor 1. See Fig. 30. The resistor and capacitor in parallel with the base circuit of Transistor 1 stabilizes the base circuit voltage. Transistor 1 acts as diode. That is, any positive voltage on the base of Transistor 1 is allowed to go to the collector and charge up Capacitor 1. See Fig. 30. Since the voltage output of the detonation sensor is a.c. voltage, the voltage impressed at the base of Transistor 1 will go from a positive voltage to a negative voltage. The positive voltage that has charged up Capacitor 1 now "wants" to go back to the detonation sensor. This voltage is blocked at the collector of Transistor 1. In this way Transistor 1 acts as a diode, allowing current flow in one direction (charging up Capacitor 1), and blocking it in the other direction.

Once a capacitor is charged up with voltage, it will keep that charge for a very long time or until it is discharged to ground. The voltage stored in Capacitor 1 is impressed at the input to the A/D converter. See Fig. 30. The logic module monitors the A/D connector every time the coil is fired. Once the engine has detonated, Capacitor 1 voltage level increases. The logic module now takes away the maximum allowed advance on just the cylinder that is detonating (as determined by the sync signal) by 4.2° up to a maximum of 2.8° when there is part throttle and a no boost condition. When the engine is in boost, 2.1° up to a maximum of 15° is taken away from total timing. After the coil has been fired the logic module monitors the A/D converter again to "see" if the voltage has been lowered at Capacitor 1. Remember it was said earlier that a capacitor will hold its charge until allowed to go to ground.

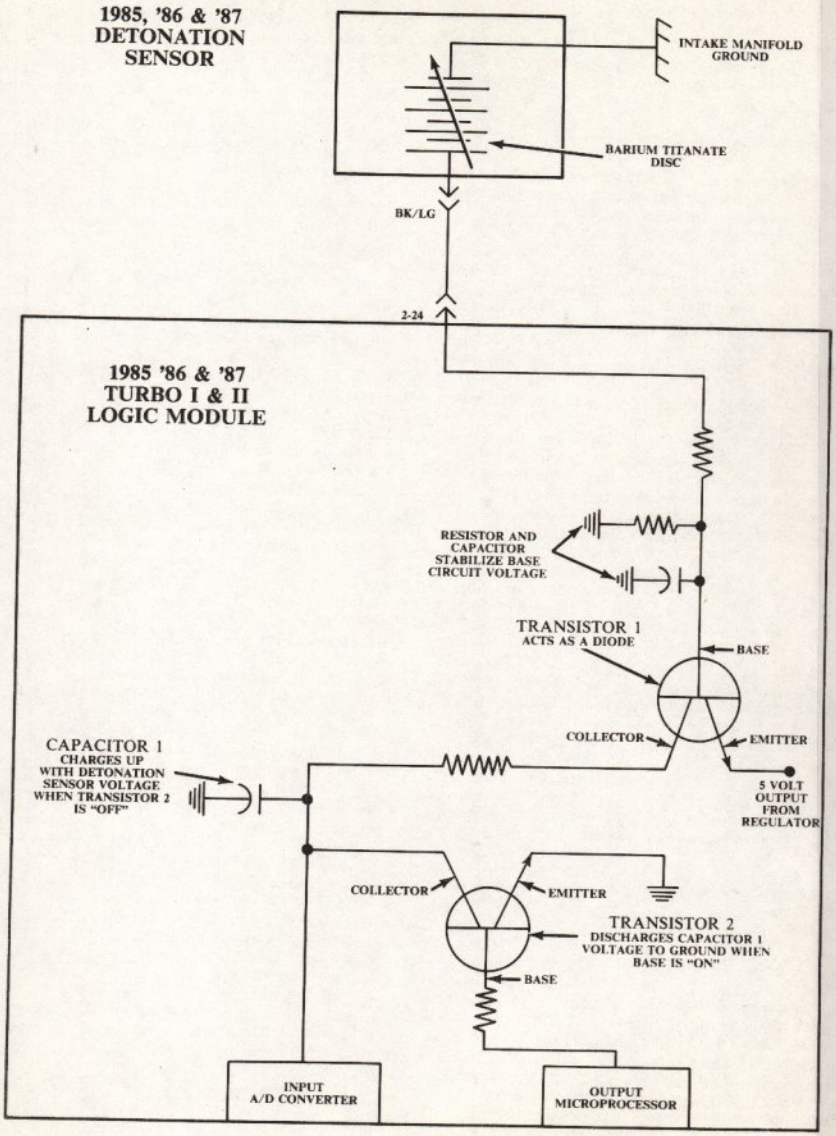

1985, '86 & '87
DETONATION
SENSOR
INTAKE MANIFOLD
GROUND
BARIUM TITANATE
DISC
BK/LG
2-24
1985 '86 & '87
TURBO I & II
LOGIC MODULE
RESISTOR AND
CAPACITOR
STABILIZE BASE
CIRCUIT VOLTAGE
TRANSISTOR 1
ACTS AS A DIODE
BASE
COLLECTOR
EMITTER
CAPACITOR 1
CHARGES UP
WITH DETONATION
SENSOR VOLTAGE
WHEN TRANSISTOR 2
IS "OFF"
5 VOLT
OUTPUT
FROM
REGULATOR
COLLECTOR
EMITTER
TRANSISTOR 2
DISCHARGES CAPACITOR 1
VOLTAGE TO GROUND WHEN
BASE IS "ON"
BASE
INPUT
A/D CONVERTER
OUTPUT
MICROPROCESSOR

Fig. 30

Let's review the current situation. Detonation has happened and Capacitor 1 has charged up to a higher voltage. The logic module takes corrective action and begins to take away timing advance. The logic module now wants to "see" if the voltage at the input to the A/D converter has come down. However; the voltage at Capacitor 1 remains high. At this time the microprocessor sends out a positive voltage to the base of Transistor 2. See Fig. 30. This turns "on" Transistor 2 and allows Capacitor 1 to discharge to ground. Once Capacitor 1 has been discharged down to a lower voltage, the microprocessor turns "off" the base of Transistor 2 and Capacitor 1 now charges up to the present detonation sensor voltage. The logic module "looks to see" if the voltage at the input to the A/D converter has come down due to the decrease in timing.

If the voltage remains high at the input to the A/D converter, the logic module will take away up to a total of 15° of maximum allowed timing of the cylinder that is detonating. All other cylinders will remain at the normal timing. The logic module will keep the timing taken away until Capacitor 1 voltage drops off back to normal.

Also, when the engine is detonating, the logic module reduces boost along with timing in order to eliminate detonation. This is done by changing the duty cycle voltage at the wastegate solenoid. When the engine is in boost, if detonation happens more than once every 5½ seconds, the MAP voltage is lowered to a maximum of 8% of what it was. If detonation happens more than once between 5½ seconds and 17 seconds, the MAP voltage is lowered at a percentage based on frequency of detonation, up to the maximum of 8%. If detonation happens only once every 18 seconds or more, there is no reduction in MAP voltage even though the engine is in boost.

Also, for this circuit, if poor engine performance is suspected due to detonation, the diagnostic readout box may be used to confirm this in the same manner as in 1984.

Detonation Sensor Fault Code 1985 Only

Anytime the engine is above 5,000 rpm, the detonation sensor must output at least .04 volts for 3 seconds. If either the time or the voltage limits are not met, the logic module will turn on the power loss/limited light.

If the detonation sensor ever sends out more than .04 volts, the logic module turns off the power loss/limited light. However; fault code 17 remains in the logic module's memory.

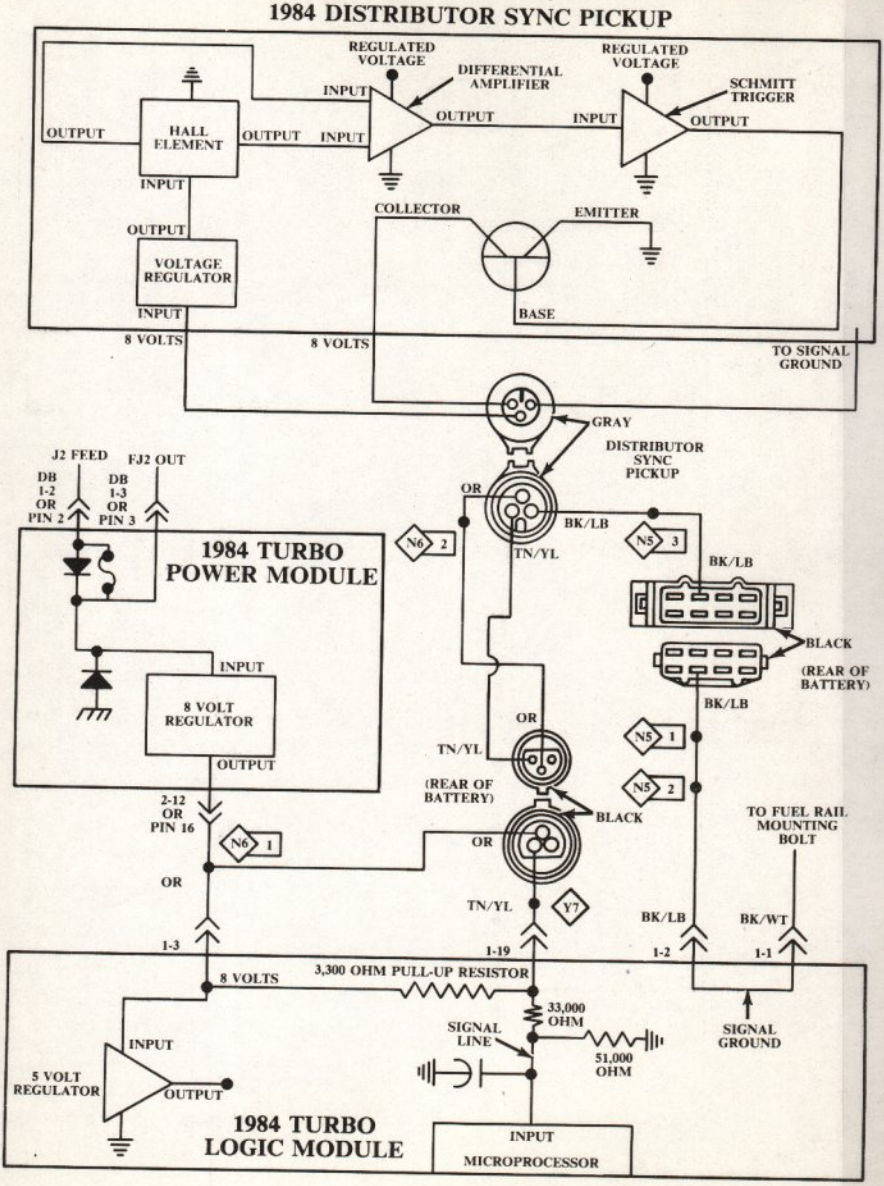

1984 DISTRIBUTOR SYNC PICKUP
REGULATED VOLTAGE
DIFFERENTIAL AMPLIFIER
REGULATED VOLTAGE
SCHMITT TRIGGER
INPUT
OUTPUT
HALL ELEMENT
OUTPUT
INPUT
OUTPUT
INPUT
OUTPUT
INPUT
OUTPUT
COLLECTOR
EMITTER
VOLTAGE REGULATOR
INPUT
BASE
8 VOLTS
8 VOLTS
TO SIGNAL GROUND
GRAY
DISTRIBUTOR SYNC PICKUP
J2 FEED
FJ2 OUT
DB 1-2 OR PIN 2
DB 1-3 OR PIN 3
OR
BK/LB
N6 2
N5 3
TN/YL
BK/LB
1984 TURBO POWER MODULE
BLACK
(REAR OF BATTERY)
INPUT
8 VOLT REGULATOR
BK/LB
OR
N5 1
TN/YL
N5 2
OUTPUT
(REAR OF BATTERY)
BLACK
TO FUEL RAIL MOUNTING BOLT
2-12 OR PIN 16
N6 1
OR
OR
TN/YL
Y7
BK/LB
BK/WT
1-3
1-19
1-2
1-1
8 VOLTS
3,300 OHM PULL-UP RESISTOR
33,000 OHM
SIGNAL LINE
51,000 OHM
SIGNAL GROUND
INPUT
5 VOLT REGULATOR
OUTPUT
1984 TURBO LOGIC MODULE
INPUT
MICROPROCESSOR

Fig. 31

Sync and Reference (Distributor) Pickups

The sync and reference pickups are each a 3 wire sensor mounted in the distributor housing. The sync pickup "gives" the logic module (the microprocessor) 1 basic piece of information:

1. Camshaft position

The reference pickup "gives" the logic module 2 basic pieces of information:

1. Crankshaft position
2. Engine RPM

The sync and reference pickups "give" the logic module this information by toggling a voltage between 0 and 8 volts or 0 and 5 volts respectively. When a change in voltage happens, this "tells" the logic module that the engine is rotating. Both the sync and reference pickups are electrically the same for 1984 - '87. After 1984 the resistor values changed in the logic module for the sync pickup circuit. The reference pickup circuit is the same for 1984 - '87. The sync and reference pickups will be discussed first.

1984, '85, '86 & '87 Sync and Reference Pickups

Internally, the pickups have 5 major components. See Fig. 31 or 32. The first major component is the voltage regulator. The second major component is the hall element. The hall element is a thin sheet of conductive material. The material is fed a regulated voltage and provided a signal ground. The outputs of the hall element are at right angles to current flow. See Fig. 31. When the hall element is exposed to a magnetic field it will generate a voltage that is proportional to the strength of the magnetic field. This voltage is very small and must be amplified. The third major component is the differential amplifier. See Fig. 31. The differential amplifier is fed a regulated voltage and provided a signal ground. It has 2 inputs and 1 output. The hall element is connected to the inputs of the differential amplifier. The voltage difference from the hall element is multiplied by the differential amplifier and then sent to its output. The output of the differential amplifier is an analog voltage and it must be converted to a digital voltage. The fourth major component is the Schmitt Trigger. See Fig. 31. It is fed a regulated voltage and provided a signal ground. The Schmitt Trigger has 1 input and 1 output. The differential amplifier feeds the input of the Schmitt Trigger. When the voltage from the differential amplifier goes above a preset value, the Schmitt Trigger turns "on" and sends a positive voltage to its output. When the voltage from the differential amplifier goes below a preset value, the Schmitt Trigger turns "off" and the voltage at its output goes low (0 volts). The fifth major component is the NPN transistor. See Fig. 31. The collector of the transistor is fed 8 volts (sync pickup) or 5 volts (reference pickup). This voltage comes out of the logic module. The emitter of the transistor is connected to signal ground. The base of the transistor is fed by the Schmitt Trigger. When the Schmitt Trigger outputs a positive voltage, this turns "on" the transistor and the 5 or 8 volts is allowed to go to ground. When the Schmitt Trigger outputs a low voltage (0 volts), this turns the transistor "off" and the circuit is opened up.

Let's review the whole pickup assembly in operation. When the rotor "window" passes between the permanent magnet and the hall element, this allows the magnetic field to cut through the hall element. See Fig. 37. The hall element now outputs a small positive voltage. This voltage goes to the differential amplifier where it is amplified to a larger positive voltage. This larger positive voltage then goes to the Schmitt Trigger. When the voltage goes above the preset limit, the Schmitt Trigger turns "on". When the Schmitt Trigger turns "on", this turns "on" the base circuit of the NPN transistor and allows the 5 or 8 volts to go to ground. Now the engine rotates and the shutter blade moves between the hall element and the permanent magnet. This makes the magnetic field pass through the soft iron shutter blade. This happens because soft iron has lower reluctance (magnetic resistance) than the air does. Now the hall element "sees" no magnetic field and outputs no voltage. This in turn lowers the differential amplifier output. The lower differential amplifier output turns the Schmitt Trigger "off". This turns the NPN transistor "off" and the collector voltage goes up to 5 or 8 volts again. In this way the pickup toggles the voltage from the logic module.

In summary, when the hall element "sees" a magnetic field, the collector voltage (GY or TN/YL wire) goes low. When the hall element "sees" no magnetic field, the collector voltage goes high (5 or 8 volts).

1984, '85, '86 & '87 Reference Pickup and Circuit

Anytime the ignition key is "on", the logic module sends out a regulated 5 volts to the 1,000 ohm pull-up resistor. See Fig. 32. The pull-up resistor is connected to the signal line. From this point the voltage goes two places. One place the 5 volts goes is out of the logic module on the GY wire to the collector of the NPN transistor. See Fig. 32. If the shutter blade is between the hall element and the permanent magnet, the transistor is "off" and the 5 volts stops at the collector of the transistor. The second place the 5 volts goes is to an inverter in the logic module. See Fig. 32. The inverter takes the 5 volts and inverts it to 0 volts. This 0 volts then goes to the microprocessor. When the distributor turns and the "window" is between the hall element and the permanent magnet, the 5 volts goes to ground through the NPN transistor. The inverter in the logic module inverts the 0 volts to 5 volts. The

1984 DISTRIBUTOR REFERENCE PICKUP

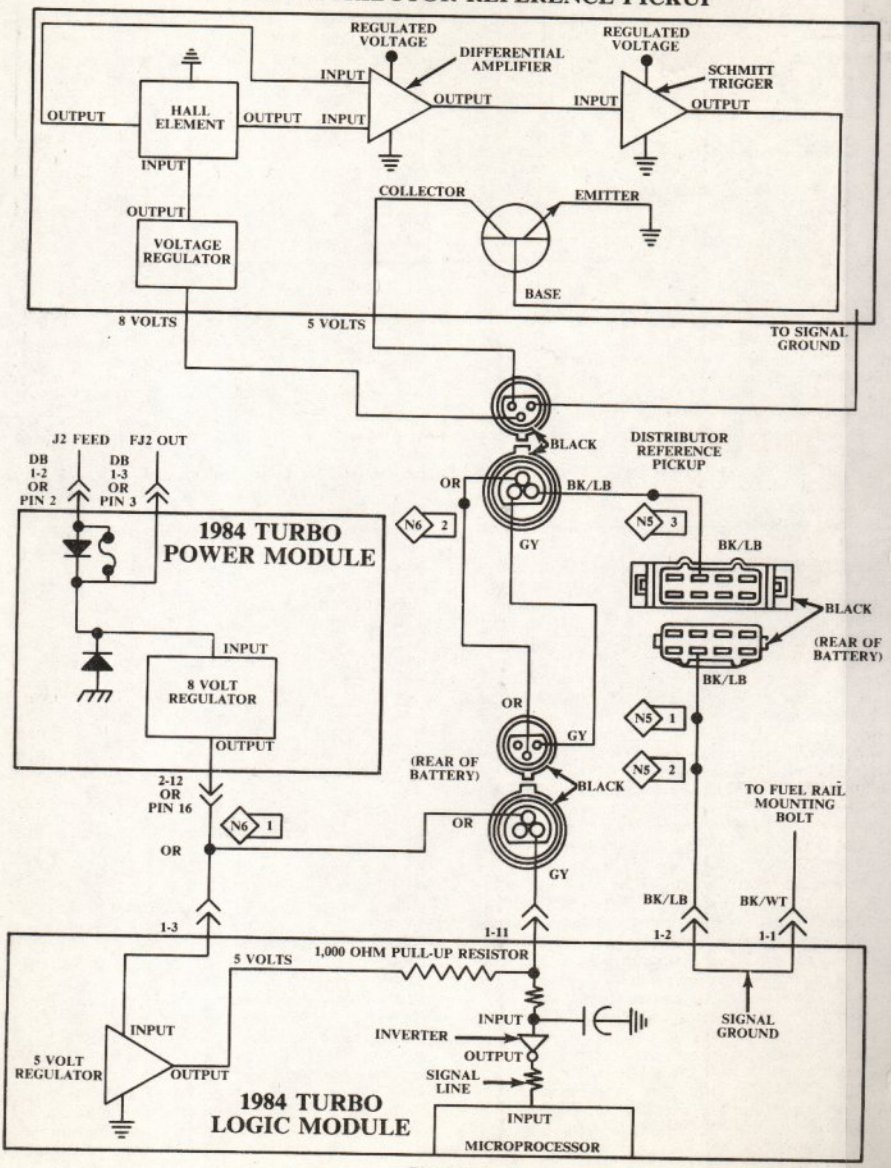


Fig. 32

microprocessor "looks" at the voltage when it is in the transition period (5 changing to 0). The longer the time between transition periods translates to a lower RPM. The opposite of this is also true. The microprocessor "knows" crankshaft position by the reference signal going from 5 volts to 0 volts. This "tells" the microprocessor that the crank is 12° before top dead center. The 12° is what the "base timing" is set to in this example.

Reference Pickup Fault Code

If no 0-5 volt pulses have been received by the logic module since the memory was last cleared, fault code 11 is stored in the logic module's memory and the engine will not start. If the logic module "sees" a transition period after that time, fault code 11 is erased from memory and the engine will start. If the engine stalls because the reference signal is lost, no fault code 11 will be stored in memory. The logic module's memory must be cleared and then the engine cranked to set fault code 11.

1984, '85, '86 & '87 Sync Pickup and Circuit

Anytime the ignition key is "on", the logic module sends a regulated 8 volts to the pull-up resistor (3,300 ohm in 1984 and 2,000 ohm in 1985 - '87). See Fig. 31 or 33. From this point the 8 volts leaves the logic module on the TN/YL wire and goes to the collector of the NPN transistor. If the shutter blade is between the hall element and the permanent magnet, the transistor is "off" and the 8 volts stops at the collector of the transistor. Because of the resistor network in the logic module, when the transistor is "off" the microprocessor or CPI "sees" less than 5 volts (4.67 volts in 1984 and 3.07 volts in 1985 - '87). See Fig. 33. The CPI is used as an interface between the signal line and the microprocessor in 1985 - '87. When the distributor turns and the "window" is between the hall element and the permanent magnet, the 8 volts goes to ground through the NPN transistor. When this happens, the microprocessor or the CPI "sees" a low voltage (0 volts). When the sync signal goes low, the microprocessor "looks" at the reference signal to determine which bank of injectors to fire (for 1985 only). In 1984 this "told" the power module which bank of injectors to turn on. If the reference signal is high, it fires one bank of injectors and if the reference signal is low, it fires the other bank of injectors.

For 1986 & '87, everytime the reference signal goes low (5 to 0 volts), the state of the sync signal is monitored. At this time the sync signal will also be low because both pickups use the same shutter blade. When the "window-in-blade" passes between the reference pickup, the reference will go low during its normal "high time". When this happens, the sync is "looked at" and must be high. This resets a counter in the logic module. If the counter counts 5 consecutive low sync states (no sync signal in 1 distributor revolution), the ASD relay is ungrounded and the engine is shut down. One pair of injectors is fired when the sync is high and the pair is fired when the sync is low, when compared to reference signal.

The microprocessor "knows" camshaft position by comparing the sync signal to the reference signal. For example, if cylinder #1 approaches TDC, the reference signal goes low and the microprocessor "knows" that a piston is near TDC. With just the reference signal, the microprocessor doesn't "know" if #1 is on the intake stroke (fire injector) or on the power stroke (fire coil). At this point the microprocessor just needs to "know" which one. So the sync signal "tells" the microprocessor which stroke that cylinder is on. By doing that, the sync signal "gives" the microprocessor camshaft position.

Sync Pickup Fault Code

If the logic module receives a reference signal but no sync signal, fault code 54 is set in the logic module's memory. In 1984, '86 or '87 the engine will stall with no sync signal. In 1985 the engine will remain running with no sync signal. At this time the logic module turns on the power loss/limited light. The system is now in "limp-in". The logic module will guess at injector timing. In the case of detonation with no sync signal, all 4 cylinders are retarded, not just the one detonating. If the sync signal returns, the power loss/limited light will be turned off however; fault code 54 remains in memory.

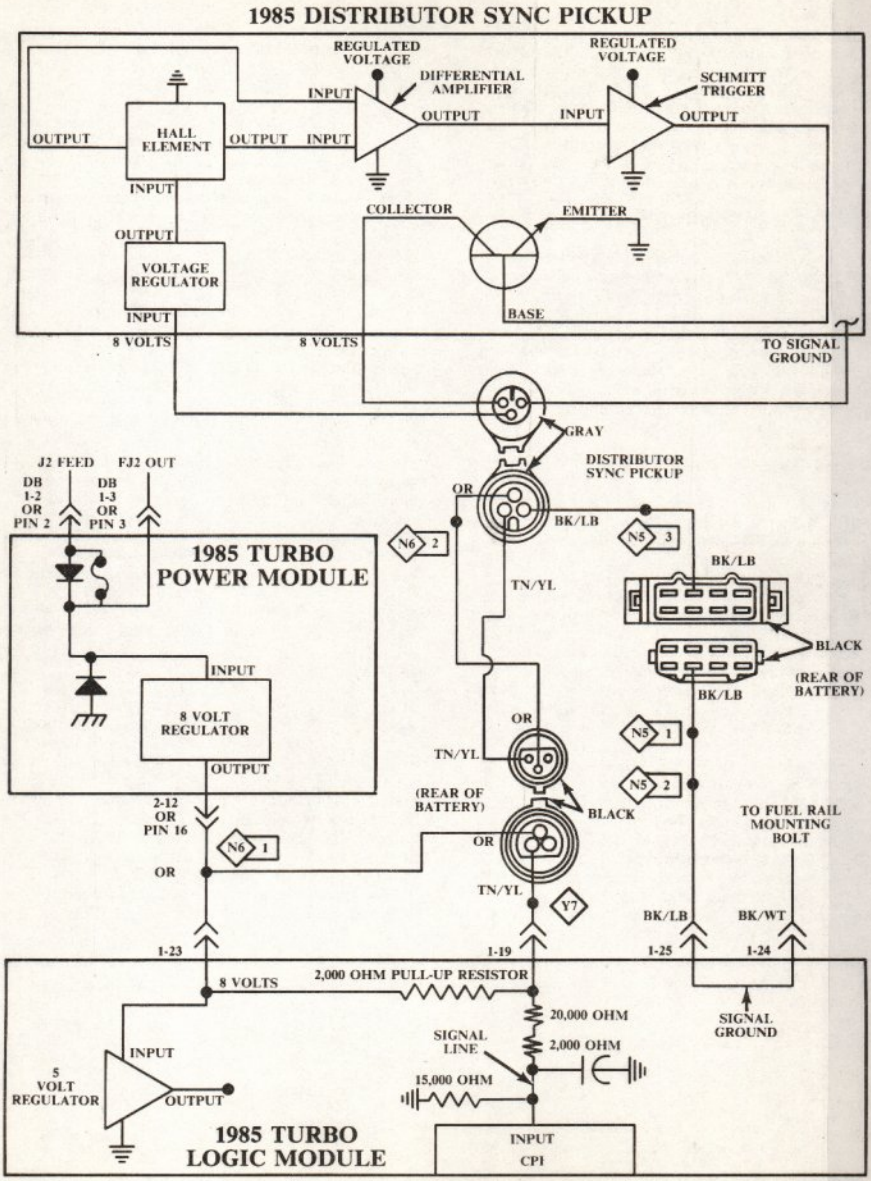

1985 DISTRIBUTOR SYNC PICKUP
REGULATED VOLTAGE
DIFFERENTIAL AMPLIFIER
INPUT
OUTPUT
HALL ELEMENT
OUTPUT
INPUT
INPUT
OUTPUT
VOLTAGE REGULATOR
INPUT
8 VOLTS
REGULATED VOLTAGE
SCHMITT TRIGGER
INPUT
OUTPUT
COLLECTOR
EMITTER
BASE
8 VOLTS
TO SIGNAL GROUND
GRAY
DISTRIBUTOR SYNC PICKUP
J2 FEED
FJ2 OUT
DB 1-2 OR PIN 2
DB 1-3 OR PIN 3
OR
BK/LB
N6 2
N5 3
BK/LB
TN/YL
BLACK
(REAR OF BATTERY)
BK/LB
1985 TURBO POWER MODULE
INPUT
8 VOLT REGULATOR
OUTPUT
OR
N5 1
TN/YL
N5 2
(REAR OF BATTERY)
BLACK
TO FUEL RAIL MOUNTING BOLT
2-12 OR PIN 16
N6 1
OR
OR
TN/YL
Y7
BK/LB
BK/WT
1-23
1-19
1-25
1-24
8 VOLTS
2,000 OHM PULL-UP RESISTOR
20,000 OHM
SIGNAL GROUND
SIGNAL LINE
2,000 OHM
15,000 OHM
INPUT
5 VOLT REGULATOR
OUTPUT
1985 TURBO LOGIC MODULE
INPUT
CPI

Fig. 33

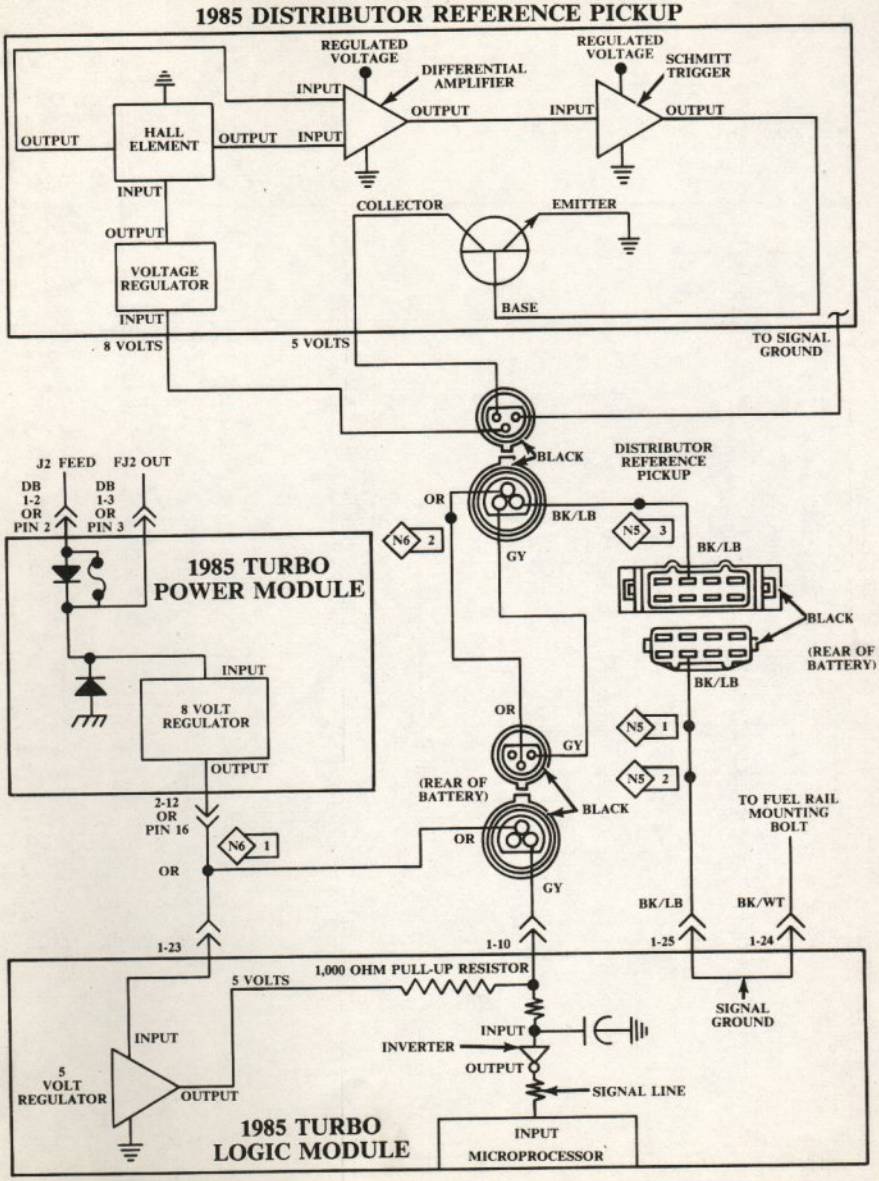

1985 DISTRIBUTOR REFERENCE PICKUP
REGULATED VOLTAGE
DIFFERENTIAL AMPLIFIER
REGULATED VOLTAGE
SCHMITT TRIGGER
INPUT
OUTPUT
INPUT
OUTPUT
OUTPUT
HALL ELEMENT
OUTPUT
INPUT
INPUT
COLLECTOR
EMITTER
OUTPUT
VOLTAGE REGULATOR
BASE
INPUT
8 VOLTS
5 VOLTS
TO SIGNAL GROUND
BLACK
DISTRIBUTOR REFERENCE PICKUP
J2 FEED
FJ2 OUT
DB 1-2 OR PIN 2
DB 1-3 OR PIN 3
OR
BK/LB
N6 2
N5 3
GY
BK/LB
1985 TURBO POWER MODULE
BLACK
(REAR OF BATTERY)
BK/LB
INPUT
8 VOLT REGULATOR
OUTPUT
OR
GY
N5 1
N5 2
(REAR OF BATTERY)
BLACK
TO FUEL RAIL MOUNTING BOLT
2-12 OR PIN 16
N6 1
OR
OR
GY
BK/LB
BK/WT
1-23
1-10
1-25
1-24
1,000 OHM PULL-UP RESISTOR
5 VOLTS
SIGNAL GROUND
INPUT
INPUT
INVERTER
5 VOLT REGULATOR
OUTPUT
OUTPUT
SIGNAL LINE
1985 TURBO LOGIC MODULE
INPUT MICROPROCESSOR

Fig. 34

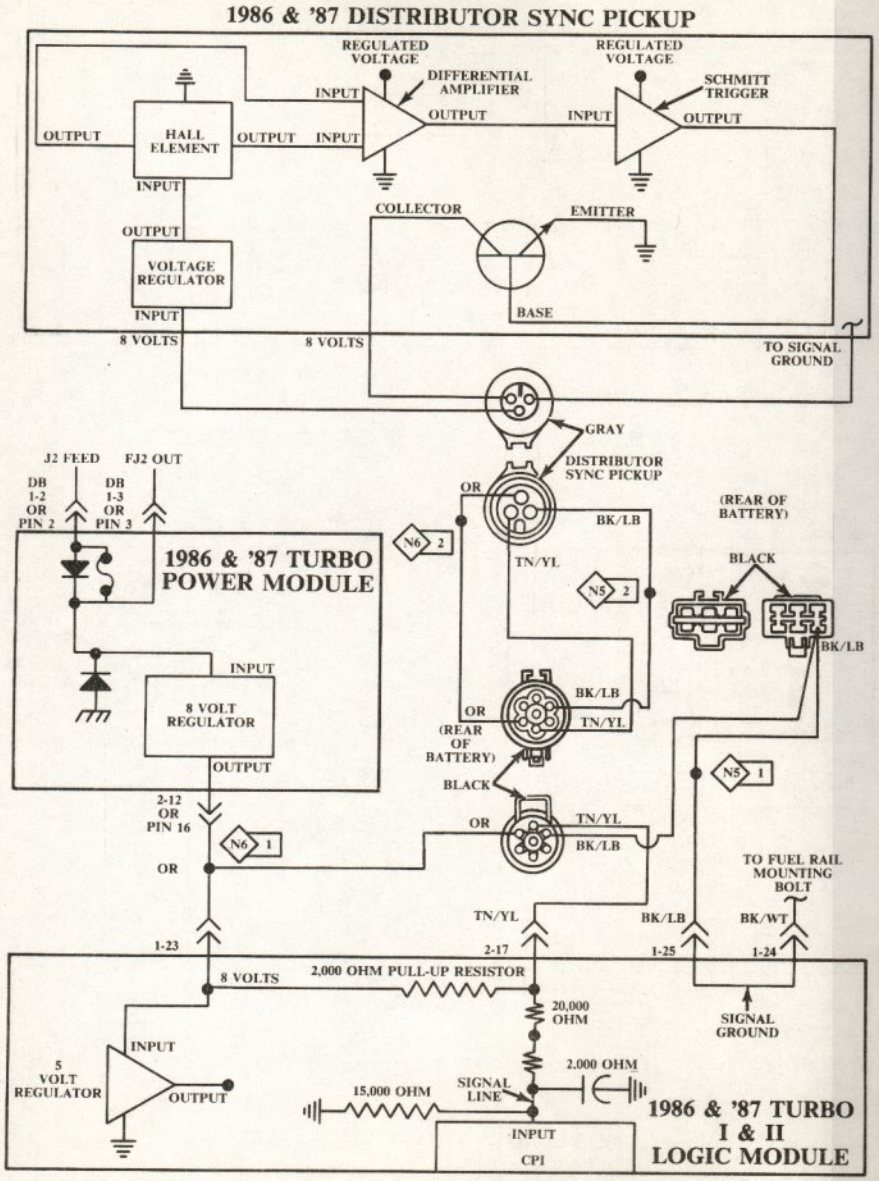

1986 & '87 DISTRIBUTOR SYNC PICKUP
REGULATED VOLTAGE
DIFFERENTIAL AMPLIFIER
REGULATED VOLTAGE
SCHMITT TRIGGER
INPUT
OUTPUT
HALL ELEMENT
OUTPUT
INPUT
INPUT
OUTPUT
INPUT
OUTPUT
COLLECTOR
EMITTER
BASE
OUTPUT
VOLTAGE REGULATOR
INPUT
8 VOLTS
8 VOLTS
TO SIGNAL GROUND
GRAY
DISTRIBUTOR SYNC PICKUP
J2 FEED
FJ2 OUT
DB 1-2 OR PIN 2
DB 1-3 OR PIN 3
OR
(REAR OF BATTERY)
BK/LB
N6 2
TN/YL
N5 2
BLACK
1986 & '87 TURBO POWER MODULE
INPUT
8 VOLT REGULATOR
OUTPUT
BK/LB
OR
(REAR OF BATTERY)
TN/YL
BK/LB
N5 1
BLACK
2-12 OR PIN 16
OR
TN/YL
N6 1
BK/LB
OR
TO FUEL RAIL MOUNTING BOLT
TN/YL
BK/LB
BK/WT
1-23
2-17
1-25
1-24
8 VOLTS
2,000 OHM PULL-UP RESISTOR
20,000 OHM
SIGNAL GROUND
INPUT
5 VOLT REGULATOR
OUTPUT
2,000 OHM
15,000 OHM
SIGNAL LINE
INPUT
CPI
1986 & '87 TURBO I & II LOGIC MODULE

Fig. 35

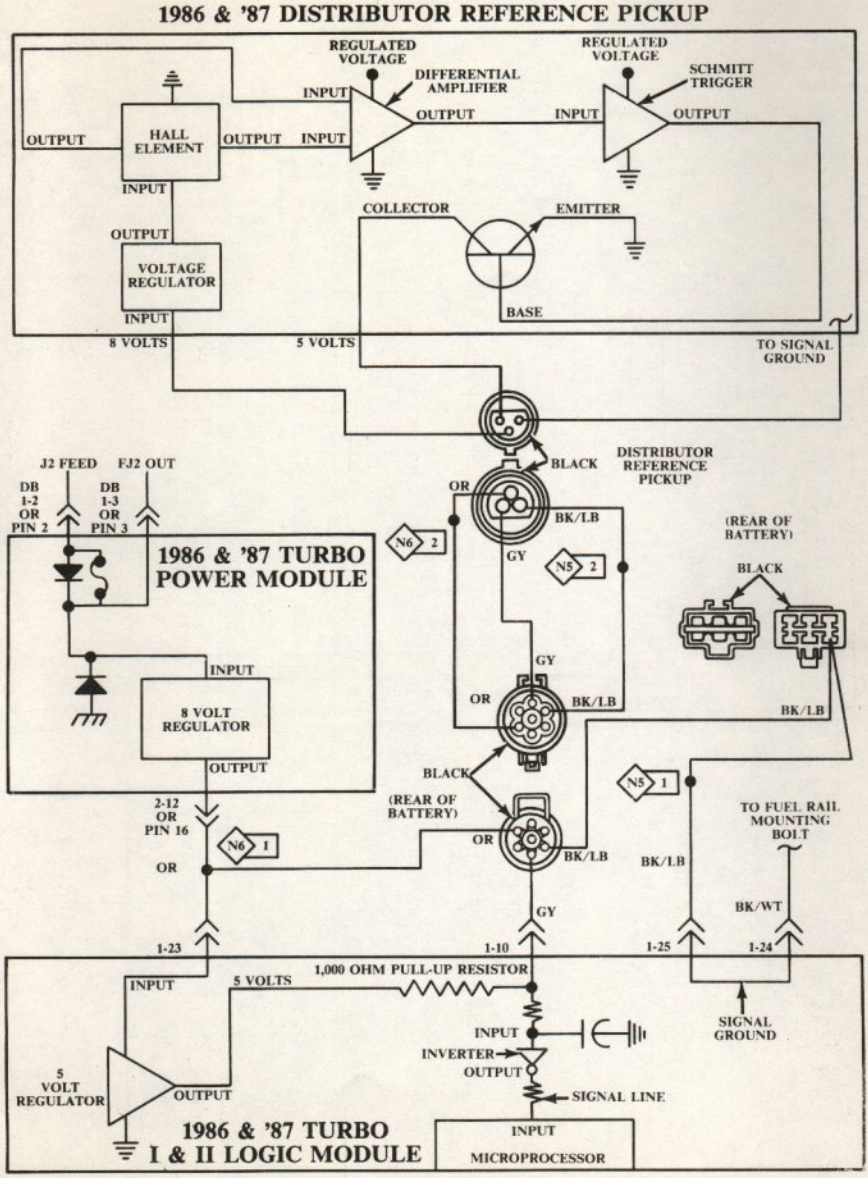

1986 & '87 DISTRIBUTOR REFERENCE PICKUP
REGULATED VOLTAGE
DIFFERENTIAL AMPLIFIER
REGULATED VOLTAGE
SCHMITT TRIGGER
INPUT
OUTPUT
INPUT
OUTPUT
INPUT
OUTPUT
OUTPUT
HALL ELEMENT
OUTPUT
INPUT
COLLECTOR
EMITTER
OUTPUT
VOLTAGE REGULATOR
INPUT
BASE
8 VOLTS
5 VOLTS
TO SIGNAL GROUND
BLACK
DISTRIBUTOR REFERENCE PICKUP
J2 FEED
FJ2 OUT
DB 1-2 OR PIN 2
DB 1-3 OR PIN 3
OR
BK/LB
N6 2
GY
N5 2
(REAR OF BATTERY)
BLACK
1986 & '87 TURBO POWER MODULE
INPUT
8 VOLT REGULATOR
OUTPUT
GY
OR
BK/LB
BK/LB
BLACK
(REAR OF BATTERY)
N5 1
TO FUEL RAIL MOUNTING BOLT
2-12 OR PIN 16
N6 1
OR
OR
BK/LB
BK/LB
GY
BK/WT
1-23
1-10
1-25
1-24
1,000 OHM PULL-UP RESISTOR
INPUT
5 VOLTS
SIGNAL GROUND
INPUT
INVERTER
OUTPUT
5 VOLT REGULATOR
OUTPUT
SIGNAL LINE
1986 & '87 TURBO I & II LOGIC MODULE
INPUT
MICROPROCESSOR

Fig. 36

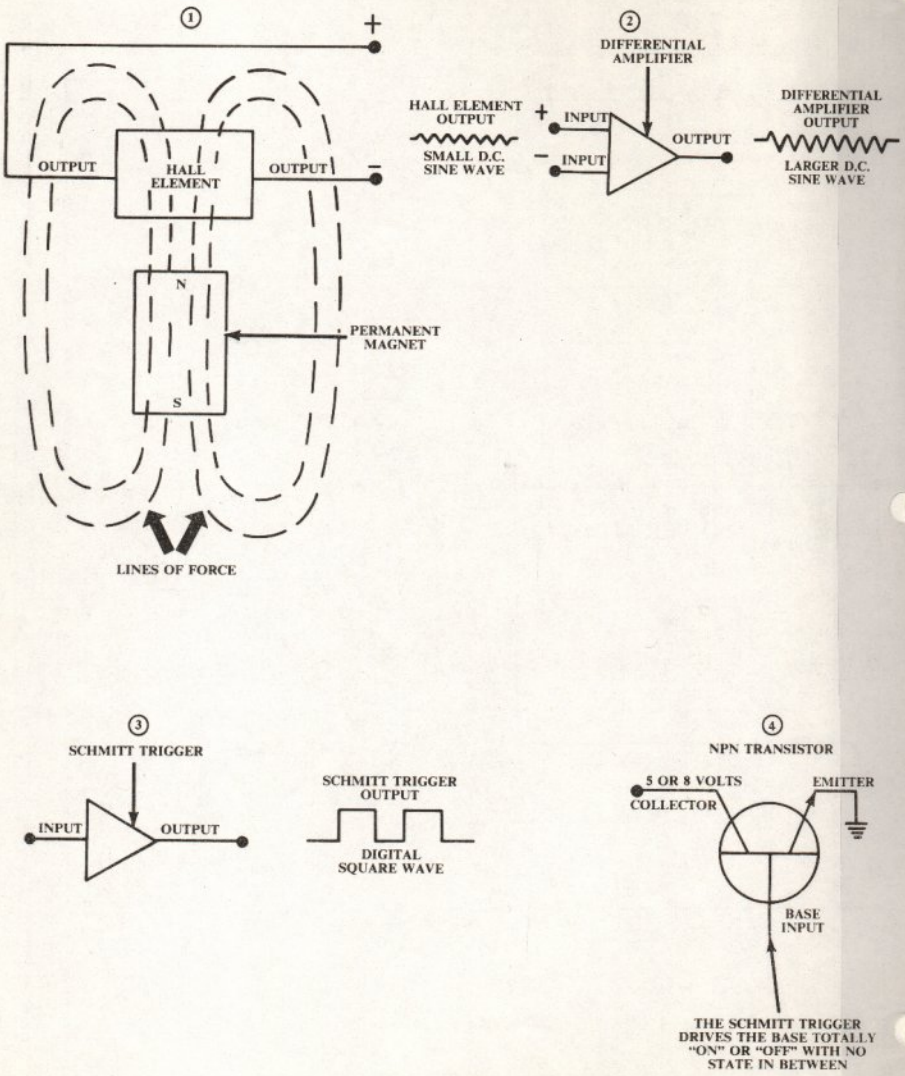


Fig. 37 Sync or Reference Pickup Component Operation

1985, '86 & '87 BATTERY TEMPERATURE SENSOR

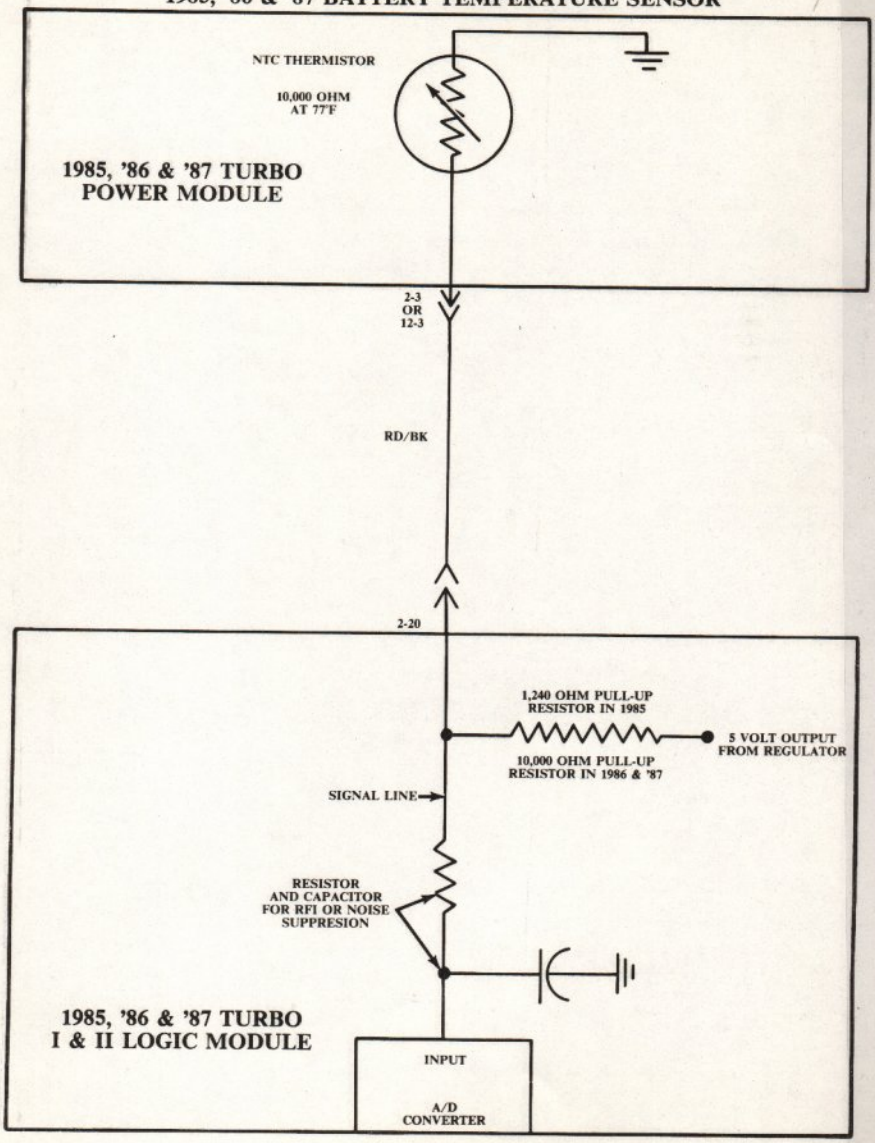


Fig. 38

Battery Temperature Sensor

The battery temperature sensor is a sensor that is mounted in the power module. The battery temperature sensor "gives" the logic module (the microprocessor) 1 basic piece of information:

1. Battery temperature

The battery temperature sensor "gives" the logic module this information by changing the voltage on the battery temperature sensor signal line inversely proportional to battery temperature. As the battery temperature increases, the battery temperature sensor signal voltage decreases. As the battery temperature decreases, the battery temperature sensor signal voltage increases. There is only one type of battery temperature sensor for 1985, '86 & '87. There is also only one type of circuit however; the pull-up resistor value changed after 1985. When the pull-up resistor value changed, this changed the temperature vs. voltage curves after 1985. See Fig. 39.

Battery Temperature Sensor and Circuit

Internally, the battery temperature sensor has a NTC thermistor rated at 10,000 ohms at 77°F. Anytime the ignition key is "on", the logic module sends a regulated 5 volts to the pull-up resistor (1,240 ohms in 1985 and 10,000 ohms in 1986 & '87). See Fig. 38. The pull-up resistor is connected to the signal line. The reason for the pull-up resistor is two fold. First, the pull-up resistor uses some of the 5 volts it is fed with when the circuit is complete and current is flowing. The voltage remaining after the pull-up resistor is the battery temperature sensor signal voltage. This voltage leaves the logic module on the RD/BK wire and goes into the power module to the battery temperature sensor. See Fig. 38. This voltage is also present at the input to the A/D converter. The resistor and capacitor at the input to the A/D converter are for RFI or noise suppression.

The second reason for the pull-up resistor is to put 5 volts at the input to the A/D converter in case of an open circuit on the RD/BK wire. This "tells" the logic module that there is an open circuit by the signal line always being high (5 volts). This happens because, when there is no current flow (open circuit), a resistor doesn't use any voltage.

Battery Temperature Sensor Fault Code

The logic module monitors the battery temperature sensor signal line (RD/BK wire) every 11 milliseconds (91 times a second), anytime the ignition key is "on". The battery temperature sensor signal voltage must not be below .04 volts or above 4.96 volts. If either of these limits are exceeded, fault code 44 is set in the logic module's memory. At this time the logic module will regulate the charging system voltage to 14.0 volts. If the voltage at the battery temperature sensor ever comes back into the acceptable range, the logic module will revert back to the normal charging rate however; fault code 44 will remain in memory.

1985 BATTERY TEMPERATURE SENSOR TEMPERATURE vs. VOLTAGE CURVE

Temperature	Voltage
302°F	0.65 V
248°F	1.19 V
194°F	2.12 V
140°F	3.34 V
86°F	4.33 V
68°F	4.55 V
50°F	4.71 V
32°F	4.82 V
14°F	4.89 V
−4°F	4.94 V
−22°F	4.97 V
−40°F	4.98 V

1986 & '87 BATTERY TEMPERATURE SENSOR TEMPERATURE vs. VOLTAGE CURVE

Temperature	Voltage
248°F	0.15 V
194°F	0.34 V
140°F	0.84 V
86°F	1.96 V
68°F	2.49 V
50°F	3.07 V
32°F	3.61 V
14°F	4.07 V
−4°F	4.35 V
−22°F	4.52 V
−40°F	4.81 V

Fig. 39

1984 PARK/NEUTRAL AND BACK-UP LIGHT SWITCH

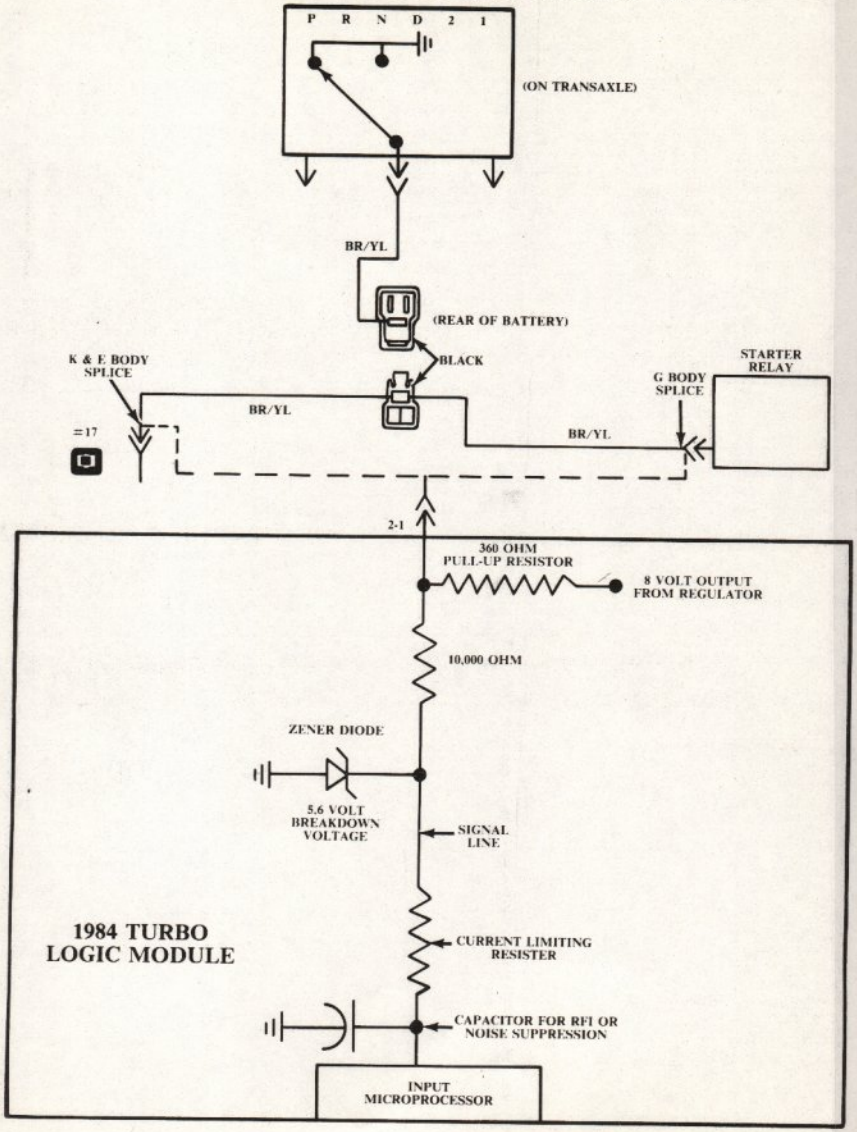


Fig. 40

Park/Neutral Switch

The park/neutral switch is a 3 wire switch mounted in the transaxle case. The park/neutral switch "gives" the logic module (the microprocessor) 1 basic piece of information:

1. When the transaxle is in gear

The park/neutral switch "gives" the logic module this information by sending a high (5.6 volts in 1984 or 3.24 volts in 1985 - '87) to the microprocessor or the CPI. The CPI is used as an interface between the signal line and the microprocessor in 1985 - '87. There is only one park/neutral switch for 1984, '85, '86 & '87. There are two types of circuits, the 1984 style and the 1985, '86 & '87 style. The 1984 style will be discussed first.

1984 Park/Neutral Switch and Circuit

Anytime the ignition key is "on", the logic module sends a regulated 8 volts to the 360 ohm pull-up resistor. See Fig. 40. If the transaxle is in park or neutral, the 8 volts goes to ground through the park/neutral switch. This puts a low voltage (0 volts) at the input of the microprocessor. When the microprocessor "sees" a low voltage, it "thinks" the transaxle is in park or neutral. When the transaxle is put into gear, this removes the ground from the circuit. At this time the voltage starts to rise on the signal line because the current flow through the pull-up resistor is reduced. When the voltage on the signal line goes above 5.6 volts, the zener diode breaks down and allows reverse current flow. What this does is keep the voltage at 5.6 volts when the transaxle is in gear. This 5.6 volts is impressed at the input of the microprocessor. See Fig. 40. When the microprocessor "sees" this 5.6 volts it "thinks" the transaxle is in gear.

1985, '86 & '87 Park/Neutral Switch and Circuit

Anytime the ignition key is "on", the logic module sends a regulated 8 volts to the 2,000 ohm pull-up resistor. See Fig. 41. If the transaxle is in park or neutral, the 8 volts goes to ground through the park/neutral switch. This puts a low voltage at the input to the CPI. When the microprocessor "sees" a low voltage at the CPI, it "thinks" the transaxle is in park or neutral. When the transaxle is put into gear, this removes the ground from the circuit. The 8 volts now goes to ground through the 20,000 ohm resistor and then the 15,000 ohm resistor. See Fig. 41. This puts a high voltage (3.24 volts) at the input to the CPI. When the microprocessor "sees" a high voltage at the CPI, it "thinks" the transaxle is in gear.

Park/Neutral Switch Fault Code

There are no fault codes for the park/neutral switch. The logic module doesn't "know" the difference between an open or short circuit and the transaxle being in or out of gear. The result is the same, either a constant high or low voltage. The logic module is capable of echoing back a response that it has "seen" a change in state (high to low or low to high voltage). The diagnostic readout box (C-4805) can be used to determine if the logic module is recognizing the change in state. With the diagnostic readout box hooked up and in the switch test mode, move the gear selector from park to reverse. The display should change from 88 to 00 or 00 to 88. This tells you that the logic module is recognizing the change in state and the circuit is OK. If there is no change in the display, there is a problem in the circuit.

1985 PARK/NEUTRAL AND BACK-UP LIGHT SWITCH

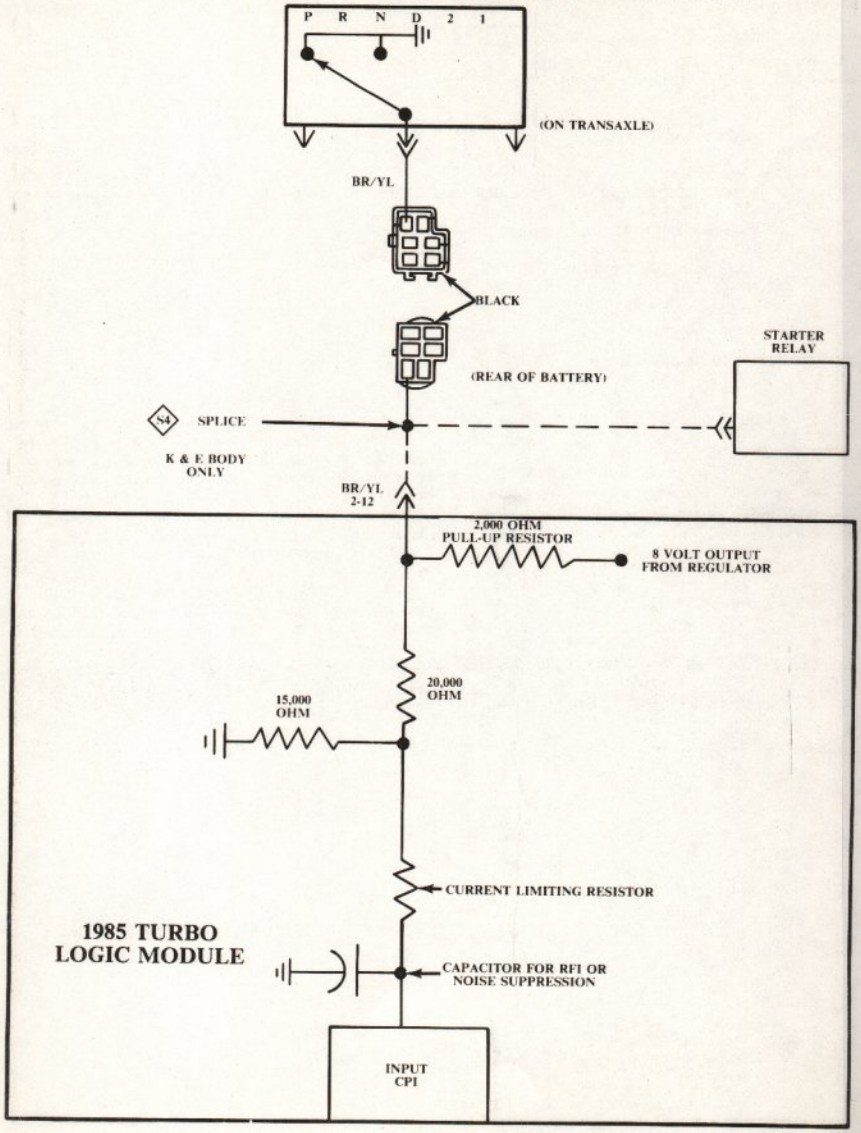


Fig. 41

1986 & '87 PARK/NEUTRAL AND BACK-UP LIGHT SWITCH

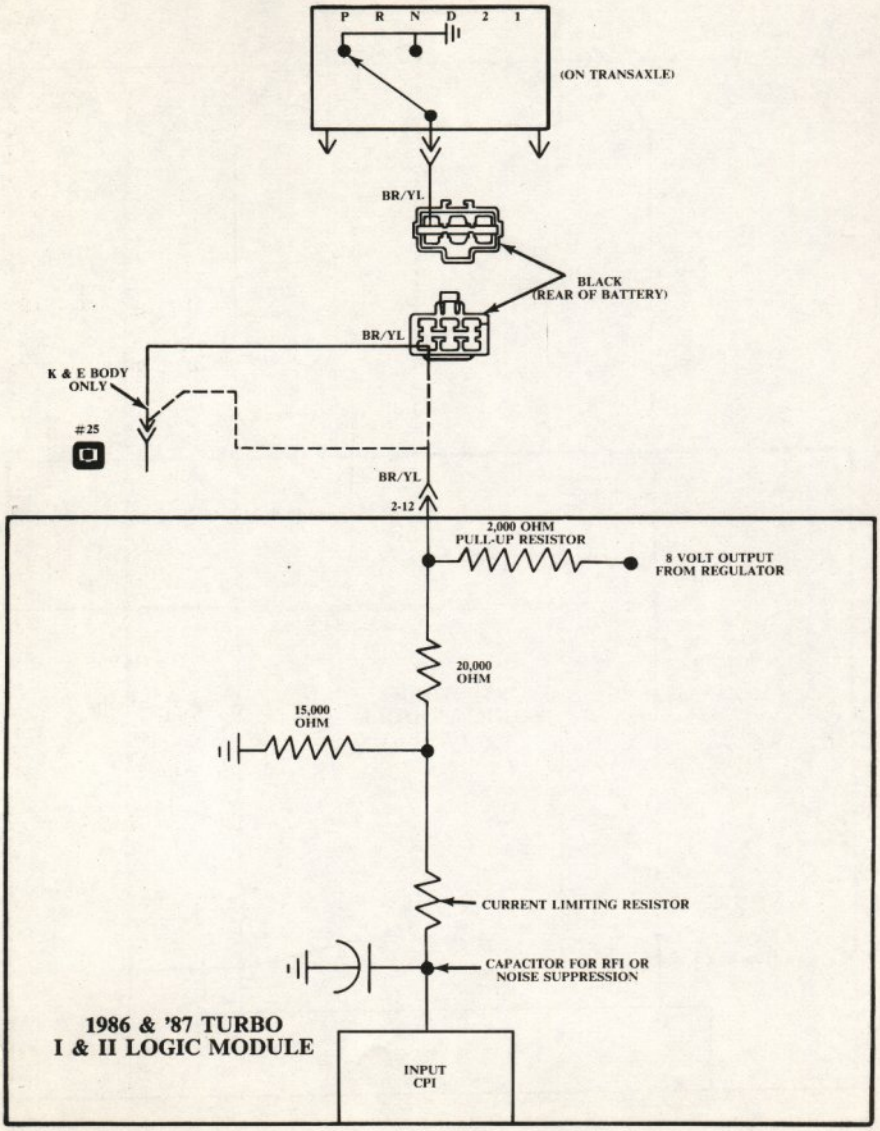


Fig. 42

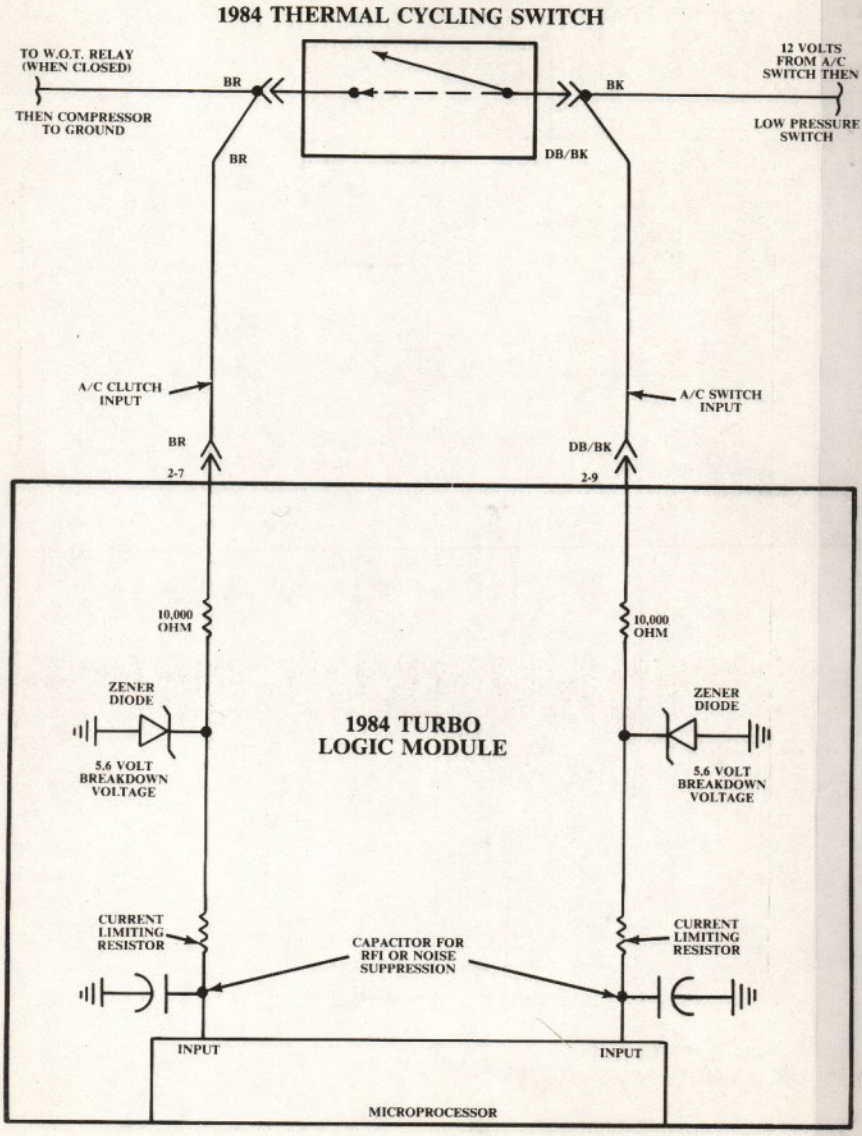

1984 THERMAL CYCLING SWITCH
TO W.O.T. RELAY (WHEN CLOSED)
THEN COMPRESSOR TO GROUND
BR
BR
BK
DB/BK
12 VOLTS FROM A/C SWITCH THEN
LOW PRESSURE SWITCH
A/C CLUTCH INPUT
A/C SWITCH INPUT
BR
DB/BK
2-7
2-9
10,000 OHM
10,000 OHM
ZENER DIODE
ZENER DIODE
1984 TURBO LOGIC MODULE
5.6 VOLT BREAKDOWN VOLTAGE
5.6 VOLT BREAKDOWN VOLTAGE
CURRENT LIMITING RESISTOR
CAPACITOR FOR RFI OR NOISE SUPPRESSION
CURRENT LIMITING RESISTOR
INPUT
INPUT
MICROPROCESSOR

Fig. 43

A/C Switch

The logic module (the microprocessor) monitors the state of the A/C circuit to determine if the compressor clutch is engaged or not. There are two different types of circuits, the 1984 style and the 1985, '86 & '87 style. The 1984 style will be discussed first.

1984 A/C Switch and Circuit

When the engine is running and the A/C button is pushed, this provides the 12 volt source which completes the circuit. With the compressor engaged, this puts 12 volts on both sides of the thermal cycling switch. The 12 volts goes into the logic module on the BR wire and the DB/BK wire. See Fig. 43. The voltage above 5.6 volts goes to ground through both zener diodes. With both the A/C clutch input and the A/C switch input high (5.6 volts), the microprocessor "thinks" the A/C button is pushed and the thermal cycling switch is closed. As the car interior cools down, the low side pressure decreases. Below a certain temperature, the thermal cycling switch opens to prevent evaporator freeze-up. This makes the A/C clutch input go low (0 volts) and the A/C switch input stays high. This "tells" the logic module that the A/C system is on and the clutch is temporarily disengaged due to low side pressure being too low. In this state the logic module is waiting for the clutch to come back on. When the A/C system is turned off, this puts a low voltage at both inputs. With both inputs low, the logic module "thinks" the A/C system is off.

1985, '86 & '87 A/C Switch and Circuit

Anytime the ignition key is "on", the logic module sends a regulated 8 volts to the 2,000 ohm pull-up resistor. See Fig. 44. With the A/C system off, the 8 volts goes to ground through the 20,000 ohm resistor to the 2,000 ohm resistor on to the 15,000 ohm resistor. See Fig. 44. This puts a high (3.07 volts) at the input to the CPI. When the microprocessor "sees" a high voltage at the CPI, it "thinks" the A/C system is off. When the engine is running and the A/C button is pushed and the blower is on, a ground is provided. This completes the circuit and the compressor is engaged. This makes the thermal cycling switch (1985) or the damped pressure switch (1986 & '87) a direct path to ground. See Fig. 44 or 45. Now the current flow through the pull-up resistor will take the path of least resistance. All the current flow will be through the pull-up resistor, out of the logic module on the BR wire to ground. This makes the input to the CPI go low (0 volts). When the microprocessor "sees" a low voltage at the CPI, it "thinks" the A/C clutch is engaged. When the thermal cycling switch or the damped pressure switch opens due to low, low side pressure, a voltage spike may be present on the signal line. This voltage spike is caused by the clutch coil magnetic field collapsing. The CPI is protected from a voltage spike by the 5.6 volt zener diode. Any voltage higher than 5.6 volts, goes to ground through the zener diode.

A/C Switch Fault Code

There are no fault codes for the A/C switch circuit. The logic module doesn't "know" the difference between an open or short circuit and the A/C system being on or off. The logic module is capable of echoing back a response that it has "seen" a change in state. The diagnostic readout box (C-4805) can be used to determine if the logic module is recognizing the change in state. With the diagnostic readout box hooked up and in the switch test mode, in 1984, push the A/C button on and off. In 1985 - '87, turn the blower on and then push the A/C button on and off. The display should change from 88 to 00 or from 00 to 88. This tells you that the logic module is recognizing the change in state and the circuit is OK. If there is no change in the display, there is a problem in the circuit.

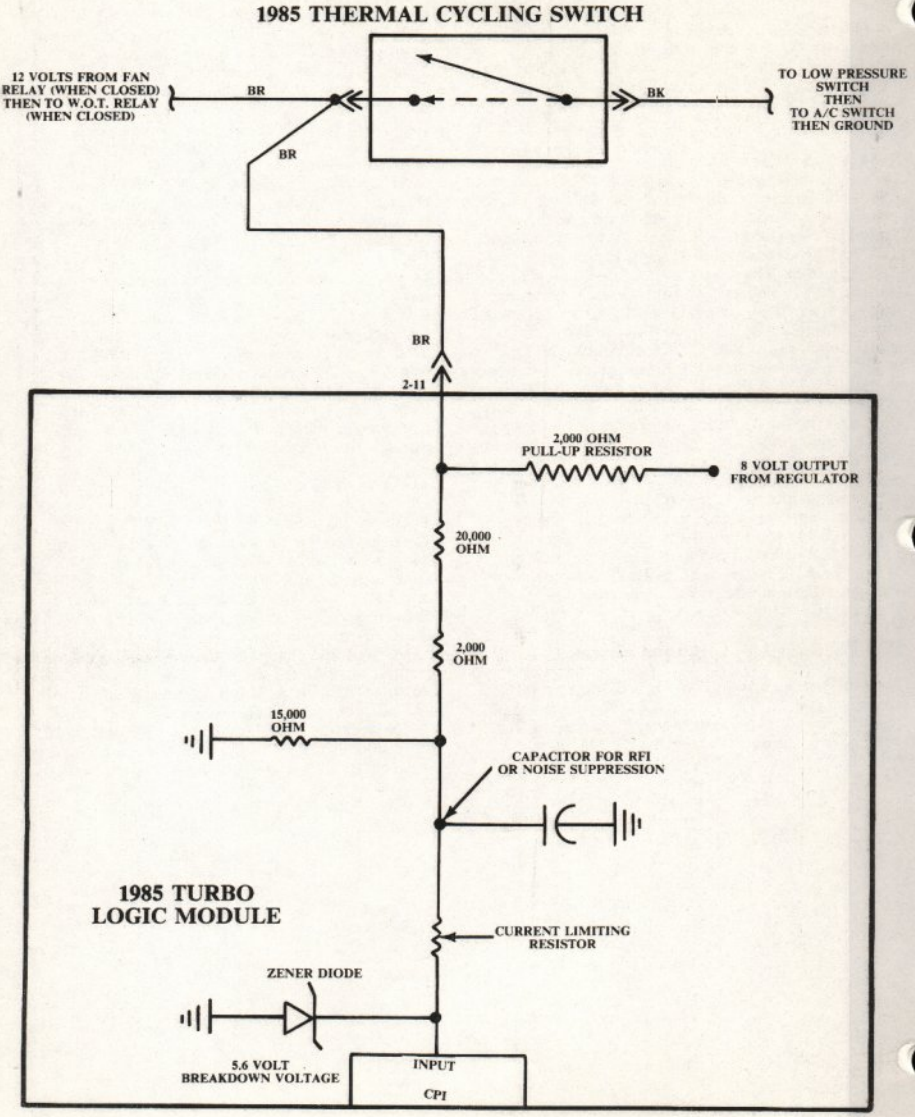

1985 THERMAL CYCLING SWITCH
12 VOLTS FROM FAN RELAY (WHEN CLOSED) THEN TO W.O.T. RELAY (WHEN CLOSED)
BR
BR
BK
TO LOW PRESSURE SWITCH THEN TO A/C SWITCH THEN GROUND
BR
2-11
2,000 OHM PULL-UP RESISTOR
8 VOLT OUTPUT FROM REGULATOR
20,000 OHM
2,000 OHM
15,000 OHM
CAPACITOR FOR RFI OR NOISE SUPPRESSION
1985 TURBO LOGIC MODULE
CURRENT LIMITING RESISTOR
ZENER DIODE
5.6 VOLT BREAKDOWN VOLTAGE
INPUT
CPI

Fig. 44

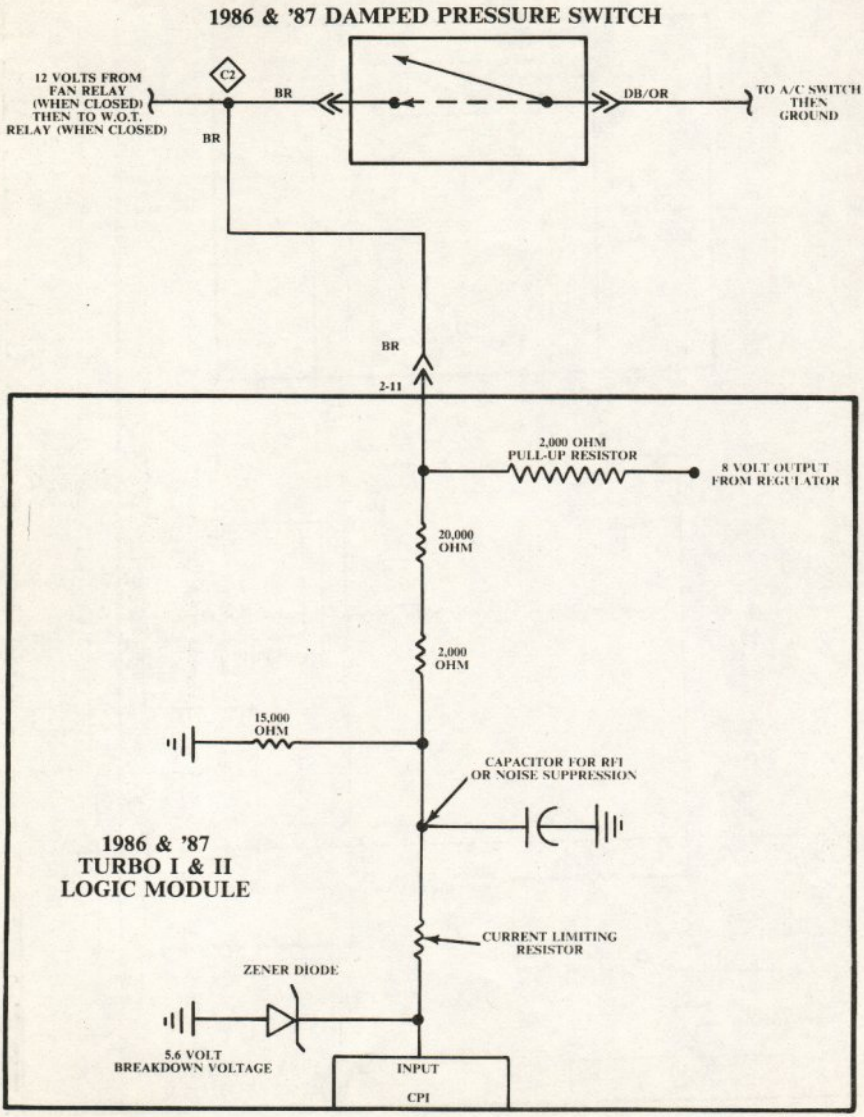

1986 & '87 DAMPED PRESSURE SWITCH
12 VOLTS FROM FAN RELAY (WHEN CLOSED) THEN TO W.O.T. RELAY (WHEN CLOSED)
C2
BR
BR
DB/OR
TO A/C SWITCH THEN GROUND
BR
2-11
2,000 OHM PULL-UP RESISTOR
8 VOLT OUTPUT FROM REGULATOR
20,000 OHM
2,000 OHM
15,000 OHM
CAPACITOR FOR RFI OR NOISE SUPPRESSION
1986 & '87 TURBO I & II LOGIC MODULE
CURRENT LIMITING RESISTOR
ZENER DIODE
5.6 VOLT BREAKDOWN VOLTAGE
INPUT
CPI

Fig. 45

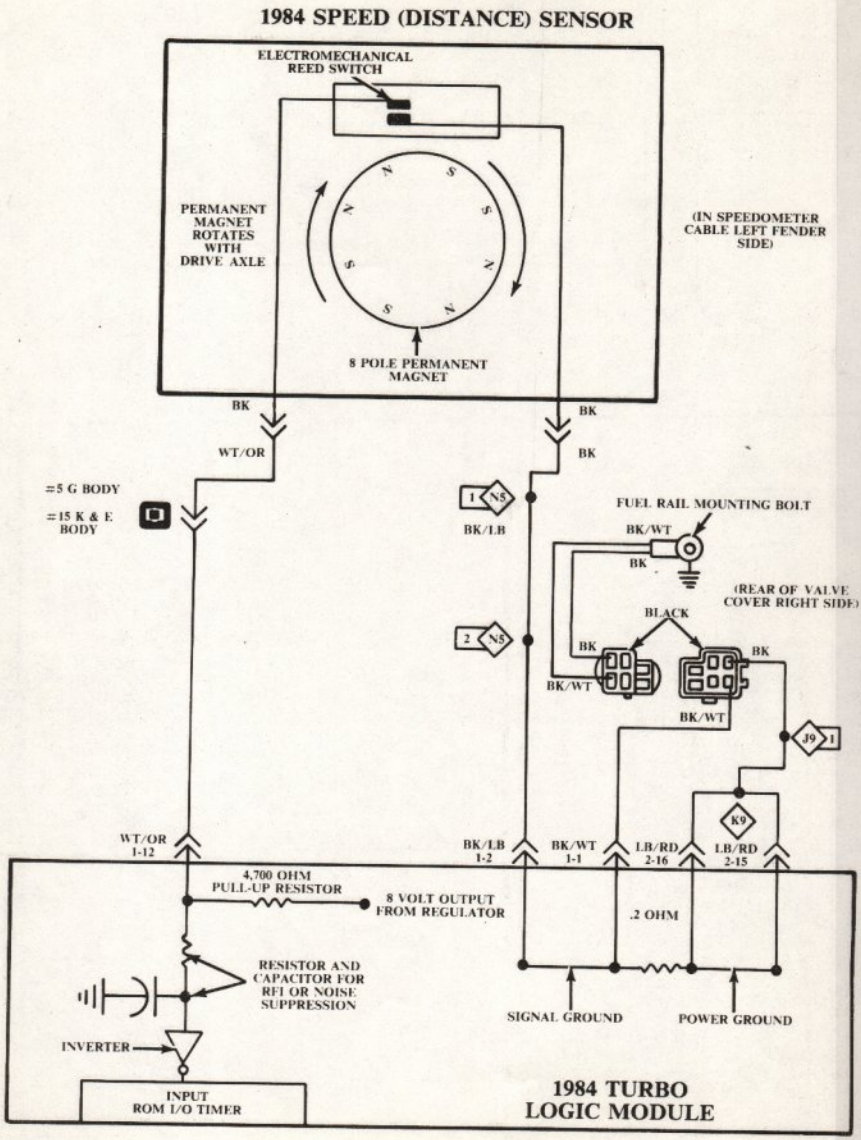

1984 SPEED (DISTANCE) SENSOR
ELECTROMECHANICAL REED SWITCH
PERMANENT MAGNET ROTATES WITH DRIVE AXLE
(IN SPEEDOMETER CABLE LEFT FENDER SIDE)
8 POLE PERMANENT MAGNET
BK
WT/OR
BK
BK
=5 G BODY
=15 K & E BODY
1 N5
BK/LB
FUEL RAIL MOUNTING BOLT
BK/WT
BK
(REAR OF VALVE COVER RIGHT SIDE)
BLACK
2 N5
BK
BK
BK/WT
BK/WT
J9 1
K9
WT/OR 1-12
BK/LB 1-2
BK/WT 1-1
LB/RD 2-16
LB/RD 2-15
4,700 OHM PULL-UP RESISTOR
8 VOLT OUTPUT FROM REGULATOR
.2 OHM
RESISTOR AND CAPACITOR FOR RFI OR NOISE SUPPRESSION
SIGNAL GROUND
POWER GROUND
INVERTER
INPUT ROM I/O TIMER
1984 TURBO LOGIC MODULE

Fig. 46

Speed (Distance) Sensor

The speed sensor is a 2 wire sensor. In 1984 the speed sensor is mounted in series with the speedometer cable. The 1985, '86 & '87 speed sensors are mounted in the transaxle extension housing. The speed sensor "gives" the logic module (the microprocessor) 1 basic piece of information:

1. Car speed

The speed sensor "gives" the logic module this information by toggling a voltage between 0 and 8 volts (1984, '85 & '86) or 0 and 5 volts (1987). When a change in voltage happens, this "tells" the logic module that the car is moving. The speed sensor will generate 8 pulses per revolution or 8,000 pulses per mile. The speed sensor is electrically the same for 1984, '85, '86 & '87. There are three different types of circuits, the 1984 style, the 1985 & '86 style and the 1987 style. The speed sensor will be discussed first.

1984, '85, '86 & '87 Speed (Distance) Sensor

Internally, the speed sensor has a round 8 pole permanent magnet. See Fig. 46. This permanent magnet rotates with the drive axle. Mounted above the permanent magnet is an electromechanical reed switch. See Fig. 46. If the upper contact of the reed switch is exposed to the north pole and the lower contact is exposed to the south pole, this causes the contacts to close. As the permanent magnet rotates, the upper and lower contacts are exposed to like poles (north and north or south and south). This causes a repelling force and the contacts open. So when the permanent magnet rotates, the contacts are opened and closed by the magnetic field. When the contacts are closed, there is continuity between the 2 wires of the speed sensor (no more than 1/2 ohm resistance). When the contacts are open, there is infinate resistance between the 2 wires of the sensor.

1984 Speed (Distance) Sensor and Circuit

Anytime the ignition key in "on", the logic module sends a regulated 8 volts to the 4,700 ohm pull-up resistor. See Fig. 46. The pull-up resistor is connected to the signal line. From this point the 8 volts goes two places. First, the 8 volts leaves the logic module on the WT/OR wire and goes to the speed sensor. If the contacts of the speed sensor are open, the 8 volts stops at the upper contact. See Fig. 46. The second place the 8 volts goes is, to the inverter in the logic module. The inverter takes the high (8 volts) at its input and inverts it to a low (0 volts). The output of the inverter feeds the I/O (Input/Output). When the car moves and the speed sensor contacts close, the 8 volts goes through the speed sensor and to the fuel rail mounting bolt for signal ground. See Fig. 46. This puts a low at the input to the inverter. This low signal then inverted to a high signal. The microprocessor uses the I/O to "read" digital signals like the speed sensor. When the microprocessor "sees" the input of the I/O go from a high to low or low to high, it "thinks" the car is moving. When the microprocessor "sees" a steady state (constant low or constant high), it "thinks" the car is stationary.

1985 & '86 Speed (Distance) Sensor and Circuit

Anytime the ignition key is "on", the logic module sends a regulated 8 volts to the 2,000 ohm pull-up resistor. See Fig. 47 or 48. If the speed sensor contacts are open, the current flows through the pull-up resistor and then through the 20,000 ohm resistor to the 15,000 ohm resistor to signal ground. This puts a high (3.24 volts) at the input to the CPI. The microprocessor "reads" the CPI the same way it "read" the I/O in 1984. When the car moves and the speed sensor contacts close, all the current flow is through the pull-up resistor and out of the logic module on the WT/OR wire to the speed sensor and the to the fuel rail mounting bolt for signal ground. This puts a low at the input of the CPI. When the microprocessor "sees" the input to the CPI change state, it "thinks" the car is moving. When the microprocessor "sees" no change in state, it "thinks" the car is stationary.

1987 Speed (Distance) Sensor and Circuit

Anytime the ignition key is "on", the logic module sends a regulated 5 volts to the 5,100 ohm pull-up resistor. See Fig. 49. The pull-up resistor is connected to the signal line. From this point the 5 volts goes two places. First the 5 volts leaves the logic module on the WT/OR wire and goes to the speed sensor. If the contacts are open, the voltage stops at the upper contact. The second place the 5 volts goes is to the inverter in the logic module. See Fig. 49. This inverter works the same as the one in 1984, that is its output is just opposite its input. The output of the inverter feeds the input of the microprocessor. When the car moves and the contacts close, this allows the 5 volts to go to the fuel rail mounting bolt for signal ground. This makes the input to the inverter low. The inverter takes this low and inverts it to a high and then sends it to the microprocessor. Just as in previous years, when the microprocessor "sees" the input change state, it "thinks" the car is moving. When there is no change in state, it "thinks" the car is stationary.

Speed (Distance) Sensor Fault Codes

During deceleration, if the throttle stays closed (minimum TPS voltage) for 7 seconds and the engine RPM is 768 RPM above target idle speed, the logic module monitors the speed sensor signal line for an input (changes in state). If no pulses are present at this time, fault code 17 is set into the logic module's memory. Any input from the speed sensor to the logic module is assumed to be correct. At this time, the logic module will "look" at the brake switch input to "determine" if the car is decelerating.

1985 SPEED (DISTANCE) SENSOR

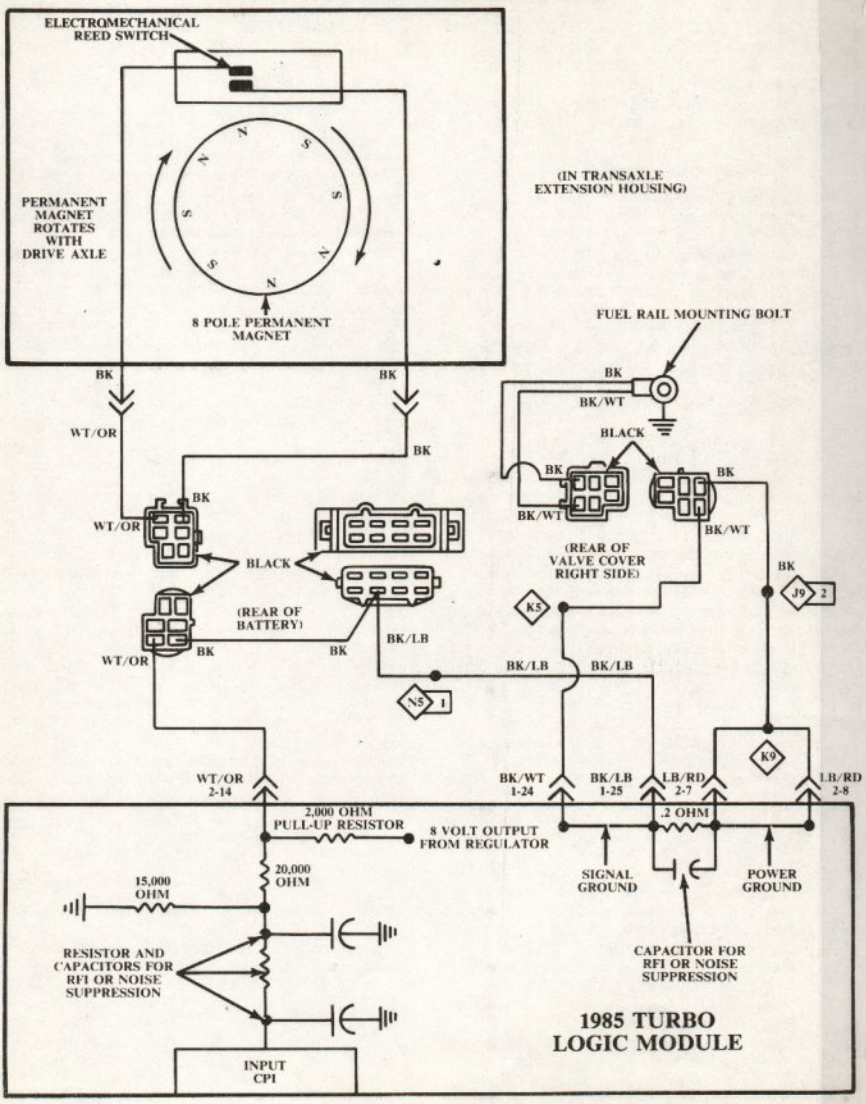


Fig. 47

1986 SPEED (DISTANCE) SENSOR

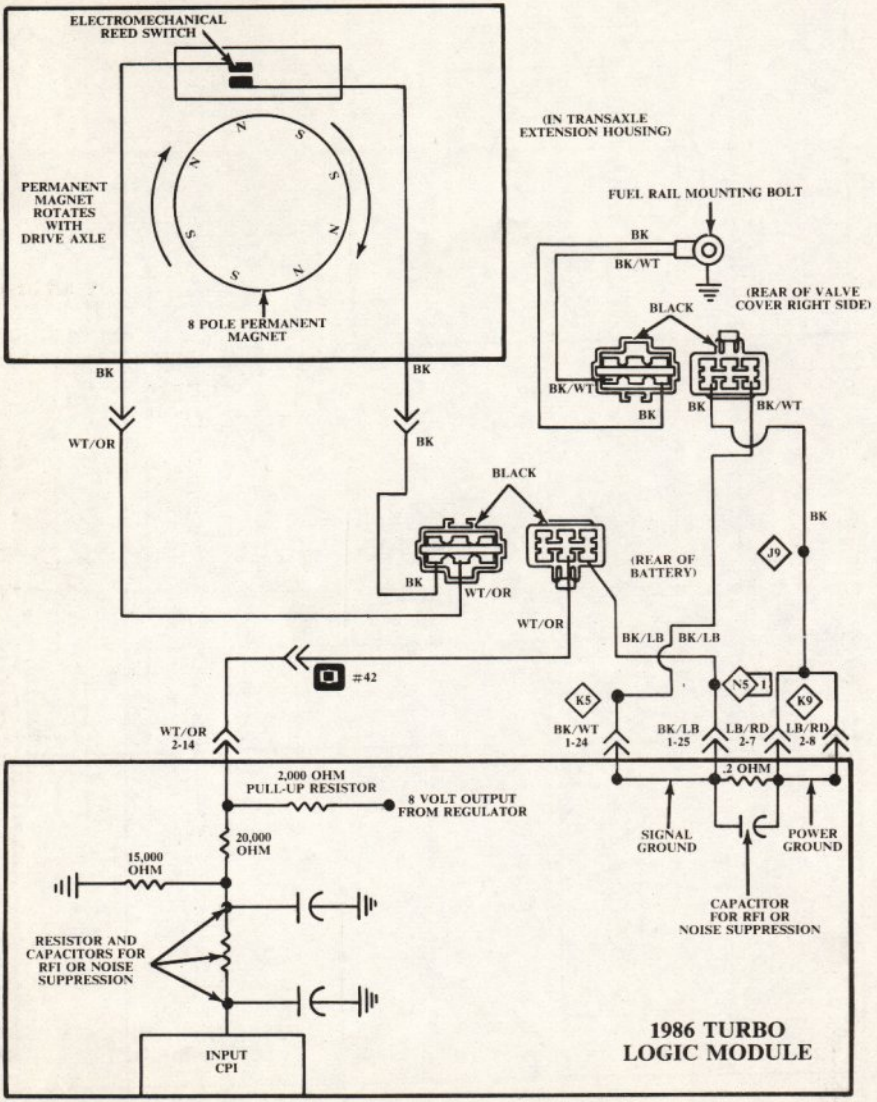


Fig. 48

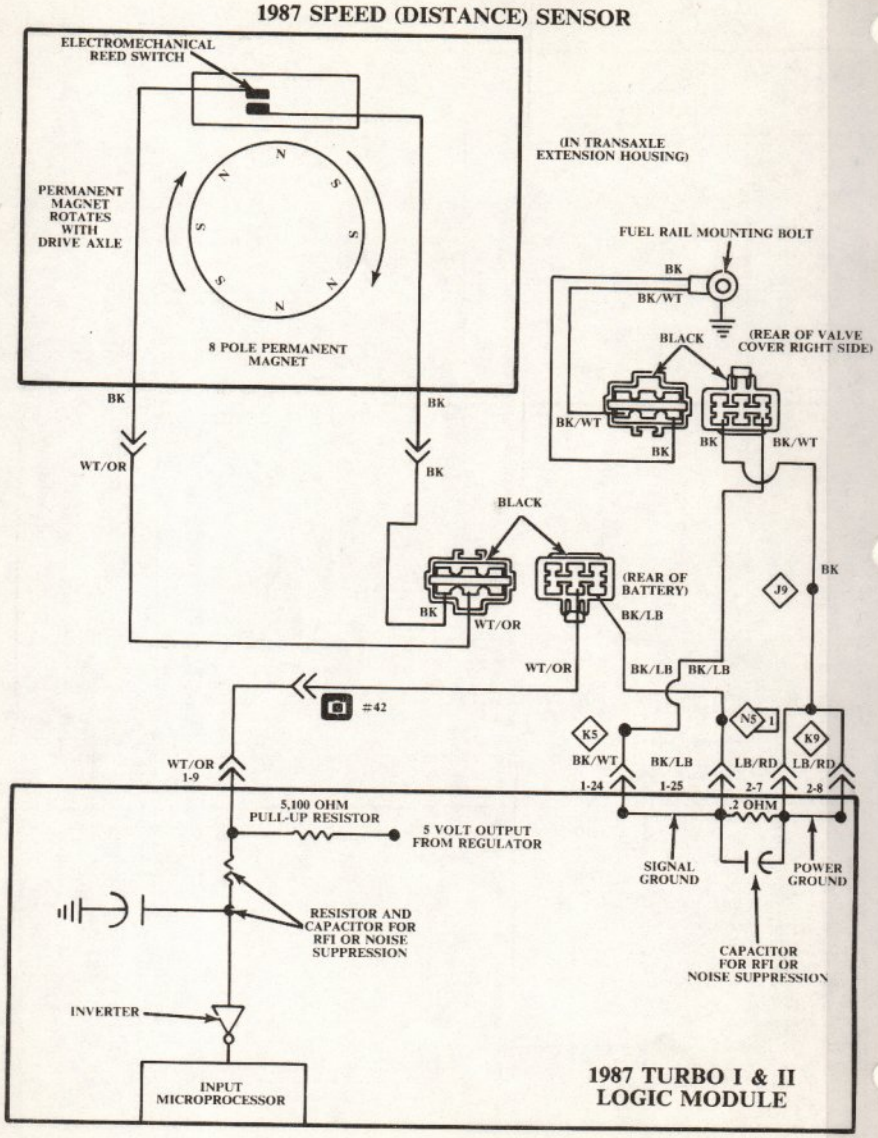

1987 SPEED (DISTANCE) SENSOR
ELECTROMECHANICAL REED SWITCH
(IN TRANSAXLE EXTENSION HOUSING)
PERMANENT MAGNET ROTATES WITH DRIVE AXLE
8 POLE PERMANENT MAGNET
FUEL RAIL MOUNTING BOLT
BK
BK/WT
BLACK
(REAR OF VALVE COVER RIGHT SIDE)
BK/WT
BK
BK
BK/WT
BK
WT/OR
BK
BLACK
(REAR OF BATTERY)
BK
WT/OR
BK/LB
J9
BK
WT/OR
BK/LB
BK/LB
#42
K5
N5
K9
WT/OR
1-9
BK/WT
1-24
BK/LB
1-25
LB/RD
2-7
LB/RD
2-8
5,100 OHM PULL-UP RESISTOR
5 VOLT OUTPUT FROM REGULATOR
.2 OHM
SIGNAL GROUND
POWER GROUND
RESISTOR AND CAPACITOR FOR RFI OR NOISE SUPPRESSION
CAPACITOR FOR RFI OR NOISE SUPPRESSION
INVERTER
INPUT MICROPROCESSOR
1987 TURBO I & II LOGIC MODULE

Fig. 49

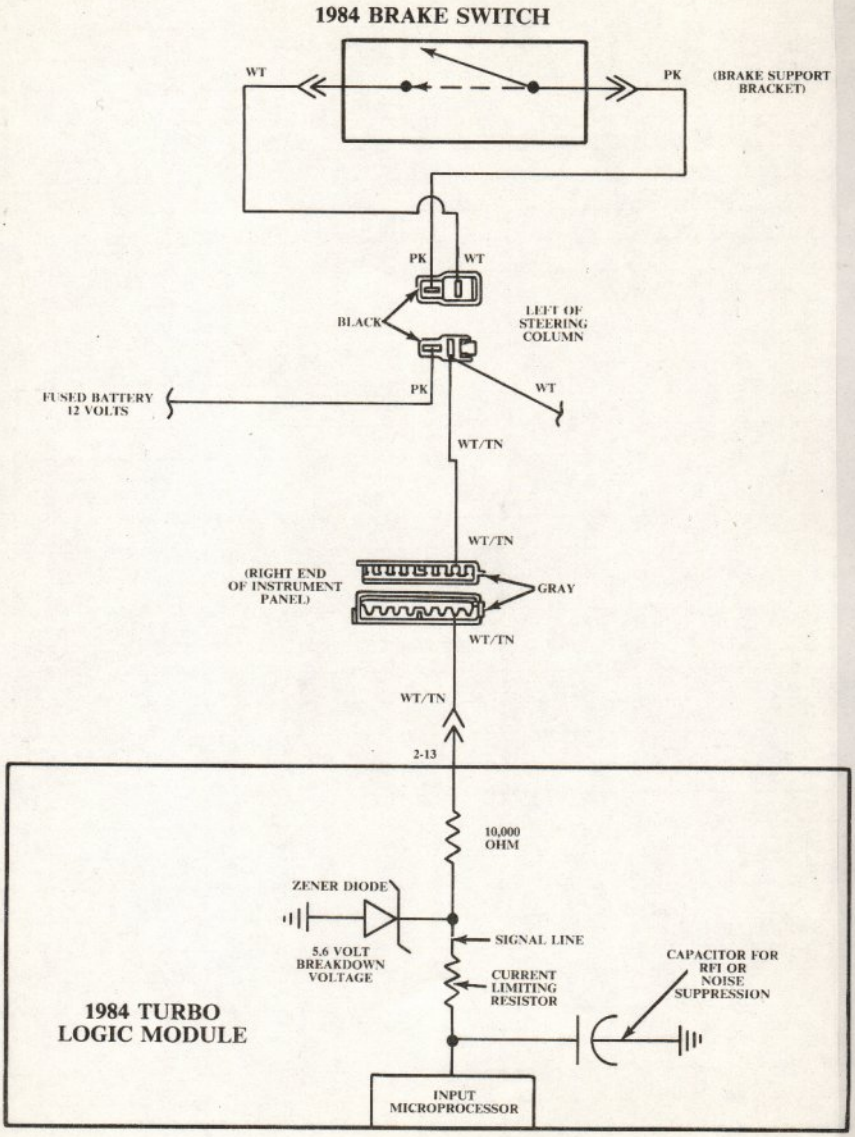

1984 BRAKE SWITCH
WT
PK
(BRAKE SUPPORT BRACKET)
PK
WT
BLACK
LEFT OF STEERING COLUMN
FUSED BATTERY 12 VOLTS
PK
WT
WT/TN
WT/TN
(RIGHT END OF INSTRUMENT PANEL)
GRAY
WT/TN
WT/TN
2-13
10,000 OHM
ZENER DIODE
5.6 VOLT BREAKDOWN VOLTAGE
SIGNAL LINE
CURRENT LIMITING RESISTOR
CAPACITOR FOR RFI OR NOISE SUPPRESSION
1984 TURBO LOGIC MODULE
INPUT MICROPROCESSOR

Fig. 50

Brake Switch

The brake switch is a 2 wire switch (non speed control) that is mounted in the brake support bracket. The brake switch "gives" the logic module (the microprocessor) 1 basic piece of information:

1. When the brakes are applied

The brake switch "gives" the logic module this information by sending a high signal (5.6 volts in 1984 or 5.14 volts in 1985 - '87) to it. When the signal goes high, this "tells" the logic module that the brakes are applied. The brake switch is electrically the same for 1984, '85, '86 & '87. There are two different types of circuits, the 1984 style and the 1985, '86 & '87 style. The 1984 style will be discussed first.

1984 Brake Switch and Circuit

Anytime the ignition key is "on" and the brake pedal is pushed, this closes the contacts in the brake switch. This sends 12 volts into the logic module on the WT/TN wire. See Fig. 50. Any voltage above 5.6 volts goes to ground through the zener diode. The zener diode limits the voltage on the signal line to 5.6 volts. See Fig. 50. This 5.6 volts then goes to the input of the microprocessor. When the microprocessor "sees" a high at its input, it "thinks" the brakes are applied. When the brake pedal is released, the switch contacts open. This puts a low at the input to the microprocessor. When the microprocessor "sees" a low at its input, it "thinks" the brakes are released.

1985, '86 & '87 Brake Switch and Circuit

Anytime the ignition key is "on", the logic module sends a regulated 8 volts to the 2,000 ohm pull-up resistor. See Fig. 51 or 52. When the brakes are released and the contacts open, all the current flow is through the pull-up resistor and to the 20,000 ohm resistor and then to the 15,000 ohm resistor to signal ground. See Fig. 51 or 52. This puts a low (3.24 volts) at the input to the CPI. When the microprocessor "sees" a low at the input to the CPI, it "thinks" the brakes are released. When the brake pedal is pushed and the contacts close, all of the current flow is into the logic module on the WT/TN wire through the 20,000 ohm resistor then to the 15,000 ohm resistor to signal ground. This puts a high (5.14 volts) at the input to the CPI. See Fig. 51 or 52. When the microprocessor "sees" a high at the input to the CPI, it "thinks" the brakes are applied.

Brake Switch Fault Code

There are no fault codes for the brake switch circuit. The logic module doesn't "know" the difference between an open or short circuit and the brake pedal being pushed or released. The logic module is capable of echoing back a response that it has "seen" a change in state. The diagnostic readout box (C-4805) can be used to determine if the logic module is recognizing the change in state. With the diagnostic readout box hooked up and in the switch test mode, push and release the brake pedal. The display should change from 00 to 88 or 88 to 00. This tells you that the logic module is recognizing the change in state and the circuit is OK. If there is no change in the display, there is a problem in the circuit.

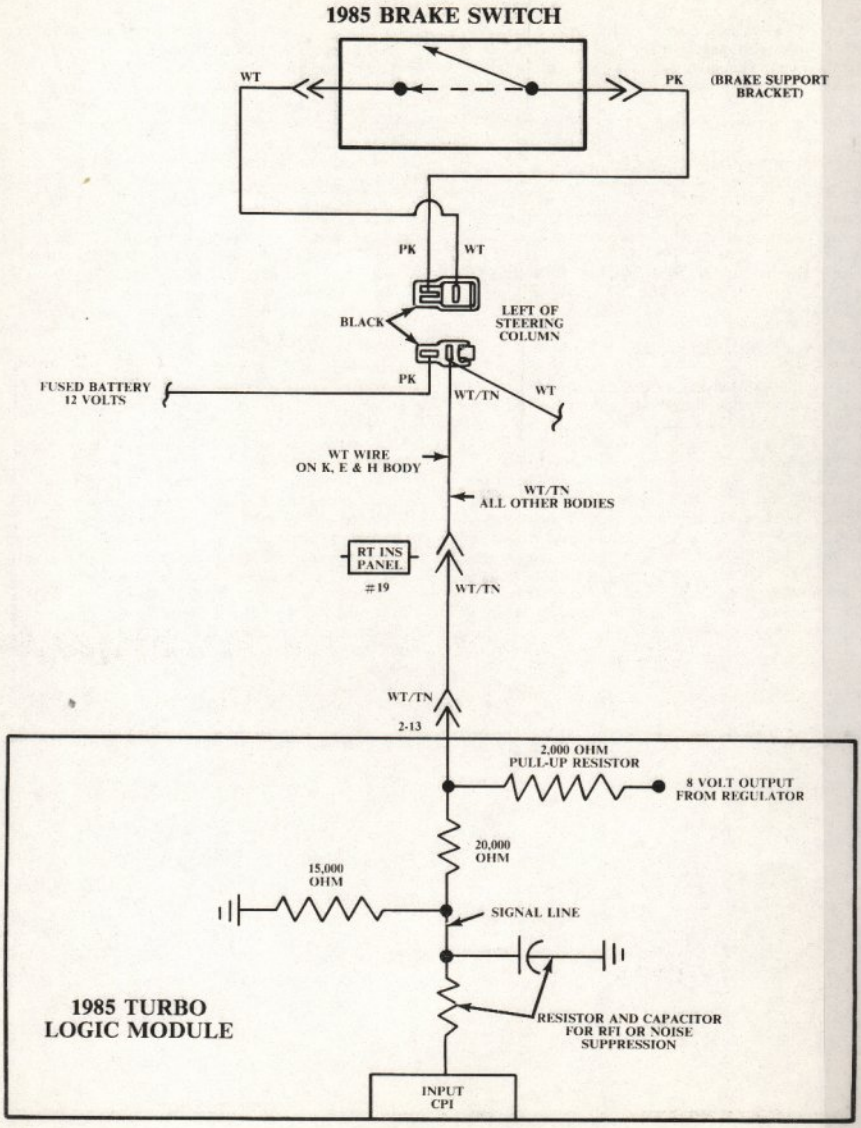

1985 BRAKE SWITCH
WT
PK
(BRAKE SUPPORT BRACKET)
PK
WT
BLACK
LEFT OF STEERING COLUMN
PK
FUSED BATTERY 12 VOLTS
WT/TN
WT
WT WIRE ON K, E & H BODY
WT/TN ALL OTHER BODIES
RT INS PANEL
#19
WT/TN
WT/TN
2-13
2,000 OHM PULL-UP RESISTOR
8 VOLT OUTPUT FROM REGULATOR
20,000 OHM
15,000 OHM
SIGNAL LINE
1985 TURBO LOGIC MODULE
RESISTOR AND CAPACITOR FOR RFI OR NOISE SUPPRESSION
INPUT CPI

Fig. 51

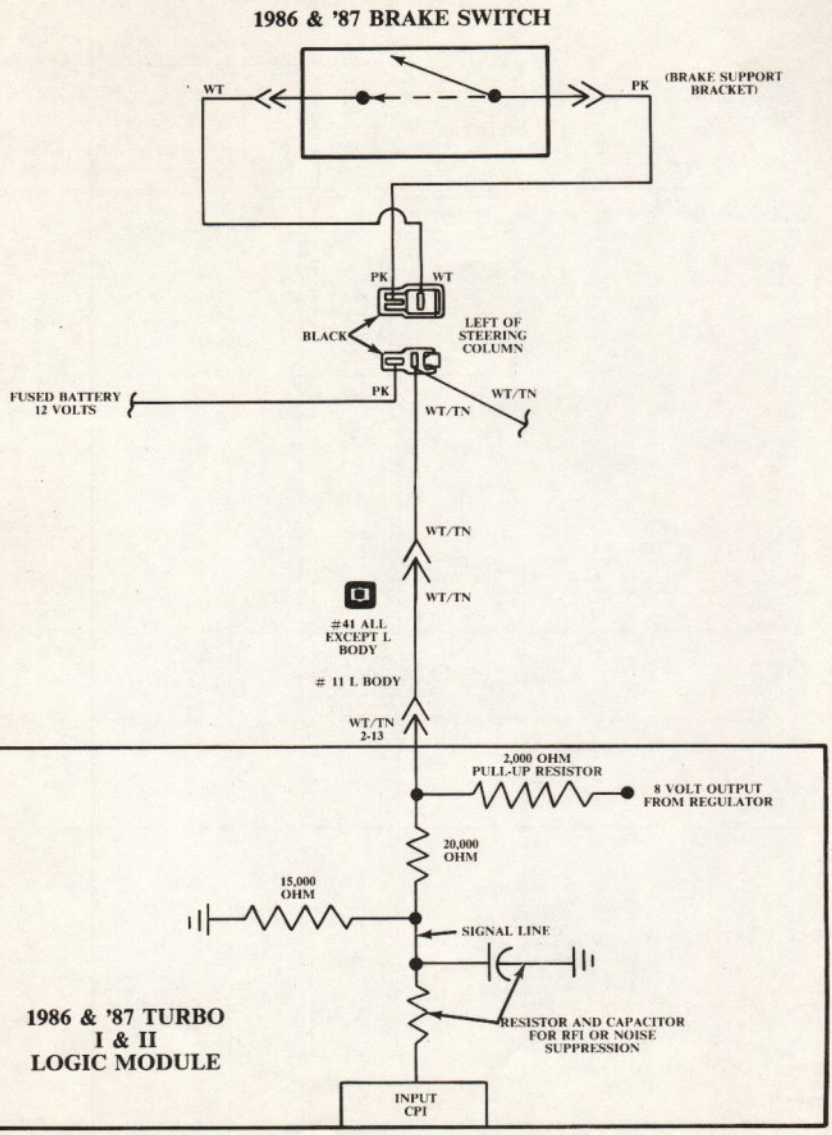

1986 & '87 BRAKE SWITCH
WT
PK
(BRAKE SUPPORT BRACKET)
PK
WT
BLACK
LEFT OF STEERING COLUMN
FUSED BATTERY 12 VOLTS
PK
WT/TN
WT/TN
WT/TN
WT/TN
#41 ALL EXCEPT L BODY
11 L BODY
WT/TN 2-13
2,000 OHM PULL-UP RESISTOR
8 VOLT OUTPUT FROM REGULATOR
20,000 OHM
15,000 OHM
SIGNAL LINE
RESISTOR AND CAPACITOR FOR RFI OR NOISE SUPPRESSION
1986 & '87 TURBO I & II LOGIC MODULE
INPUT CPI

Fig. 52

1984 HEATED REAR WINDOW SWITCH

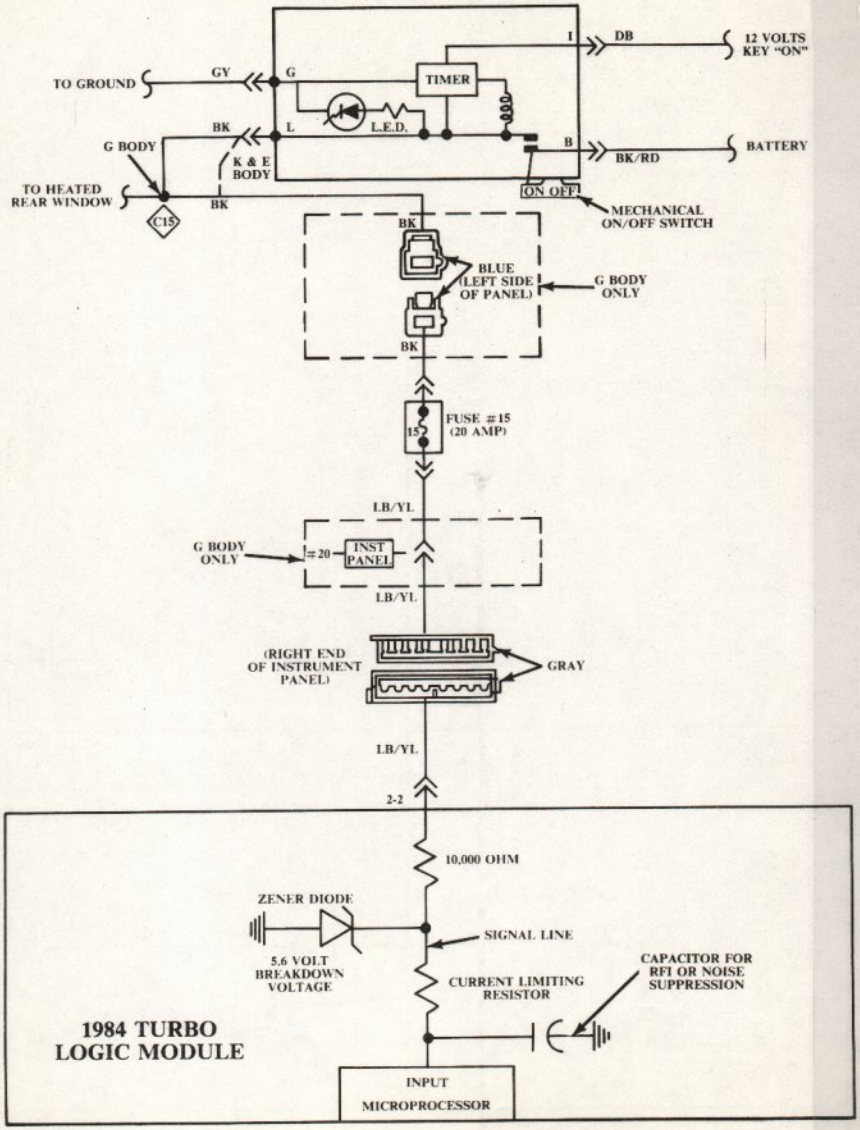


Fig. 53

Heated Rear Window Switch 1984 Only

The microprocessor monitors the state of the heated rear window circuit to determine whether it is on. Anytime the ignition key is "on" and the heated rear window switch is pushed, this puts 12 volts to the heated rear window. The 12 volts also goes into the logic module on the LB/YL wire. See Fig. 53. Any voltage above 5.6 volts goes to ground through the zener diode. The zener diode limits the voltage on the signal line to 5.6 volts. This 5.6 volts goes to the input of the microprocessor. When the microprocessor "sees" a high at its input, it "thinks" the heated rear window is on. When the heated rear window turns off, this removes the 12 volts and puts a low (0 volts) at the input to the microprocessor. When the microprocessor "sees" a low at its input, it "thinks" the heated rear window is off.

Heated Rear Window Fault Code

There are no fault codes for the heated rear window circuit. The logic module doesn't "know" the difference between an open or short circuit and the heated rear window being on or off. The logic module is capable of echoing back a response that it has "seen" a change in state. The diagnostic readout box (C-4805) can be used to determine if the logic module is recognizing the change in state. With the diagnostic readout box hooked up and in the switch test mode, push the heated rear window switch on and off. The display should change from 00 to 88 or 88 to 00. This tells you that the logic module is recognizing the change in state and the circuit is OK. If there is no change in the display, there is a problem in the circuit.

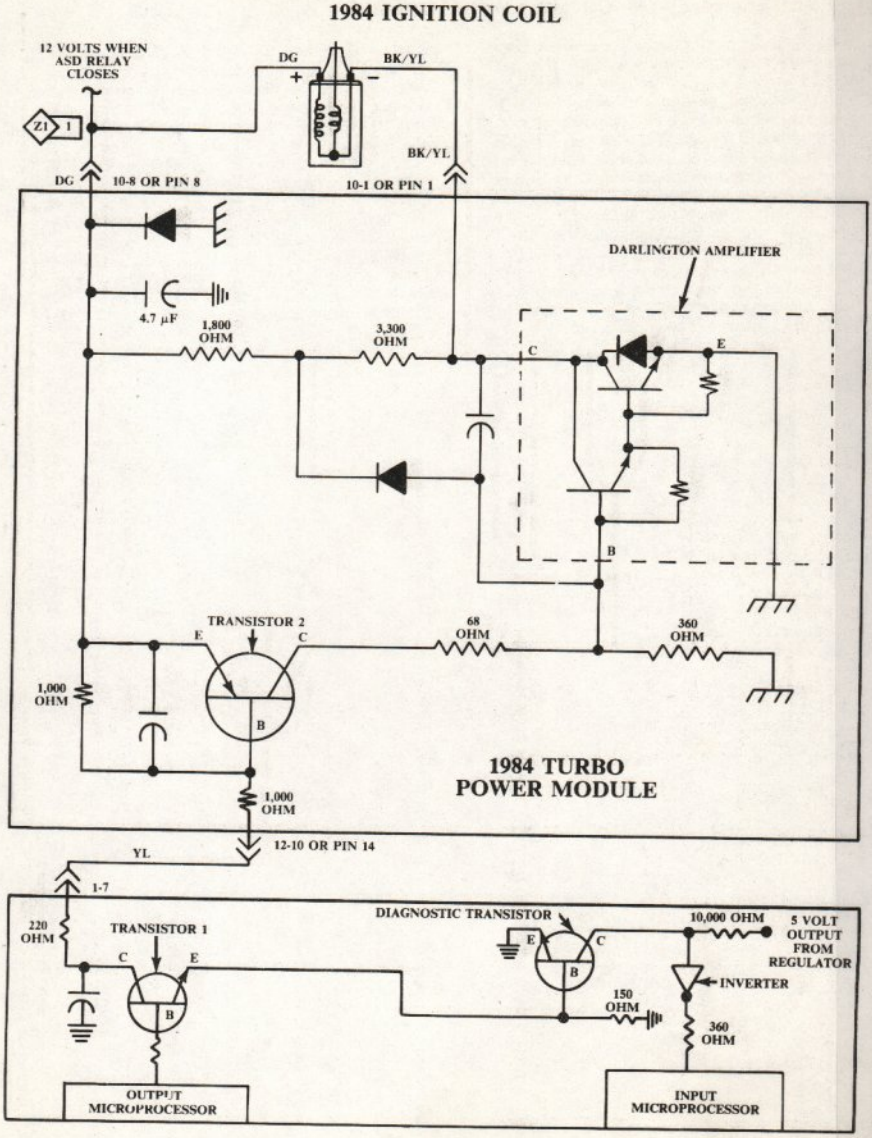

1984 IGNITION COIL
12 VOLTS WHEN ASD RELAY CLOSES
DG
BK/YL
+
−
Z1
1
BK/YL
DG
10-8 OR PIN 8
10-1 OR PIN 1
DARLINGTON AMPLIFIER
4.7 µF
1,800 OHM
3,300 OHM
C
E
B
TRANSISTOR 2
E
C
68 OHM
360 OHM
1,000 OHM
B
1,000 OHM
1984 TURBO POWER MODULE
YL
12-10 OR PIN 14
1-7
220 OHM
TRANSISTOR 1
C
E
B
DIAGNOSTIC TRANSISTOR
E
C
B
10,000 OHM
5 VOLT OUTPUT FROM REGULATOR
INVERTER
150 OHM
360 OHM
OUTPUT MICROPROCESSOR
INPUT MICROPROCESSOR

Fig. 54

1984 Ignition Coil and Circuit

Anytime the ASD relay is closed, 12 volts is sent to the positive (+) side of the coil. See Fig. 54. The 12 volts goes through the primary winding and leaves the coil on the negative (−) side "looking" for a ground. This 12 volts goes into the power module on the BK/YL wire and stops at the collector of the Darlington Amplifier. See Fig. 54. This happens because the Darlington Amplifier is "off". When the logic module "wants" to turn the primary "on", the microprocessor puts 5 volts at the base of Transistor 1 which turns it "on". See Fig. 54. When Transistor 1 is "on", this puts approximately 6 volts at the base of Transistor 2. This turns "on" Transistor 2. With Transistor 2 "on", this puts approximately 10 volts at the base of the Darlington Amplifier. That turns the Darlington Amplifier "on" and allows the coil negative to be connected to power ground. See Fig. 54. Now the primary is "on" and the magnetic field is building up. When the logic module receives a signal (5 to 0 volts) from the reference pickup, it then calculates the spark delay. Once the correct spark delay is achieved, the microprocessor removes the voltage from the base of Transistor 1. That puts 0 volts at the base of Transistor 1 which turns it "off". With Transistor 1 "off", this puts 12 volts at the base of Transistor 2 which shuts it "off". With Transistor 2 "off", that removes the 10 volts from the base of the Darlington Amplifier which turns it "off". With the Darlington Amplifier "off", that removes the ground from the coil negative and the magnetic field collapses inducing a high voltage in the secondary winding. This high voltage then crosses the spark plug gap. This "on" "off" sequence between the transistors happens as long as the reference pickup is changing states.

In summary, the logic module provides the ground for the power module which provides the ground for the coil.

1984 Ignition Coil Fault Code

Anytime the ignition key is "on" and the reference pickup is changing states (0 to 5 or 5 to 0 volts), the logic module monitors the circuit for proper switching. If the logic module receives a 0 to 5 volts reference signal, the microprocessor provides the voltage to the base of Transistor 1. This turns "on" Transistors 1 & 2, the Darlington Amplifier and the primary. This also turns "on" the Diagnostic Transistor. See Fig. 54. This makes the input to the I/O go high (after being inverted). So in this way the microprocessor can send out a command (turn "on" Transistor 1) and it "looks to see' the proper response (I/O input going high). If the response doesn't match the command, the engine will stall and fault code 43 is stored in the logic module's memory.

1985, '86 & '87 Ignition Coil and Circuit

Anytime the ASD relay is closed, 12 volts is sent to the positive (+) side of the coil. See Fig. 55. This 12 volts goes through the primary winding and leaves the coil on the negative (−) side "looking" for a ground. This 12 volts goes into the power module on the BK/YL wire and stops at the collector of Transistor 3. See Fig. 55. This happens because Transistor 3 is "off". When the logic module "wants" to turn the primary "on", the microprocessor provides a ground to the base of Transistor 1. See Fig. 55. This puts 0 volts at the base of Transistor 1 which turns it "off". When Transistor 1 is "off", this puts 5 volts at the base of Transistor 2 which turns it "on". When Transistor 2 is "on", this puts approximately 8.5 volts at the base of Transistor 3 which turns it "on". When Transistor 3 is "on" this allows the coil negative to be connected with power ground. Now the primary is "on" and the magnetic field is building up. When the logic module receives the signal from the reference pickup and the spark delay is calculated, the microprocessor removes the ground from Transistor 1. This turns Transistor 1 "on" and puts a low (.44 volts in 1985 & '86 or 1.48 volts in 1987) at the base of Transistor 2 which turns it "off". With Transistor 2 "off", this removes the 8.5 volts from the base of Transistor 3 which turns it "off". With Transistor 3 "off", this removes the ground from the coil negative collapsing the magnetic field.

In summary, the logic module provides the open circuit (Transistor 1 "off") for the power module which provides the ground for coil.

1985, '86 & '87 Ignition Coil Fault Code

Anytime the ignition key is "on" and the reference pickup is changing states, the logic module monitors the circuit for proper switching. If the logic module receives a 0 to 5 volts reference signal, the microprocessor provides a ground at the base of Transistor 1. This turns Transistor 1 "off" which turns "on" Transistors 2 & 3 and the primary. This turns the Diagnostic Transistor "off". See Fig. 55. This makes the input to the microprocessor go high. So in this way the microprocessor can send out a command (turn "off" Transistor 1) and it "looks to see" the proper response (microprocessor input going high). If the response doesn't match the command, the engine will stall and fault code 43 is stored in the logic module's memory.

1985, '86 & '87 IGNITION COIL

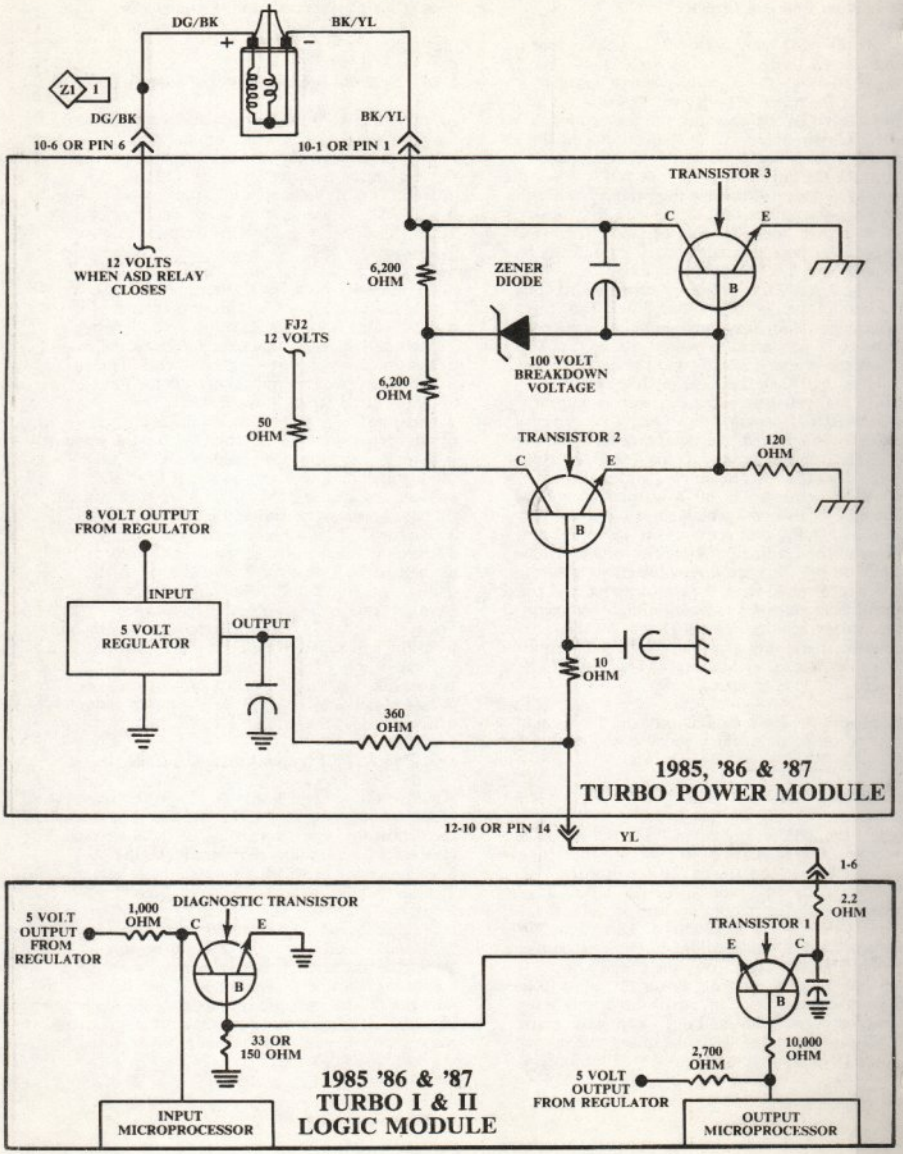


Fig. 55

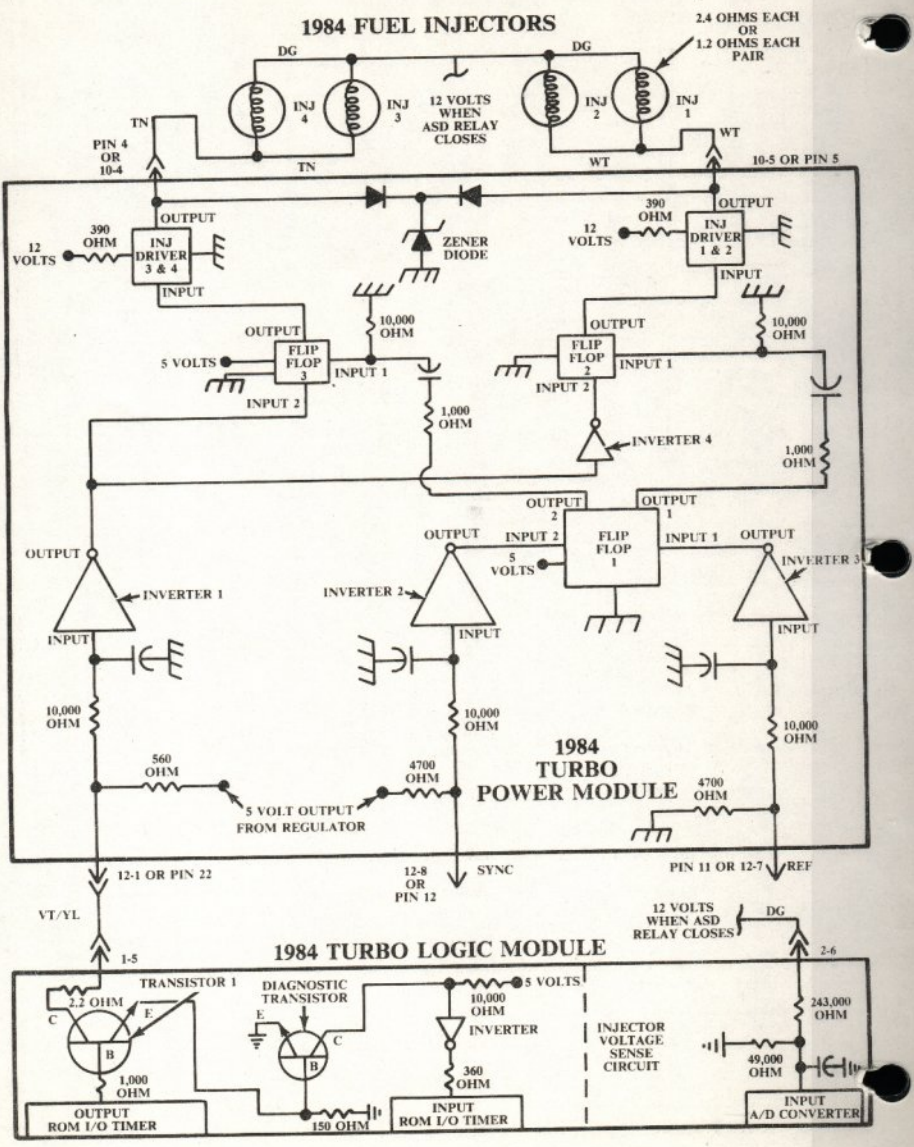

1984 FUEL INJECTORS
2.4 OHMS EACH OR 1.2 OHMS EACH PAIR
DG
DG
INJ 4
INJ 3
INJ 2
INJ 1
12 VOLTS WHEN ASD RELAY CLOSES
TN
TN
WT
WT
PIN 4 OR 10-4
10-5 OR PIN 5
12 VOLTS
390 OHM
INJ DRIVER 3 & 4
OUTPUT
INPUT
ZENER DIODE
12 VOLTS
390 OHM
INJ DRIVER 1 & 2
OUTPUT
INPUT
10,000 OHM
10,000 OHM
5 VOLTS
FLIP FLOP 3
INPUT 1
INPUT 2
FLIP FLOP 2
INPUT 1
INPUT 2
1,000 OHM
INVERTER 4
1,000 OHM
OUTPUT 2
OUTPUT 1
OUTPUT
INPUT 2
FLIP FLOP 1
INPUT 1
OUTPUT
5 VOLTS
INVERTER 3
INVERTER 1
INVERTER 2
INPUT
INPUT
INPUT
10,000 OHM
10,000 OHM
10,000 OHM
560 OHM
4700 OHM
4700 OHM
1984 TURBO POWER MODULE
5 VOLT OUTPUT FROM REGULATOR
12-1 OR PIN 22
12-8 OR PIN 12
SYNC
PIN 11 OR 12-7
REF
VT/YL
12 VOLTS WHEN ASD RELAY CLOSES
DG
1-5
1984 TURBO LOGIC MODULE
2-6
TRANSISTOR 1
2.2 OHM
C
E
B
1,000 OHM
DIAGNOSTIC TRANSISTOR
E
C
B
5 VOLTS
10,000 OHM
INVERTER
360 OHM
INJECTOR VOLTAGE SENSE CIRCUIT
243,000 OHM
49,000 OHM
OUTPUT ROM I/O TIMER
150 OHM
INPUT ROM I/O TIMER
INPUT A/D CONVERTER

Fig. 56

1984 Fuel Injectors and Circuit

Anytime the ASD relay is closed, 12 volts is sent to all four injectors. For the injectors to turn "on", they need a ground. The injectors are grounded or fired in pairs, 1 & 2 or 3 & 4. The power module provides the ground at the instruction of the sync and reference inputs. The power module removes the ground at the instruction of the logic module. In other words, the injectors are turned on by hardware and turned off by software.

When the power module receives the sync and reference signals, these signals are inverted then sent to Flip Flop 1. See Fig. 56. Depending on which signal is high or low, Flip Flop 1 will output a voltage "blip" (momentary high voltage) to Flip Flop 2 or 3. See Fig. 56. If Flip Flop 2 receives a voltage from Flip Flop 1 at input 1, Flip Flop 2 will turn "on" and output a voltage to Injector Driver 1 & 2. This turns Injector Driver 1 & 2 "on", which then provides a ground for injectors 1 & 2 at the output of the Driver.

After the logic module has calculated the pulse width (injector "on" time), the I/O in the logic module outputs a voltage to the base of Transistor 1. This turns Transistor 1 "on" which changes the voltage at the input to Inverter 1 from a high (5 volts) to a low (0 volts). Inverter 1 takes the low and inverts it to a high. This high then goes to Inverter 4. See Fig. 56. Inverter 4 takes the high and inverts it to a low. This low is then sent to input 2 of Flip Flop 2. See Fig. 56. This causes Flip Flop 2 to shut "off" the voltage from its output which shuts "off" Injector Driver 1 & 2. With the Injector Driver "off", this removes the ground from the injector pair causing them to turn "off" (close). Flip Flop 3 also gets the injector "off" signal along with Flip Flop 2, however; Injector Driver 3 & 4 is already "off". When the injectors turn "off", there may be a voltage spike (high voltage) from the magnetic field collapsing in the injector windings. The Injector Drivers are protected from voltage spikes by a zener diode. See Fig. 56. If a voltage spike happens the zener diode breaks down and allows the high voltage to go to ground.

When the sync changes state, this changes the input to Inverter 2. This "tells" Flip Flop 1 to send a voltage "blip" Flip Flop 3. When Flip Flop 3 turns "on", this turns "on" Injector Driver 3 & 4 and provides a ground for injectors 3 & 4. After the pulse width is calculated, the I/O outputs a positive voltage turning "on" Transistor 1. With Transistor 1 "on", the input to Inverter 1 goes low. Inverter 1 inverts this low to a high and sends it to Flip Flop 3. This shuts "off" the voltage output of Flip Flop 3 which shuts "off" Injector Driver 3 & 4. Flip Flop 2 also gets the injector turn "off" signal, however; that pair of injectors is already "off". This sequence of firing one pair of injectors then the other continues as long as the engine is running.

Each injector is only fired once in every 720°. Since the injectors are fired in pairs, 1 & 3 are fired on the intake stroke and 2 & 4 are fired on the exhaust stroke (intake valve closed).

There is also a injector voltage sense circuit in the logic module. See Fig. 56. This circuit "tells" the logic module how much voltage is available at the injectors. The logic module then uses this information to compensate the injector pulse width during a low voltage condition.

1984 Fuel Injector Fault Code

Anytime the ignition key is "on" and the logic module is receiving a sync and reference input, the logic module monitors the injector "off" signal (VT/YL wire) for an open circuit or a short to ground. If there is an open or short under these conditions, fault code 43 is stored in the logic module's memory and the engine will stall or never start. When the microprocessor "tells" the I/O to turn "on" Transistor 1, this shuts "off" a pair of injectors. This also turns "on" the Diagnostic Transistor which changes the input to the microprocessor from a low to a high. See Fig. 56. If Transistor 1 is turned "on" and there is no change at the input to the microprocessor, the logic module assumes an open or short condition in the injector "off" circuit. In this way the logic module can send out a command and then "look to see" if it has been carried out.

1985, '86 & '87 Fuel Injectors and Circuit

Anytime the ASD relay is closed, 12 volts is sent to all four injectors. As in 1984, for the injectors to be turned "on", they need a ground. These injectors are also fired in pairs, 1 & 2 or 3 & 4. This system is different from 1984 in that the hardware that controlled the injector circuit in the power module (the Flip Flops) has been reduced and replaced by increased software and hardware in the logic module. See Fig. 57. In this system the logic module turns the injectors "on" as well as "off".

After the logic module receives a sync signal, at the next reference signal a pair of injectors are fired. Let's assume that injectors 1 & 2 need to be fired. At this time, the logic module will turn "on" Transistor 1. See Fig. 57. This causes the voltage at the input of Inverter 1 to go low (0 volts). Inverter 1 then takes this low and inverts it to a high. This high is then sent to the input of Injector Driver 1 & 2. See Fig. 57. When the Injector Drivers receive a high at their input, they turn "on" and provide a ground at their output. See Fig. 57. This turns "on" injectors 1 & 2. Once the pulse width time has run out, the microprocessor turns "off" Transistor 1. This puts a high (12 volts) at Inverter 1. Inverter 1 takes the high and inverts it to low. This low is then sent to Injector Driver 1 & 2 which turns it "off". Again as in 1984, the Injector Drivers are protected

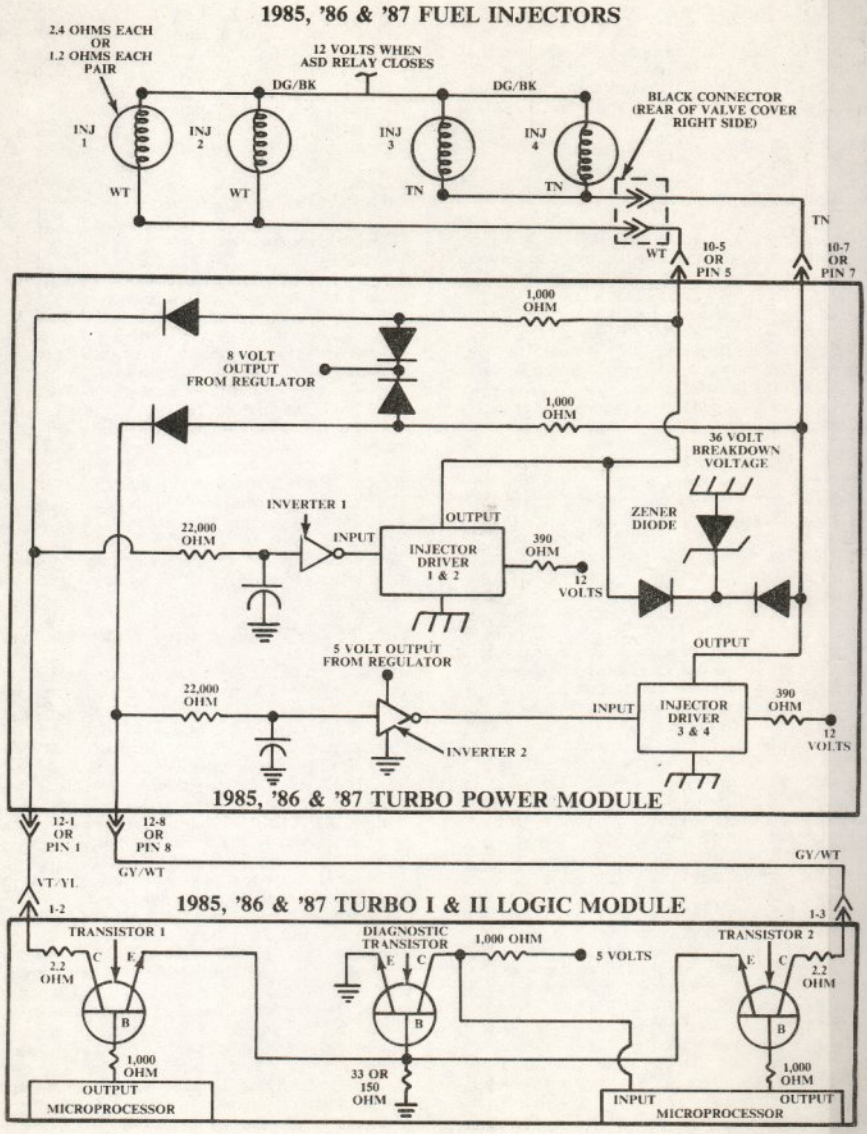

1985, '86 & '87 FUEL INJECTORS
2.4 OHMS EACH OR 1.2 OHMS EACH PAIR
12 VOLTS WHEN ASD RELAY CLOSES
DG/BK
DG/BK
BLACK CONNECTOR (REAR OF VALVE COVER RIGHT SIDE)
INJ 1
INJ 2
INJ 3
INJ 4
WT
WT
TN
TN
TN
WT
10-5 OR PIN 5
10-7 OR PIN 7
1,000 OHM
8 VOLT OUTPUT FROM REGULATOR
1,000 OHM
36 VOLT BREAKDOWN VOLTAGE
ZENER DIODE
INVERTER 1
22,000 OHM
INPUT
OUTPUT
INJECTOR DRIVER 1 & 2
390 OHM
12 VOLTS
5 VOLT OUTPUT FROM REGULATOR
OUTPUT
22,000 OHM
INPUT
INJECTOR DRIVER 3 & 4
390 OHM
12 VOLTS
INVERTER 2
1985, '86 & '87 TURBO POWER MODULE
12-1 OR PIN 1
12-8 OR PIN 8
GY/WT
VT/YL
GY/WT
1-2
1-3
1985, '86 & '87 TURBO I & II LOGIC MODULE
TRANSISTOR 1
2.2 OHM
C
E
B
1,000 OHM
OUTPUT MICROPROCESSOR
DIAGNOSTIC TRANSISTOR
E
C
B
1,000 OHM
5 VOLTS
33 OR 150 OHM
TRANSISTOR 2
E
C
2.2 OHM
B
1,000 OHM
INPUT
OUTPUT
MICROPROCESSOR

Fig. 57

from voltage spikes by a zener diode. See Fig. 57. If injectors 3 & 4 need to be turned "on", the same sequence of events happen, except that Transistor 2 starts the sequence.

1985, '86 & '87 Fuel Injector Fault Codes

During cranking, if the injectors don't switch "on" and "off" properly, fault code 26 or 27 is stored in the logic module's memory. The engine may run in this condition, however; it is more likely that the engine will stall and not restart. When the microprocessor turns "on" Transistor 1, this also turns "on" the Diagnostic Transistor. See Fig. 57. This puts a low (0 volts) at the input to the microprocessor. When the microprocessor turns Transistor 1 "off", this turns the Diagnostic Transistor "off" which makes the input to the microprocessor go high (5 volts). See Fig. 57. As the microprocessor is turning Transistor 1 "on" and "off", it "looks to see" if its input from the Diagnostic Transistor is going high and low. If the microprocessor doesn't "see" any response at its input, fault code 26 (injector 1 & 2 circuit) is stored in the logic module's memory. If the same responses don't happen when Transistor 2 is turned "on" and "off", fault code 27 (injectors 3 & 4 circuit) is stored in the logic module's memory.

1984 AIS MOTOR

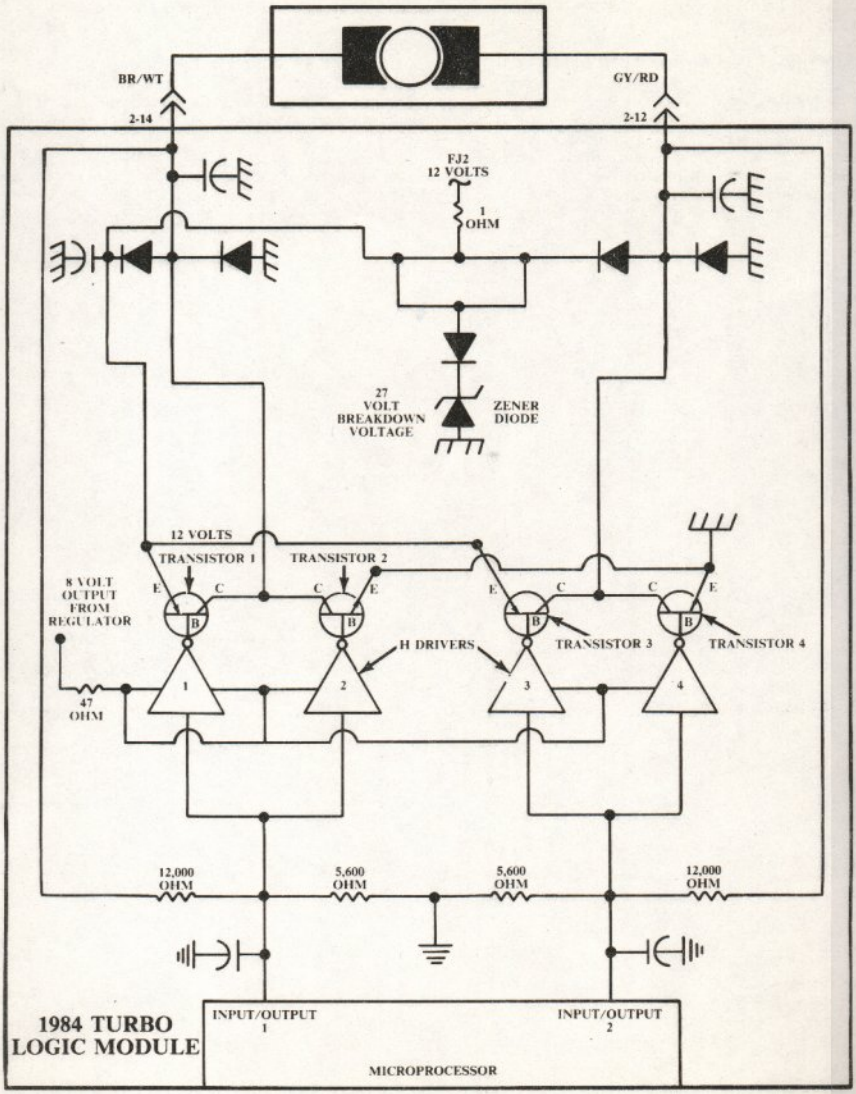


Fig. 58

1984 AIS Motor and Circuit

Anytime there is a speed error from target idle speed or at a closed throttle during decel, the logic module drives the AIS motor open or closed as necessary. To drive the AIS motor open (higher RPM), the logic module needs to put 12 volts on the GY/RD wire and provide a ground on the BR/WT wire. See Fig. 58. To do this the microprocessor commands the H Drivers to turn "on" Transistors 1 & 4 and turn "off" Transistors 2 & 3. This allows the FJ2 (12 volts) to be connected across the emitter collector of Transistor 1. See Fig. 58. The 12 volts then leaves the logic module on the BR/WT wire and goes to the AIS motor. The 12 volts then leaves the AIS motor "looking" for a ground. This then goes into the logic module on the GY/RD wire and goes to power ground through the collector emitter of Transistor 4. See Fig. 58. When the microprocessor "wants" to close the AIS motor, it has to reverse the polarity of the two wires. It does this by sending opposite commands to the H Drivers. This turns "on" Transistors 2 & 3 and turns "off" Transistors 1 & 4. See Fig. 58. The end result is 12 volts on the GY/RD wire and a ground on the BR/WT wire.

1985, '86 & '87 Turbo I AIS Motor and Circuit

This is the same AIS motor as in 1984 and a similar circuit. In 1984 there were four H Drivers to handle the current flow. The extra current flow capabilities was not needed and therefore eliminated. There are only two H Drivers in this circuit. See Fig. 59. These H Drivers work the same as in 1984, however; the transistors are internal and not shown in the schematic. When the microprocessor "wants" to drive the AIS motor, it "tells" the CPI to output a command (voltage) to the H Drivers. One of the H Drivers will output 12 volts while the other one outputs a ground. This is reversed to get the AIS motor to move in the opposite direction.

1987 Turbo II AIS Stepper Motor and Circuit

The Turbo II AIS stepper motor is the same style as what's used on the 1986 & '87 low pressure EFI. This style of AIS motor has two electromagnets (windings) and one permanent magnet. The permanent magnet is the "nut" with threads on the inside. The AIS motor pintle is attached to a shaft with threads on it (the "bolt"). When one electromagnet is energized, it causes the permanent magnet to be attracted to it. The permanent magnet now rotates to line up with the opposite poles. Now the other winding is energized making a new set of magnetic poles that the permanent magnet is attracted to. The permanent magnet then rotates to line up with the opposite poles again. As the permanent magnet rotates, the pintle is driven in or out.

The AIS stepper circuit is the same as 1985-'87 with one exception. Since there are two windings and four connections, there are four H Drivers in the logic module. See Fig. 60. When the microprocessor "wants" to drive the AIS stepper motor it "tells" the CPI to output a command (voltage) to H Drivers 1 & 4. This provides 12 volts on one wire and a ground on the other wire. The AIS moves one step and stops. The microprocessor now "tells" the CPI to turn "off" H Drivers 1 & 4 and to turn "on" H Drivers 2 & 3. This causes the AIS to move one more step. This oscillating back and forth from winding to winding happens so fast, (How fast does it move?), so fast that the AIS stepper motor appears to be running continuously.

AIS Motor Fault Code

Anytime the AIS motor is driven open or close, the microprocessor momentarily stops the output command and "looks" at the voltage present on that line. The voltage is fed back through diagnostic resistors (12,000 and 5,600 ohm in 1984 and 27,000 and 12,000 ohm in 1985-'87). See Fig. 58 or 59. In this way the output line is now an input line. This happens for only a few microseconds (millionths of second) so we don't notice the AIS motor stopping. When the microprocessor or CPI commands the H Drivers to drive the AIS motor, if the proper voltage is not fed back, the microprocessor assumes that there is an open in the H Driver circuit. If there is voltage fed back when there is no command for them to be "on", the microprocessor assumes that the H Drivers turned "on" by themselves. In either case fault code 25 is stored in the logic module's memory. An open circuit in the harness or the AIS motor will not set fault code 25. This is true because when the microprocessor commands the H Drivers to turn "on", they turn "on" and feed back the proper voltage inside the logic module. So an open outside the logic module is not detected. See Fig. 58 or 59.

1985, '86 & '87 TURBO I AIS MOTOR

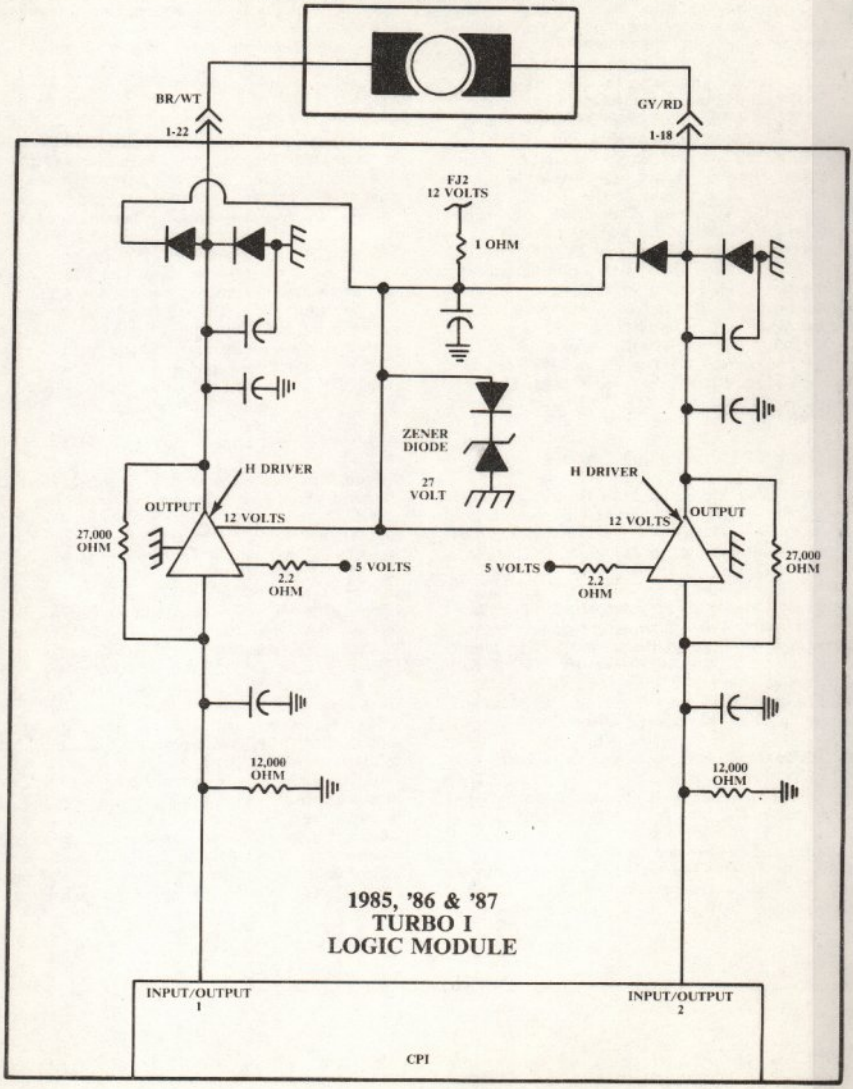


Fig. 59

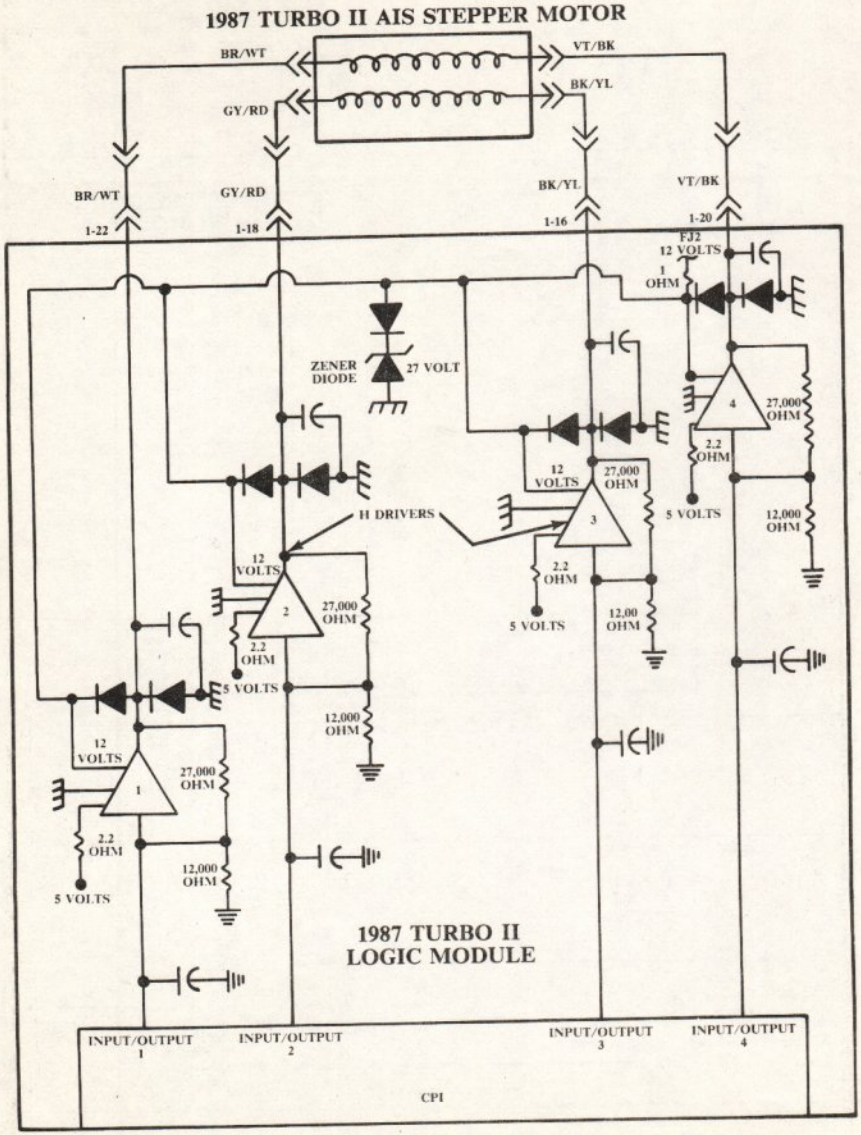

1987 TURBO II AIS STEPPER MOTOR
BR/WT
GY/RD
VT/BK
BK/YL
1-22
1-18
1-16
1-20
FJ2
12 VOLTS
1 OHM
ZENER DIODE
27 VOLT
H DRIVERS
12 VOLTS
27,000 OHM
2.2 OHM
5 VOLTS
12,000 OHM
1
2
3
4
1987 TURBO II
LOGIC MODULE
INPUT/OUTPUT 1
INPUT/OUTPUT 2
INPUT/OUTPUT 3
INPUT/OUTPUT 4
CPI

Fig. 60

1985, '86 & '87 ALTERNATOR FIELD

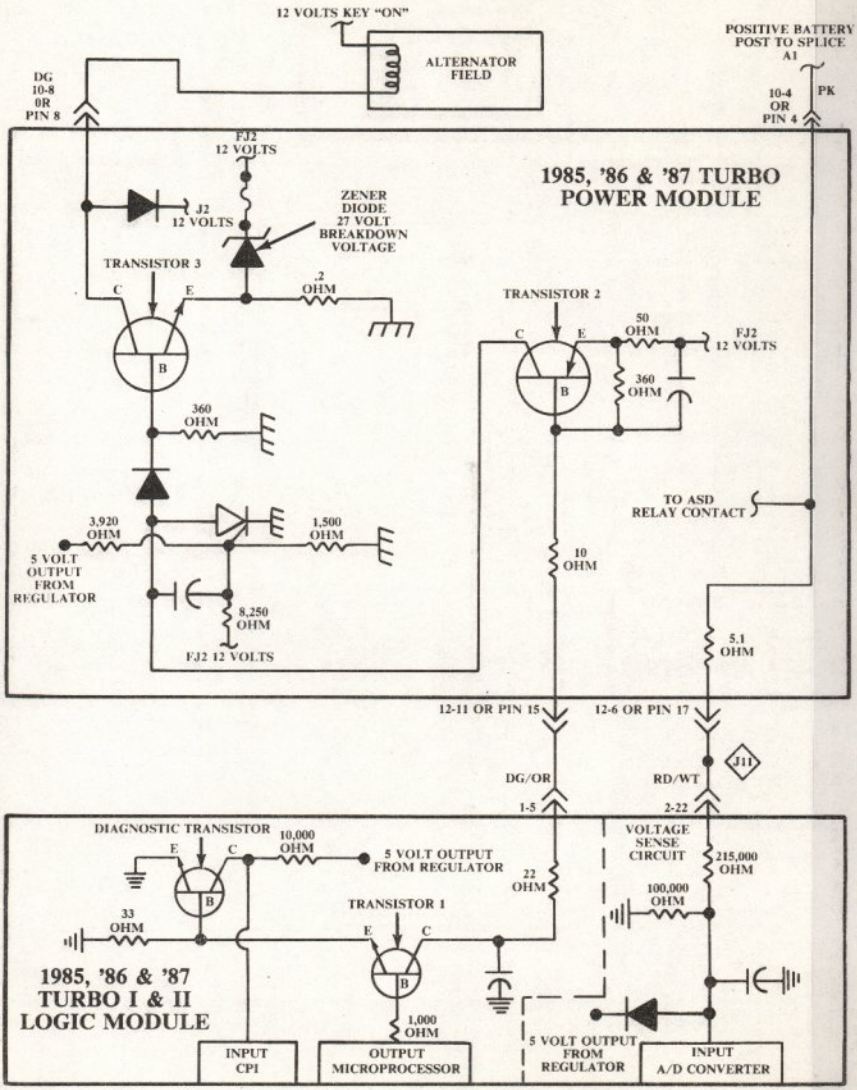


Fig. 61

Alternator Field and Circuit

In 1985 and later there is no external voltage regulator. The logic and power modules now do this job. Anytime the ignition key is "on", 12 volts is fed to one end of the alternator field. The other end of the alternator field is connected to the collector of Transistor 3 in the power module. See Fig. 61. If the base of Transistor 3 has a low voltage, it is "off" and the 12 volts stops at the collector. When the microprocessor "wants" the field "on", it outputs a duty cycle voltage to the base of Transistor 1 (some percentage of full "on"). This turns Transistor 1 "on" and which lowers the voltage at the base of Transistor 2 which turns it "on". With Transistor 2 "on", this puts a positive voltage at the base of Transistor 3. See Fig. 61. This turns Transistor 3 "on" and allows the field to go to ground (some percentage of full field). If the logic module senses low charging system voltage (sensed through the voltage sense circuit), it will increase the duty cycle at the base of Transistor 1. This increases the output of Transistor 2 which increases the output of Transistor 3. The result is more of a grounded field which increases alternator output. So the logic module "looks" at the voltage sense circuit to determine the duty cycle of the field. The logic module also "looks" at the battery temperature sensor to determine the maximum voltage charge rate.

Alternator Field or Charging System Fault Codes

Fault Code 16 (Open Circuit Voltage Sense Wire)

60 seconds after engine start-up, the logic module begins to monitor the voltage sense line for an open circuit. If after 60 seconds the voltage on the sense circuit is below 4 volts, the power loss/limited light is turned on. The system is now in "limp-in". Fault code 16 is stored in the logic module's memory. At this time the logic module will drive the alternator field at a 25% duty cycle to prevent overcharging and to get the car home.

If the voltage is between 4 and 8½ volts, the logic module will full field the alternator for 20 seconds. If after the 20 seconds, the voltage is between 7½ and 8½ volts, the power loss/limited light is turned on. The system is now in "limp-in". Fault code 16 is stored in the logic module's memory, however it cannot be read using the diagnostic readout box (C-4805). The alternator is driven at the "limp-in" duty cycle of 25%. If after the 20 seconds of full fielding the alternator, the voltage comes up above 8½ volts, the system reverts back to normal operation and the power loss/limited light is turned off.

The logic module is fed 12 volts (battery feed) from one wire (RD/WT) out of the power module. This wire divides at the J11 splice and feeds two different circuits in the logic module. One circuit is the voltage sense circuit. The second circuit is the 12 volt feed for the second 5 volt regulator in the logic module. This regulator powers up the logic module's memory and keeps it powered up when the ignition key is "off". This also allows you to read fault codes. This second circuit is tied to the 8 volt feed to the logic module through a diode inside the logic module.

If the 12 volt feed circuit is open on the power module side of the J11 splice, 8 volts will feed back through the diode in the logic module. This would put 8 volts on both circuits (voltage sense and memory) all the time. This is the reason for the 7½ to 8½ voltage "window" in the software. If the voltage doesn't come up after the alternator has been full fielded, the logic module assumes that the 12 volt feed is open on the power module side of the J11 splice and the 8 volts present on the voltage sense circuit is not the actual charging system voltage. In this case the power loss/limited light is turned on, however; no fault code is stored in memory. This happens because, when the ignition key is turned "off" to start the diagnostic sequence, this shuts down the 8 volt regulator in the power module. This clears the logic module's memory because the 8 volts is the only thing keeping the memory "alive". If the engine is restarted, after 1 minute and 20 seconds, the power loss/limited light will come back on again.

If the 12 volt feed from the power module is open on the logic module side of the J11 splice (voltage sense pin 2-22) the power loss/limited light is turned on 60 seconds after engine start-up and fault code 16 is stored in the logic module's memory.

If the 12 volt feed from the power module is open on the logic module side of the J11 splice (memory pin 2-2) the power loss/limited light is not turned on at all and there are no fault codes in memory.

Fault Code 41 (Improper Switching in Control Circuit)

The microprocessor drives the base of Transistor 1 "on" and "off" to regulate alternator output. When Transistor 1 turns "on", this also turns "on" the base of the Diagnostic Transistor. See Fig. 61. This causes the input to the CPI to go low (0 volts). In this way the microprocessor commands the base of Transistor 1 "on". If Transistors 1 & 2 are switching properly, the microprocessor receives a response "saying" that the command was carried out. If the response (CPI going high or low) doesn't match the command (the microprocessor driving the base of Transistor 1 high or low), Fault code 41 is stored in the logic module's memory. The power loss/limited light is not turned on. If this condition continues, the overcharging or undercharging will set another fault code.

Fault Code 46 (Voltage Too High Too Long)

If the charging system voltage is more than 1 volt above the desired charging rate (as determined by the battery temperature sensor) for more than 20 seconds, the power loss/limited light is turned on. Fault code 46 is stored in the logic module's memory. The power loss/limited light will stay on as long as the voltage is above the desired charging rate. If the voltage falls down into the desired control level, the power loss/limited light will turn off, however; fault code 46 remains in memory.

Fault Code 47 (Voltage Too Low Too Long)

Six minutes after engine start-up and a minimum coolant temperature of 160°F, and the engine RPM is above 1500 RPM and the voltage sense line indicates the charging system voltage 1 volt below desired level for more than 20 seconds, fault code 47 is stored in the logic module's memory. In this case the power loss/limited light is not turned on.

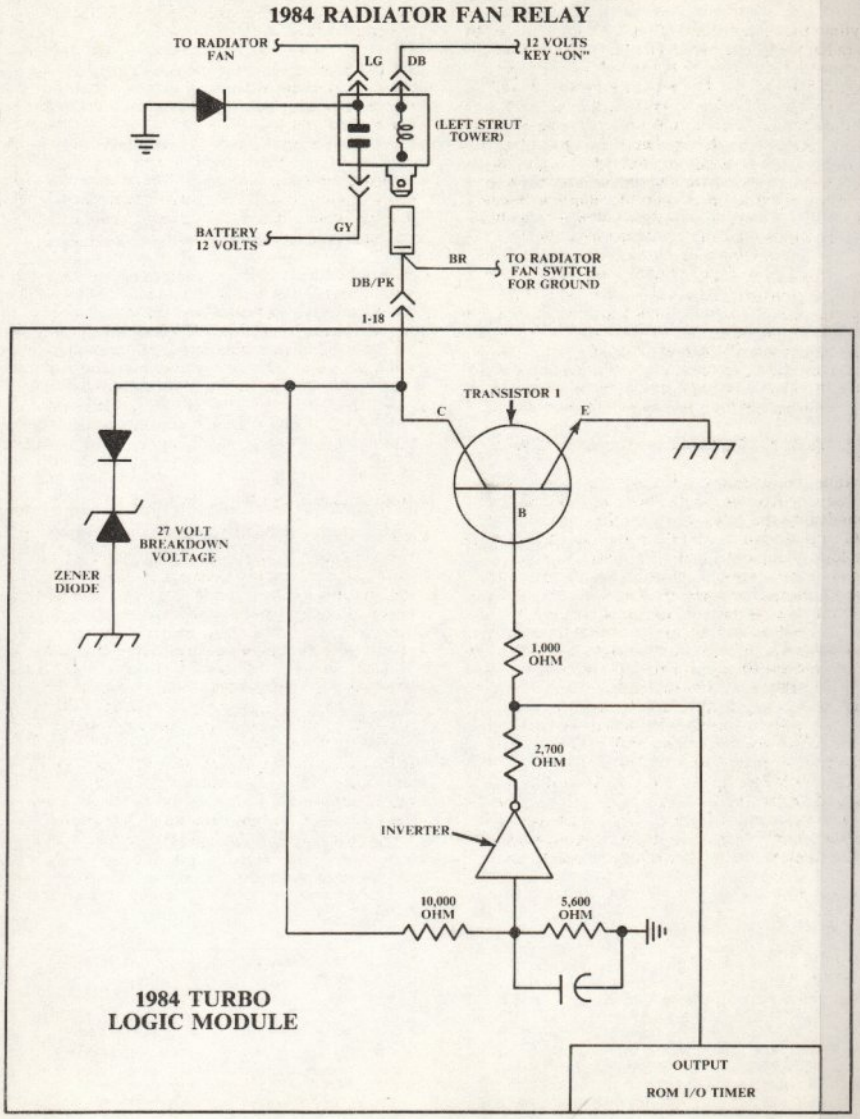

1984 RADIATOR FAN RELAY
TO RADIATOR FAN
LG
DB
12 VOLTS KEY "ON"
(LEFT STRUT TOWER)
BATTERY 12 VOLTS
GY
BR
TO RADIATOR FAN SWITCH FOR GROUND
DB/PK
I-18
TRANSISTOR 1
C
E
B
27 VOLT BREAKDOWN VOLTAGE
ZENER DIODE
1,000 OHM
2,700 OHM
INVERTER
10,000 OHM
5,600 OHM
1984 TURBO LOGIC MODULE
OUTPUT
ROM I/O TIMER

Fig. 62

1984 Radiator Fan Relay and Circuit

Anytime the ignition key is "on", 12 volts is sent to the radiator fan relay coil. The relay coil can be grounded by the radiator fan switch or the logic module. See Fig. 62. The radiator fan switch will ground when the radiator temperature is above 210°F and unground when the temperature is below 180°F. The logic module grounds the relay coil when the A/C clutch is engaged. This puts a high (5.6 volts) at the input to the microprocessor. See Fig. 43. When the input goes high, the microprocessor "tells" the I/O to put a positive voltage at the base of Transistor 1. See Fig. 62. This turns "on" Transistor 1 which allows the relay coil to go to ground. When the A/C clutch cycles out, the input to the microprocessor goes low, this "tells" the microprocessor that the clutch has cycled out and to turn the fan off. This is done by the I/O removing the voltage from the base of Transistor 1. It should be noted that in this circuit a condition could exist where by the A/C compressor is engaged and the fan is not on, due to a problem in the fan circuit.

1985, '86 & '87 Radiator Fan Relay and Circuit

Anytime the ignition key is "on", 12 volts is fed to the radiator fan relay coil. The relay coil can be grounded by the A/C switch on the dash (if the A/C cutout relay is closed), or by the logic module. The logic module will close the relay based on engine temperature (coolant sensor) and car speed (speed sensor). The logic module will turn the fan on at any speed when the temperature is above 240°F. The logic module will turn the fan off at any speed when the temperature is below 210°F. See Fig. 63. In between these temperatures the fan may be on or off depending on the coolant temperature vs. car speed. For example, if the coolant temperature is 215°F and the car speed is 55 MPH, the fan is off. If the car slows down below 45 MPH the fan turns on. If the coolant temperature is 230°F and the car speed is 30 MPH the fan is on. If the car speeds up above 35 MPH, the fan turns off. This system also has an anti-steaming feature. This feature prevents moisture from rising up from the radiator when the car is stopped. When the coolant temperature (as indicated by the coolant sensor) is below 100°F and the ambient air temperature (indicated by the battery temperature sensor in the power module) is below 60°F, this "tells" the logic module that conditions are right for radiator steaming to happen. If the engine is started under these temperature conditions, the logic module will start a 2 minute timer when the coolant temperature is 180°F. After 180°F, if the car is stopped (as determined by the speed sensor) while the 2 minute timer is running, the fan is turned on for a maximum of 2 minutes or till the car starts to move. If the car is driven the whole 2 minutes while the timer is running, ram air will evaporate the condensation on the radiator.

When the logic module "wants" to turn "on" the relay coil, the microprocessor "tells" the CPI to output a voltage which turns "on" the Solenoid Driver. See Fig. 64. When the Solenoid Driver is "on", it allows the relay coil to go to ground. When the microprocessor "wants" the relay coil "off", it "tells" the CPI to remove the voltage from the Solenoid Driver which turns it "off". It should be noted that by the A/C contacts in parallel with the fan contacts, the A/C will never be on when the fan is off.

1984 Radiator Fan Relay Fault Code

When the coolant temperature is below 150°F, the logic module "looks to see" if the fan is off with the A/C clutch engaged or the fan is on with the A/C off. If either of these conditions exist, fault code 35 is stored in the logic module's memory.

1985, '86 & '87 Radiator Fan Relay Fault Code

Anytime the logic module "wants" the fan on, the CPI sends out a voltage to turn the Solenoid Driver "on". The CPI momentarily (microseconds) interrupts the output and "looks" at the voltage that is fed back through the 27,000 ohm resistor. See Fig. 64. This is the input function of the CPI. If an "on" command is sent, it "looks" for an "on" response. If the response is not the same as the command, fault code 35 is stored in the logic module's memory.

1985, '86 & '87 RADIATOR FAN TURN ON & OFF TEMPERATURE vs. CAR SPEED CHART

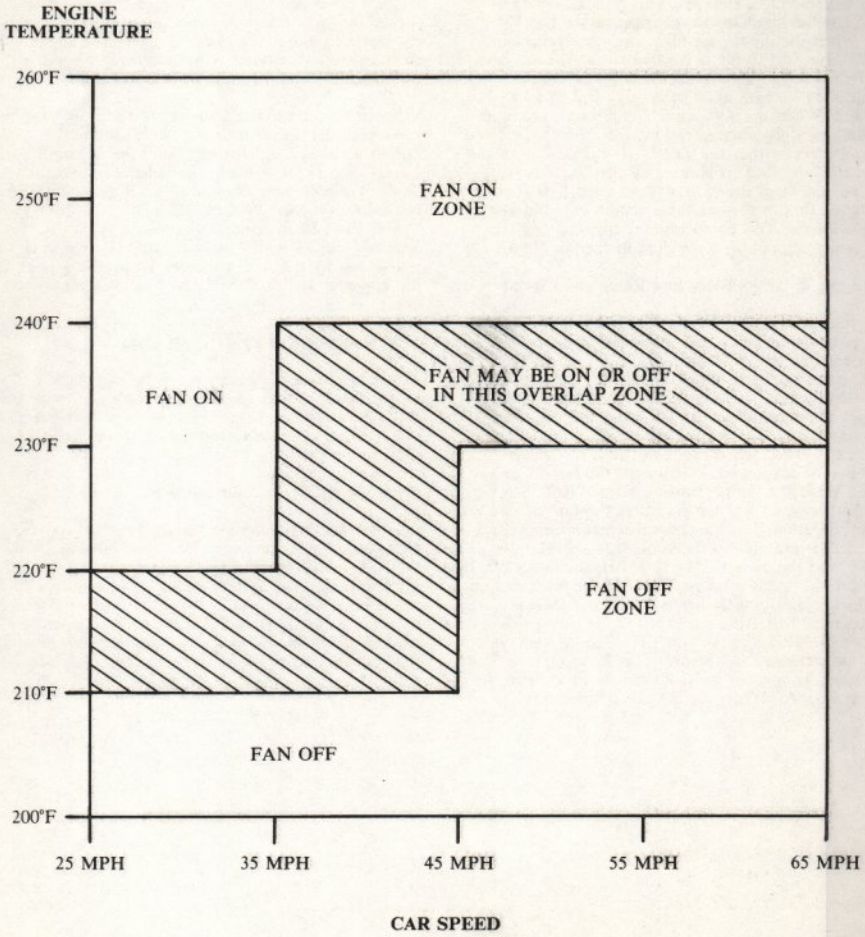


Fig. 63

1985, '86 & '87 RADIATOR FAN RELAY

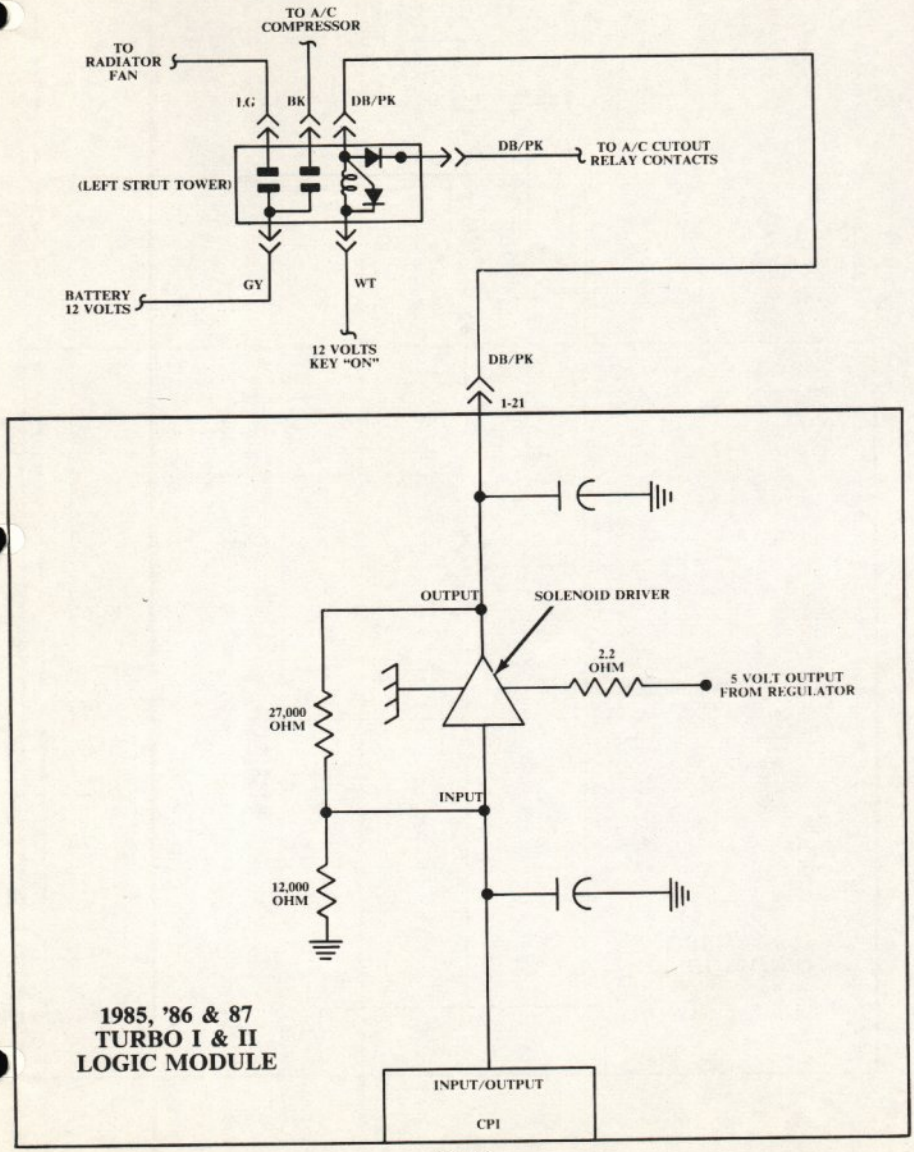


Fig. 64

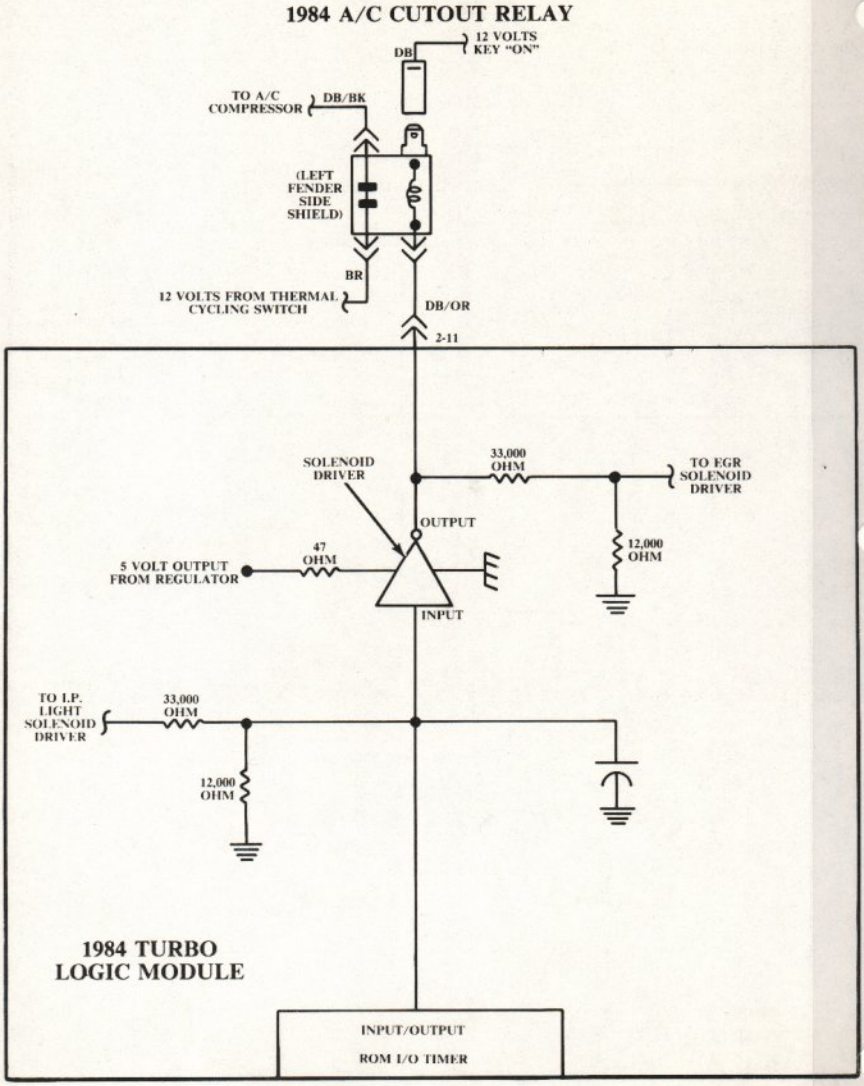

1984 A/C CUTOUT RELAY
12 VOLTS
KEY "ON"
DB
TO A/C
COMPRESSOR
DB/BK
(LEFT
FENDER
SIDE
SHIELD)
BR
12 VOLTS FROM THERMAL
CYCLING SWITCH
DB/OR
2-11
SOLENOID
DRIVER
33,000
OHM
TO EGR
SOLENOID
DRIVER
OUTPUT
47
OHM
5 VOLT OUTPUT
FROM REGULATOR
12,000
OHM
INPUT
TO I.P.
LIGHT
SOLENOID
DRIVER
33,000
OHM
12,000
OHM
1984 TURBO
LOGIC MODULE
INPUT/OUTPUT
ROM I/O TIMER

Fig. 65

A/C Cutout Relay and Circuit

Anytime the ignition key is "on", 12 volts is sent to the A/C cutout relay coil. For the relay coil to contacts to close, the logic module must provide the ground. See Fig. 65 or Fig. 66. When the relay contacts are open (no ground), the A/C clutch circuit is opened up. The logic module will remove the ground to the relay coil under these conditions:

1. After engine start-up for 10 seconds
2. Below 500 RPM
3. If TPS voltage indicates W.O.T.

At any other time when the engine is running, the logic module provides the ground for the relay coil. When the logic module "wants" the relay coil "on", the microprocessor "tells" the I/O or the CPI to output a voltage which turns "on" the Solenoid Driver. See Fig. 65 or Fig. 66. When the Solenoid Driver is "on", it allows the voltage to go to ground. When the logic module "wants" the relay coil "off", the microprocessor "tells" the I/O or the CPI to remove the voltage from the Solenoid Driver which turns it "off". This removes the ground from the relay coil and the contacts open.

A/C Cutout Relay Fault Code

Anytime the logic module is turning the relay coil on and off, the microprocessor is "looking" for the proper response. When the microprocessor "tells" the I/O or the CPI to command the Solenoid Driver to turn "on", the output signal is momentarily (microseconds) interrupted and the voltage is "looked" at to "see" if the response is the correct answer to the command. This is the input function. If the response is wrong, fault code 33 is stored in the logic module's memory. Also, any car that is not equipped with A/C will always have this fault code in memory. In this case it should be disreguarded.

1985, '86 & '87 A/C CUTOUT RELAY

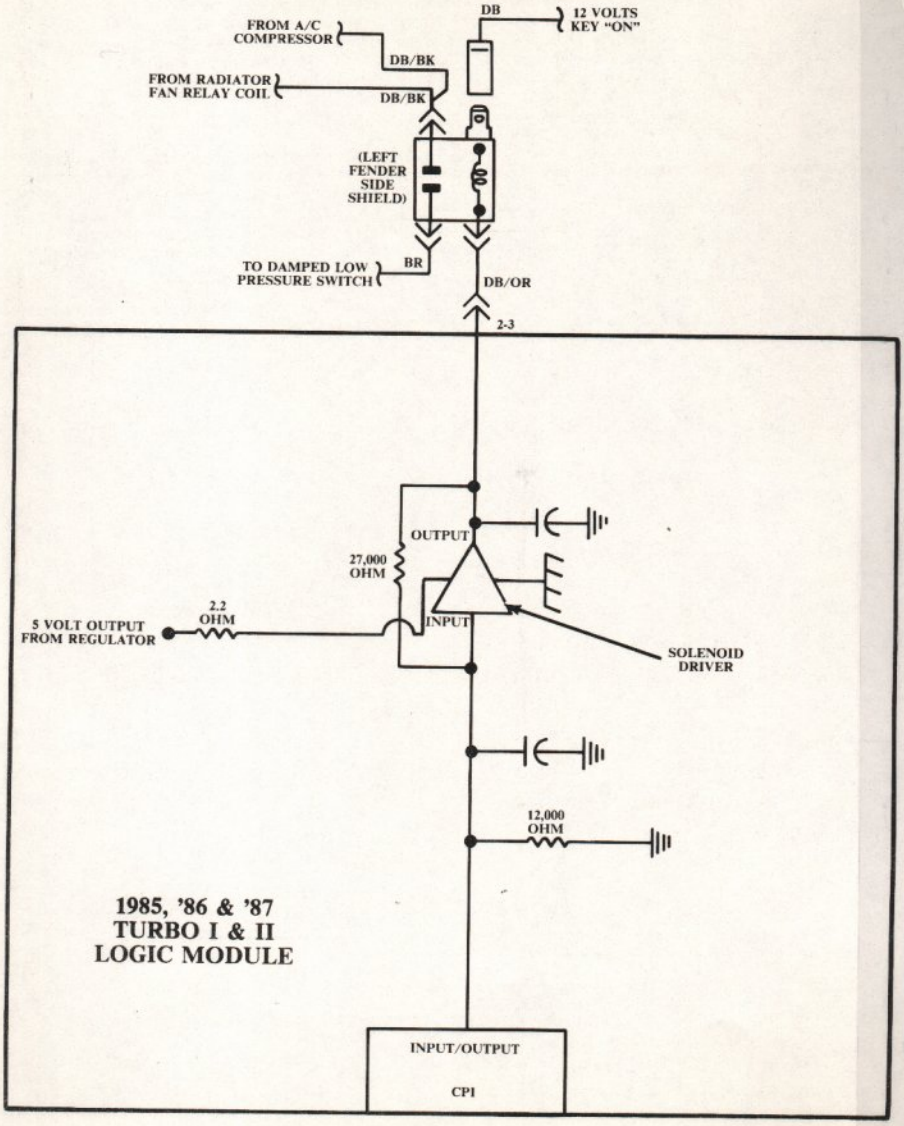


Fig. 66

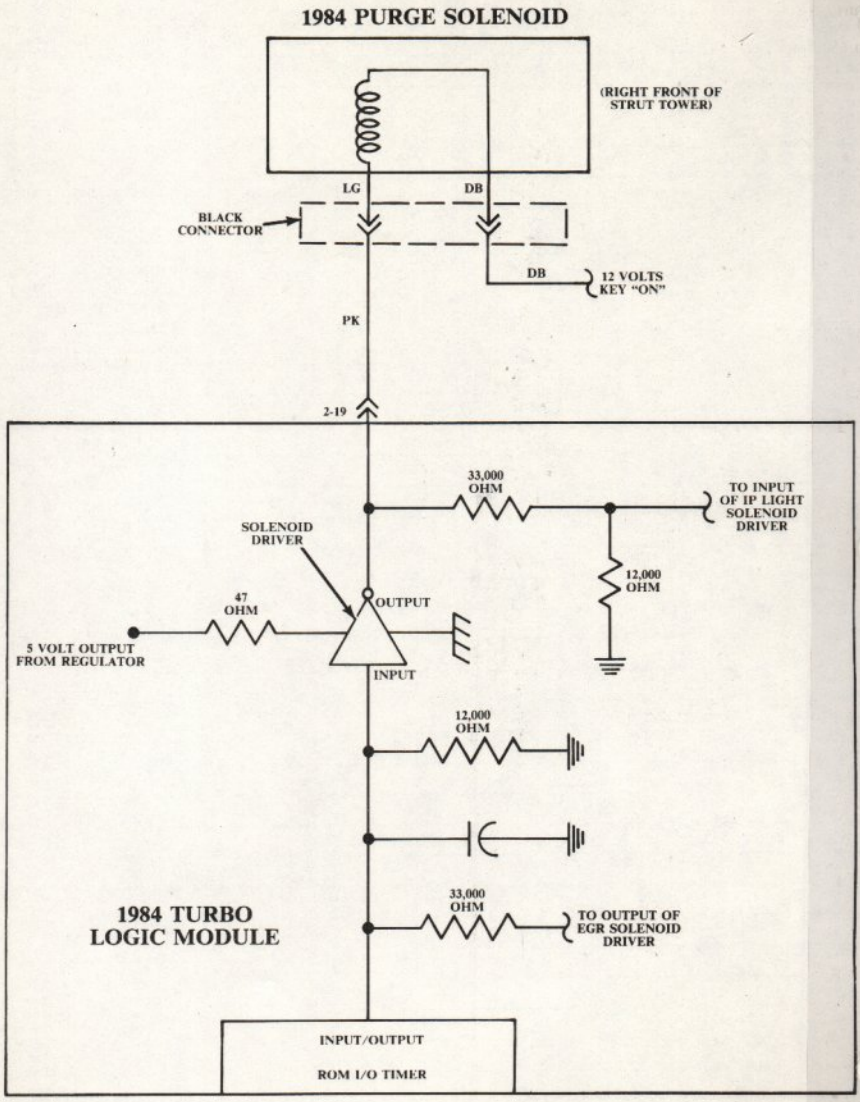

1984 PURGE SOLENOID
(RIGHT FRONT OF STRUT TOWER)
LG
DB
BLACK CONNECTOR
DB
12 VOLTS KEY "ON"
PK
2-19
33,000 OHM
TO INPUT OF IP LIGHT SOLENOID DRIVER
SOLENOID DRIVER
12,000 OHM
47 OHM
OUTPUT
5 VOLT OUTPUT FROM REGULATOR
INPUT
12,000 OHM
1984 TURBO LOGIC MODULE
33,000 OHM
TO OUTPUT OF EGR SOLENOID DRIVER
INPUT/OUTPUT
ROM I/O TIMER

Fig. 67

Purge/EGR/Wastegate/Baro Read Solenoids

In 1984 there are only two solenoids, the purge and the EGR. In 1985 & '86 there are four solenoids, the purge, EGR, wastegate and baro read. In 1987 there are three solenoids, the combination purge & EGR, the wastegate and the baro read. All the solenoids electrically operate the same way. All the solenoids are spring loaded (off position) so that the bottom port is connected to the side port. This connects manifold pressure to the component. This is true for all except the wastegate solenoid. In this case the bottom port is the vent which is connected to the wastegate actuator through the side port. When the solenoids are grounded, (all except wastegate) this vents manifold pressure. In the case of the wastegate solenoid, when it is grounded, manifold pressure is connected to the wastegate actuator.

All the solenoids are fed 12 volts key "on". When the microprocessor "wants" the solenoids turned on, it "tells" the I/O or the CPI to output a voltage to the input of the Solenoid Driver. See Fig. 67. When the Solenoid Driver receives a voltage at its input, it turns "on" and allows the current to go to ground through its output. When the microprocessor "wants" the solenoid off, it "tells" the I/O or the CPI to remove the voltage from the Solenoid Driver which shuts it "off" and removes the ground from the solenoid. The solenoid now returns to its spring loaded position.

Solenoid Fault Codes

Anytime the microprocessor is turning the solenoids on or off, it "looks to see" if the proper voltage is being fed back through the resistors. These Solenoid Drivers like the previous drivers, when they are turned "on" or "off", the microprocessor momentarily (microseconds) interrupts the output voltage and then "looks to see" if the command has been carried out. If the proper response (voltage) is not "seen", fault code 31 (purge) or 34 (EGR) or 36 (wastegate) or 37 (baro read) is stored in the logic module's memory. The power loss/limited light is turned on only if the wastegate fault code is in memory. Since boost can no longer be electronically controlled, the fuel shut off point (overboost limit) is lowered to 1.8 psig above the desired boost level. This is determined by the MAP voltage. It should be noted that if the electrical wastegate circuit is open, the manifold pressure will overcome the spring tension in the wastegate solenoid and open the wastegate actuator.

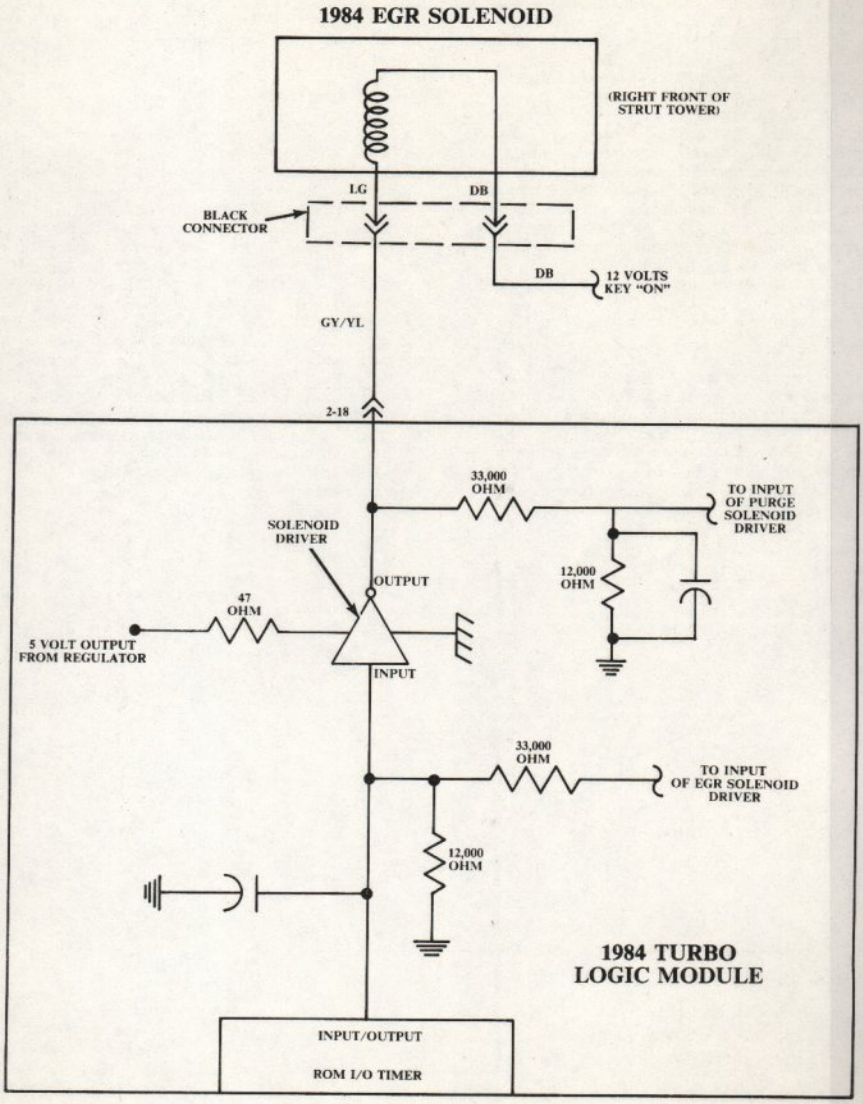

1984 EGR SOLENOID
(RIGHT FRONT OF STRUT TOWER)
LG
DB
BLACK CONNECTOR
DB
12 VOLTS KEY "ON"
GY/YL
2-18
33,000 OHM
TO INPUT OF PURGE SOLENOID DRIVER
SOLENOID DRIVER
OUTPUT
12,000 OHM
47 OHM
5 VOLT OUTPUT FROM REGULATOR
INPUT
33,000 OHM
TO INPUT OF EGR SOLENOID DRIVER
12,000 OHM
1984 TURBO LOGIC MODULE
INPUT/OUTPUT
ROM I/O TIMER

Fig. 68

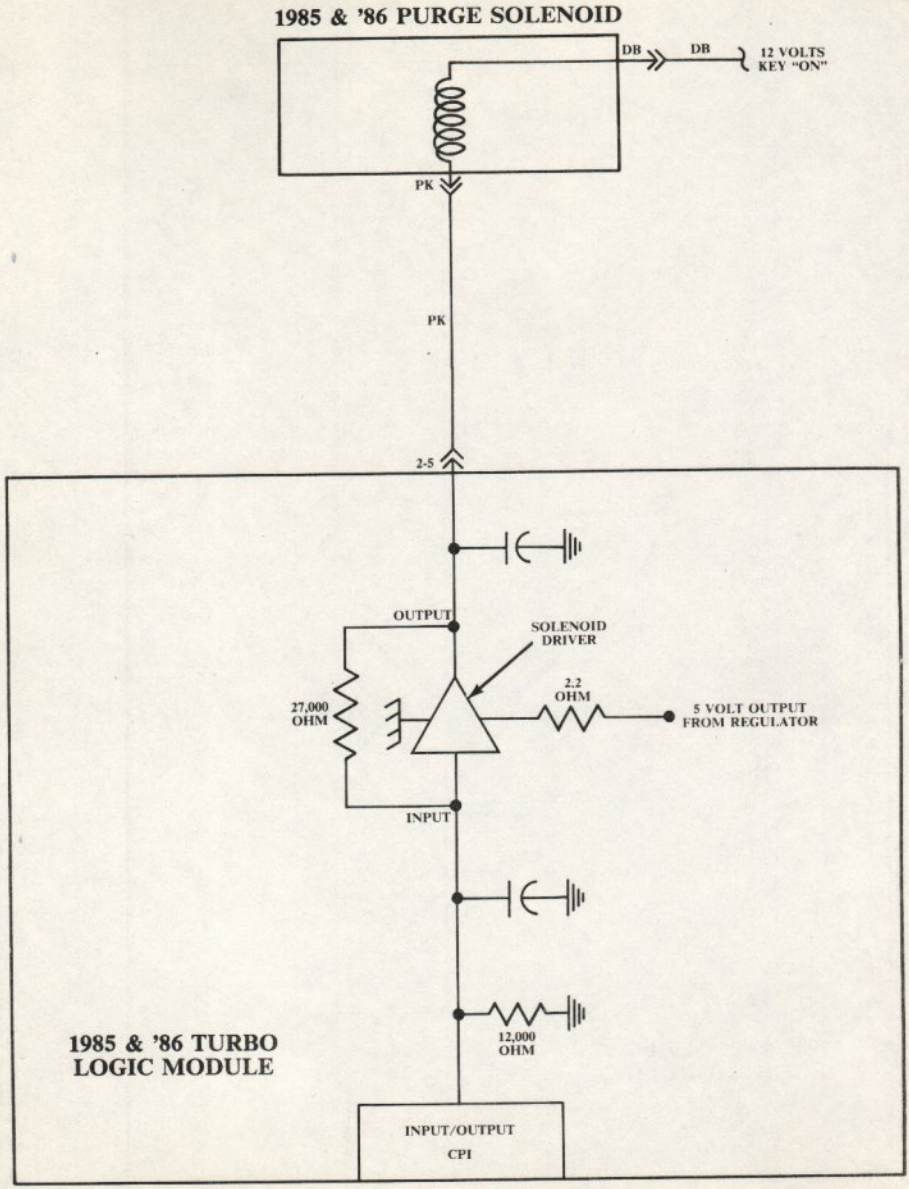

1985 & '86 PURGE SOLENOID
DB
DB
12 VOLTS
KEY "ON"
PK
PK
2-5
OUTPUT
SOLENOID
DRIVER
2.2
OHM
27,000
OHM
5 VOLT OUTPUT
FROM REGULATOR
INPUT
12,000
OHM
1985 & '86 TURBO
LOGIC MODULE
INPUT/OUTPUT
CPI

Fig. 69

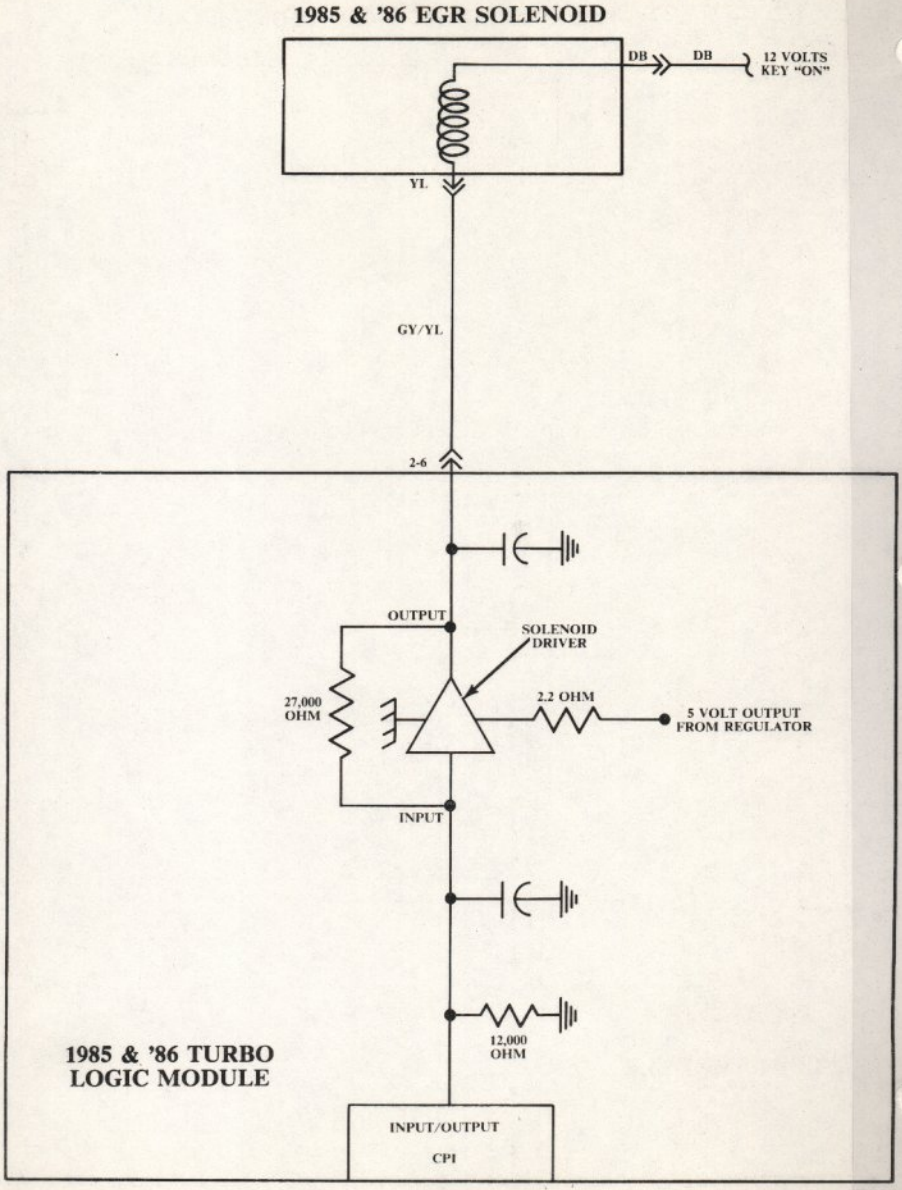

1985 & '86 EGR SOLENOID
DB
DB
12 VOLTS
KEY "ON"
YL
GY/YL
2-6
OUTPUT
SOLENOID
DRIVER
27,000
OHM
2.2 OHM
5 VOLT OUTPUT
FROM REGULATOR
INPUT
12,000
OHM
1985 & '86 TURBO
LOGIC MODULE
INPUT/OUTPUT
CPI

Fig. 70

1987 COMBINATION PURGE & EGR SOLENOID

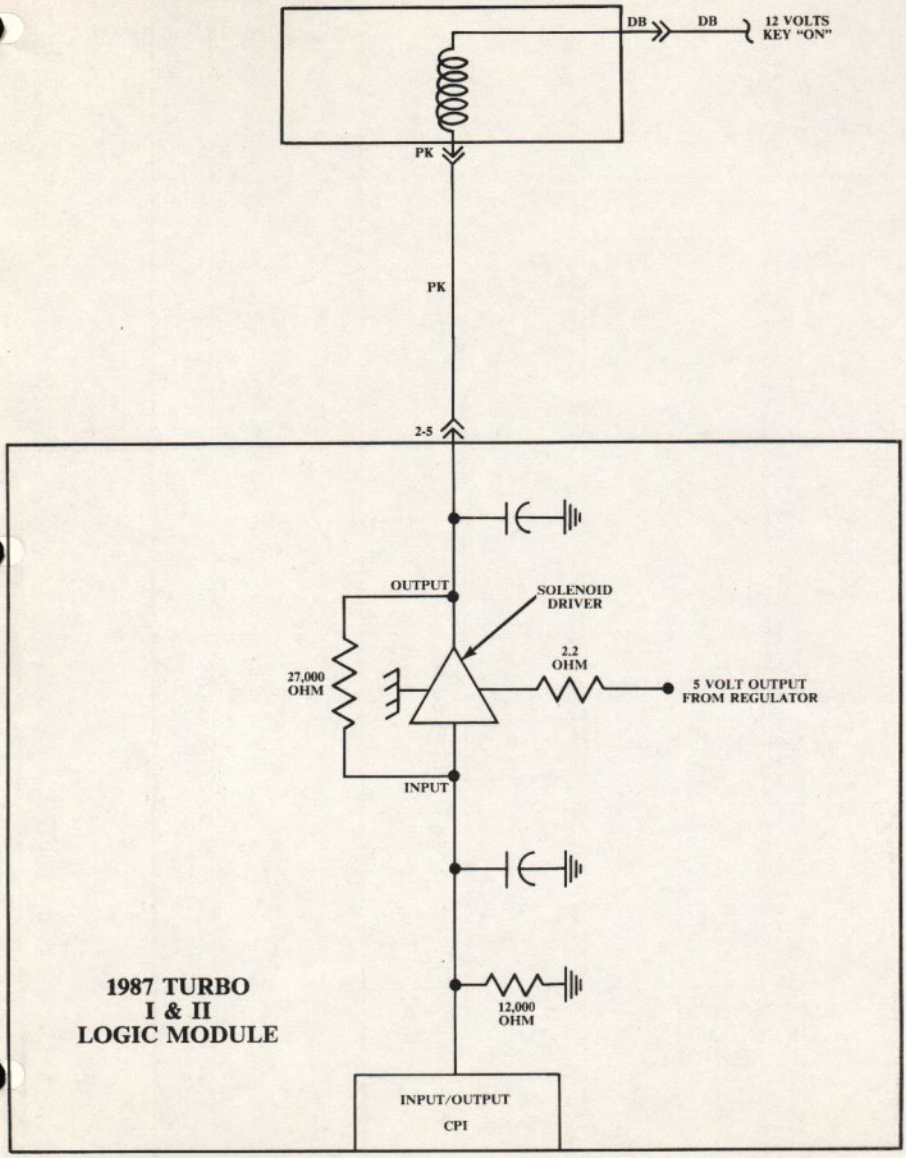


Fig. 71

1985, '86 & '87 BARO READ SOLENOID

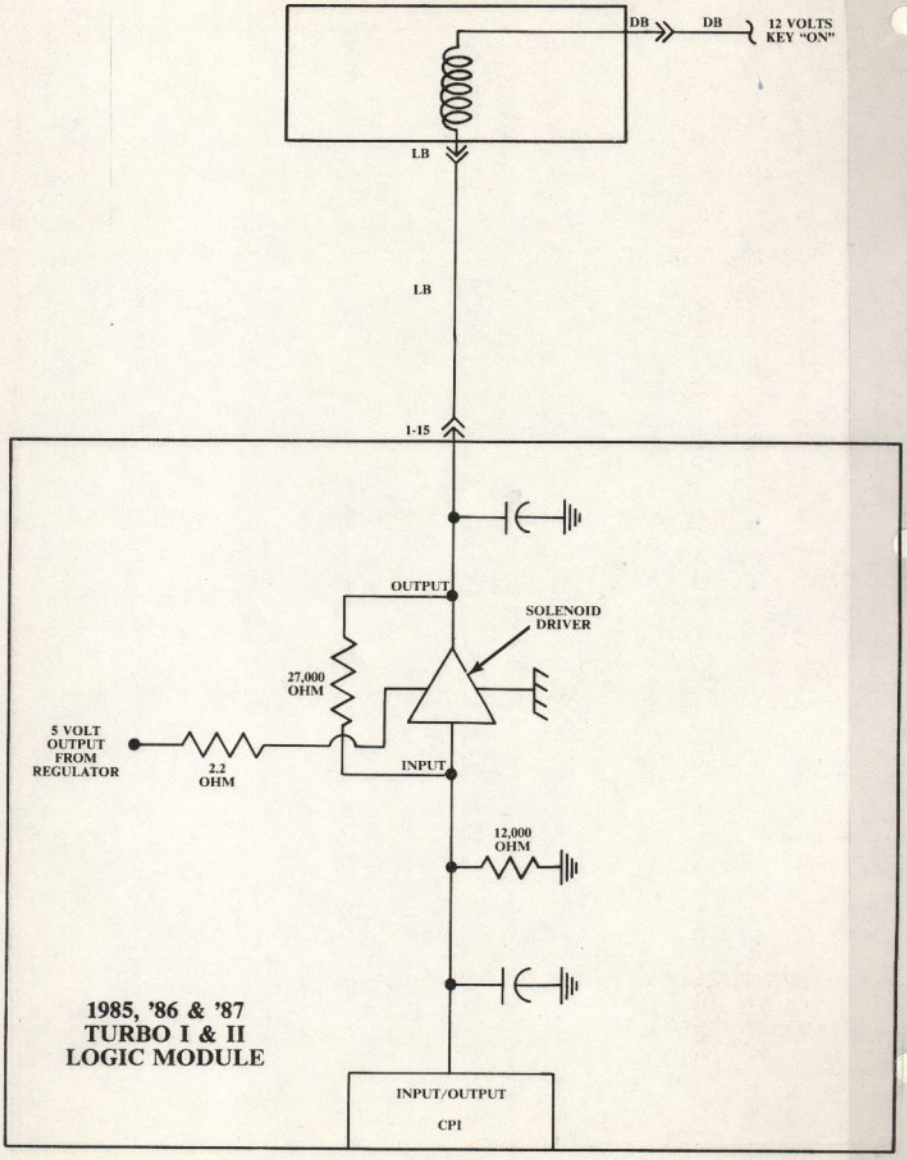


Fig. 72

1985, '86 & '87 WASTEGATE SOLENOID

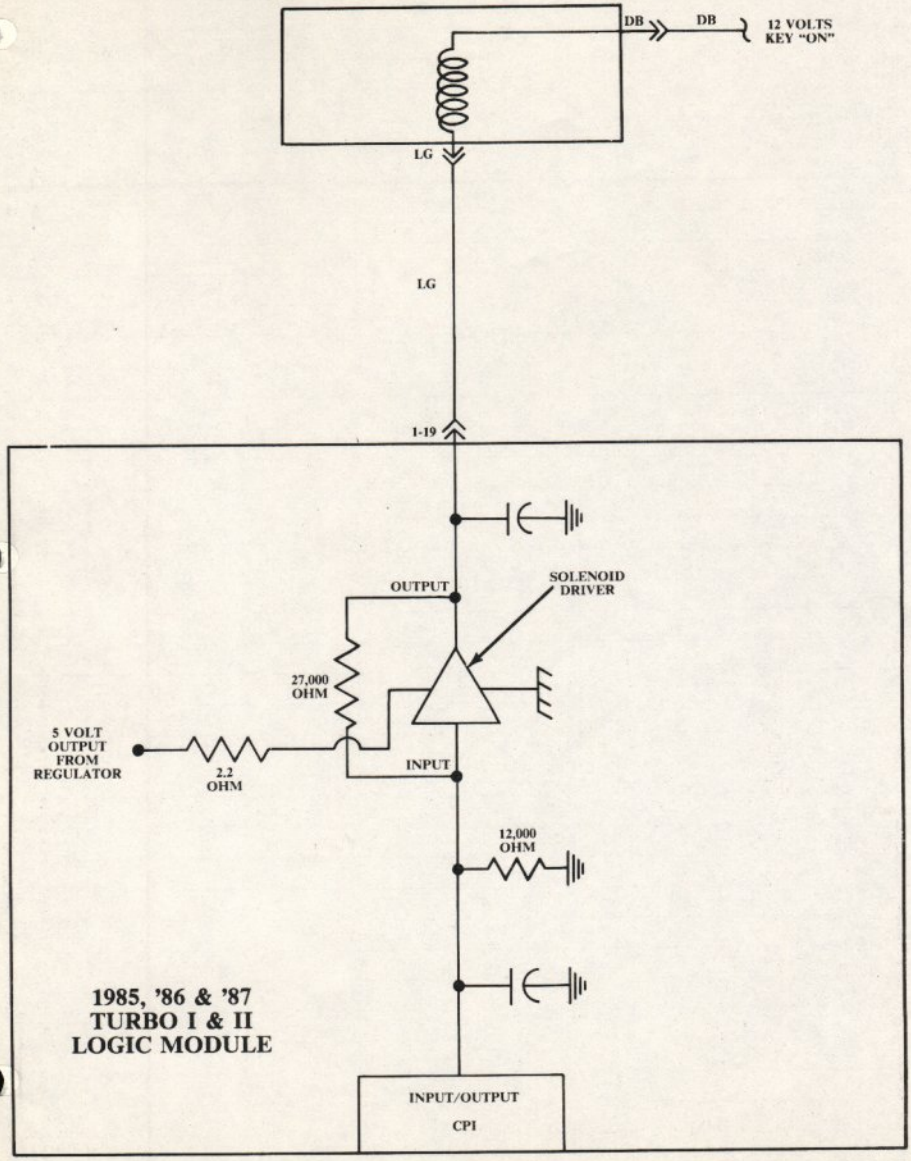


Fig. 73

Logic Module

The main components of the logic module are:

1. MPU - Microprocessor Unit (1984 - '87)
2. A/D Converter - Analog to Digital Converter (1984 - '87)
3. PROM - Programmable Read Only Memory (1984 - '87)
4. RIOT - Read Only Memory + Input/Output Function + Timer (1984 only)
5. CPI - Cemos Peripheral Interface (1985 - '87)
6. PTM - Programmable Timer Module (1985 - '87)
7. Decoder (1984 - '87)
8. Latch (1984 - '87)

MPU - Microprocessor Unit

The MPU is the basic core of the logic module. The 1984 unit was basically the same as in 1985-'87, but was simply less capable. The 1984 system used the power module to do some of the work that the logic module now does. For instance, the sync signal on the turbo system went to the power module, and the power module would turn the injectors on. With the later system, the signal goes to the logic module, and it turns the injectors on, through the power module.

The MPU is the main chip. It executes programs. The programs it executes are, for the most part, received from the peripherical chips. The peripheral chips are A/D converter, PROM, CPI, RIOT, PTM, Decoder, and Latch.

The MPU receives input and elicits outputs through the Binary Code. The Binary Code is an eight digit number made up entirely of "0's" and "1's". The order in which the "0's" and "1's" appear determines the meaning of the code to the MPU. Each chip has a certain number of "addresses" (information circuits). When the MPU receives a signal, its programming tells it to go to the PROM to get instructions on what to do about the signal. This is called "addressing" the PROM. The PROM will then give the instructions to the MPU in the binary code. The MPU will then carry them out.

The MPU also calls up addresses on the various chips to receive information, or to give commands. Each address on each chip contains eight "bits" of binary code.

A/D Converter - Analog to Digital Converter

The A/D converter converts voltage signals to digital (binary) code that can be recognized by the MPU. The A/D converter uses a bank of resistors (called an R2R ladder) to break all inputs down to 5 volts. This 5 volts is then "split". The binary code sent on to the MPU is a result of this "split". If the signal is then above 2.5 volts, it becomes a "1" in the binary code. If the signal is less than 2.5 volts, it becomes a "0". Eight "splits" results in one binary code sent to the MPU. The MPU, through its programming, then "decides" which chip (peripheral) to address to get instructions.

It should be noted that the MPU must address the PROM in order to get instructions to go to the A/D converter in the first place. It can easily be seen that the MPU is dependant upon the PROM to operate at all. Keep in mind, the MPU must get instructions to do anything.

PROM - Programmable Read Only Memory

In the 1984 system an F PROM was used. The F stands for fusible link. The 1984 system simply blew fusible links inside the module when there was any kind of current problem. This was very inefficient and, in 1985 the system was changed to use an E PROM. The E stands for erasable. This allowed faulty memory to be erased, and allowed for a great deal more programming.

The PROM contains the instructions needed by the MPU to do its job. These instructions detail various strategies to be used on different vehicle applications (i.e. spark curve on a manual trans. vs. auto trans.). The PROM is what tells the MPU what spark curve to use.

RIOT - Read Only Memory + Input/Output Function + Timer

This chip had several functions. It contains some memory. It is used by the MPU to switch from reading inputs to sending outputs, and has a timer function to control duty cycles. This chip was only used in 1984.

CPI - Cemos Peripheral Interface

Cemos Peripheral Interface is a type of semiconductor technology. The CPI has been used since 1985 and replaced the RIOT. The CPI has a larger capacity to do all of the jobs the RIOT did in 1984. It allows more switching than before, and has more memory and timers.

There are a number of timers that can be used for various purposes. For instance, one timer can be used to control the fuel pulse width, and another can be used to control spark timing. However, it should be noted that the MPU programs the timer for on time. The timer then turns itself off. Therefore, the MPU must get instructions from the PROM concerning how long the timer should be on, then must address the CPI to turn the timer on. The

timer can be programmed to tell the MPU when it is going off so the MPU can check with the PROM to find out if the timer should be reprogrammed to stay on longer.

The Read Only Memory function of the CPI is simply more instructions that are available to the MPU if it needs them. The ROM is like an extension of the PROM.

PTM - Programmable Timer Module

This chip is used on Turbo applications only. It contains three timers used by the MPU to control the Wastegate Solenoid. Only one of the three timers is used. The chip is necessary because all of the timer space was used up in the CPI.

Decoder

The MPU uses the Decoder to select the address it wants on a given chip. The Decoder turns the rest of the peripheral chips off to the MPU and turns the selected one on. The Decoder is needed because the MPU can only read one address at a time. The MPU operates the Decoder through a binary code.

Latch

The Latch chip is the high tech chip in the logic module. As was stated, the MPU cannot call-up more than one address at a time. Therefore, it can't call-up an address on the PROM and read the instructions at the same time. The Latch allows the MPU to do this.

For example, the MPU has 16 outputs for addressing. These are called "address lines". The PROM had 65,000 + addresses. Each address has 8 "data" lines in it. Each data line produces a "0" or a "1". When all 8 data lines are tied to the MPU it reads a binary code. The Latch chip allows the MPU to look up an address line and read the data lines in it at the same time. This is done through "multiplexing" the address line. The Latch holds the address line open to the MPU while the MPU reads the data (code) at that address.

Logic Module Operation

The following is generalized example of how the MPU interacts with the various peripheral chips to receive and output information. This routine varies with different inputs, but is basically the same for most actions inside the logic module.

The MPU has programming inside of it that causes it to call-up certain addresses on a cyclical basis. This means that the MPU automatically looks at some addresses.

During the routine the MPU will call-up an address in the PROM. This might be an instructional address. The MPU must go through the Decoder to call-up **any** address. The Decoder the blocks all other inputs to the MPU and allows it to listen to only the address being asked for at that time.

The MPU cannot address a chip and look at the information at that address at the same time, so the information at the address will be "loaded" into the Latch chip. The Latch chip will hold the address (through multiplexing) while the MPU reads the information.

If, for example, the information at the requested PROM address is an instruction to the MPU to look at the voltage signal being generated by the MAP sensor, the MPU will go through the Decoder to address the A/D converter. The A/D converter will convert the MAP voltage signal to a binary signal. The information coming from the address on the A/D converter will then be loaded into the Latch chip and held for the MPU to look at.

The information from the MAP sensor would cause the MPU to address the PROM for instructions. This would, again, be done through the Decoder. The information from the PROM would be stored on the Latch chip and read by the MPU.

If the instructions from the PROM tell the MPU to change the fuel pulse width, the MPU would then go through the Decoder to address the CPI Input/Output function, and output a command to one of the Timers in the CPI for a given length of time. The MPU might also program the Timer to let it know when it is about to time out. The Timer, left unchecked, would then turn off as programmed.

It should be kept in mind that this example is of one sequence of events. All of the action described above takes place in a matter of microseconds, and the MPU can complete hundreds of these actions in a matter of milliseconds. Thus it appears to be doing everything, at the same time.

Turbocharger

Before entering into a discussion concerning the turbocharged engine system, we should pause and review the basic characteristics of both the internal combustion engine and the turbocharger.

The internal combustion engine is classified as an air breathing machine. This means that the amount of power that can be obtained from a given displacement engine is determined by the amount of air that it breathes in a certain period of time and not by the amount of fuel that is used. This is because the fuel that is burned requires air with which it can mix to complete the combustion cycle. Once the air/fuel ratio reaches a certain point, the addition of more fuel will not produce more power, only black smoke. The more dense the smoke, the more the engine is being overfueled. Therefore, increasing the fuel delivery beyond the air/fuel ratio limit results only in excessive fuel consumption.

Turbochargers are installed on an engine to put more and denser air into the engine combustion chambers. Because of this increased volume and weight of compressed air, more fuel can be scheduled to produce more horsepower from a given size engine. The turbocharged version of an engine will also maintain a higher level of power output than the non-turbocharged version when operated at altitudes above sea level.

A turbocharger is an air pump designed to operate on the normally wasted energy in engine exhaust gas. These gases drive the turbine (hot) wheel and shaft which is coupled to a compressor (cold) wheel which when rotating provides a high volume of air to the engine combustion chambers.

The turbocharger, although precision built, is basically a very simple but durable machine. It does, however, require maintenance and care as does any other piece of working machinery, but a positive flow of clean lubricating oil is critical.

The heat energy and pressure in the engine exhaust gas are what drives the turbine wheel. The speed of the rotating assembly and output of the compressor wheel are controlled by the design and sizing of the turbine wheel and turbine housing. The housing acts as a nozzle to direct the exhaust gas flow to the turbine wheel blades which drive the shaft wheel assembly. Since the compressor wheel is directly coupled to the same shaft, it rotates at the same speed as the turbine wheel. Clean air from the air filter is drawn into the compressor housing and wheel where it is compressed and delivered through a crossover pipe to the engine air intake manifold. Each engine size must be properly matched, because the amount of air pressure rise and volume delivered to the engine from the compressor outlet is determined by wheel size, housing size, and performance matching of the turbocharger to a given engine.

There are quite a number of benefits to be gained by turbocharging. Combustion of the fuel more complete, cleaner, and takes place within the engine cylinders where its work is accomplished, because the turbocharger delivers an abundance of air to the engine. The positive air pressure (above atmospheric pressure) that is maintained in the engine intake manifold benefits the engine in several ways. During engine valve overlap (before the intake stroke starts), clean air is pushed across the combustion chamber scavenging all remaining burned gases, cooling the cylinder heads, pistons, valves and the exhaust gas. The cleaner burning of the fuel plus the engine cooling which results helps to extend engine life.

It is possible to safely increase the power output of some engines by as much as 40 to 50%, with little or no change in engine components, with the correct selection and/or matching of a turbocharger.

Causes of Turbocharger Failures and Maintenance Practices

There are many and varied causes for turbocharger failures. They can be grouped into five major categories:

1. Lack of lubrication and/or oil lag.
2. Foreign material in the lubricating system.
3. Oil oxidation or breakdown.
4. Foreign material in either the exhaust or air induction systems.
5. Material and workmanship.

Good maintenance practices should be observed, particularly regarding air and oil filtration, to maintain the service life and performance of a turbocharger. A larger percentage of turbocharger failures are caused by oil lag, restriction or lack of oil flow and dirt in the oil. The second largest percentage is caused by foreign objects entering the compressor and/or turbine wheels.

Dust or sand entering the turbocharger compressor housing from a leaky air inlet system can seriously erode the compressor wheel blades and will result in the deterioration of turbocharger and engine performance. The wearing away of the blades, if uneven, can induce a shaft motion which will pound out and eventually fail the turbocharger shaft bearings. Ingestion of sand or dust will also cause excessive wear on engine parts, such as pistons, rings etc. Entrance of large or heavy objects, bolts, nuts,

rocks, tools, etc., will completely destroy the turbocharger and in may instances cause severe damage to the engine.

Plugged or restricted air cleaner systems, resulting from poor maintenance procedures, will reduce air pressure and volume at the compressor air inlet and cause the turbocharger to lose performance. The restricted air cleaner and the resultant air pressure drop between cleaner and turbocharger can, during engine idle periods, cause oil pullover at the compressor end of the turbocharger. This would be a compressor end oil seal leak without a failure of seal parts. Proper servicing of the air cleaner system can prevent and correct the above problems.

The turbocharger operates at shaft speeds in excess of 60,000 rpm and utilize full floating bearings which have correspondingly high rotational speeds. Adequate clean oil supply is required for cooling and lubrication to maintain the turbocharger bearing system.

Dirt or foreign material, when introduced into the turbocharger bearing system by the lube oil, creates wear primarily on the center housing bearing bore surfaces. Contaminants imbed in the bearing surfaces and act as an abrasive cutting tool and eventually wear through. When bearing and bore wear becomes excessive, the shaft hub and either or both wheels will start to rub on the housings, causing the rotating assembly to turn slower. Turbocharger and engine performance will rapidly deteriorate from this point, and such indications as engine power loss, excessive smoke, excessive noise and the appearance of oil at either or both ends of the turbocharger could be noted. Contaminated and dirty oil is prevented when the oil is changed at the proper intervals.

A turbocharger should never be operated under engine load conditions with less than 30 psi oil pressure. A turbocharger is much more sensitive to a limited oil supply than an engine, due to the high rotational speed of the shaft and the relatively small area of the bearing surfaces.

Oil pressure and flow lag during engine starting can have detrimental effects on the turbocharger bearings. During normal engine starting, this should not be a problem. There are, of course, abnormal starting conditions. Oil lag conditions will most often occur during the first engine start after engine oil and filter change when the lubricating oil system is empty. Similar conditions can also exist if an engine has not been operated for a prolonged period of time because engine lube systems have tendency to bleed down. Before allowing the engine to start, the engine should be cranked over until a steady oil pressure reading is observed, priming the lubricating system. The same starting procedure should be followed when starting an engine in cold weather as the engine oil can be congealed and take a longer period of time to flow. Turbocharger bearing damage can occur if the oil delay is in excess of 30 seconds, and much sooner if the engine is allowed to accelerate much beyond idle rpm.

The Intercooler

On the 1987 Turbo II, an intercooler is used. An intercooler is a air to air heat exchanger that is mounted next to the radiator. It can be thought of as a "radiator" that cools the intake air. The reason the intercooler is used is, because as the turbocharger compresses the air, its temperature rises. Now the hot (250° F) air is fed into the engine (Turbo I). Because the air is forced into the engine, more air can get in than a non turbo, however; since its temperature is so high, the air isn't very dense. The intercooler takes the hot compressed air and cools it down (140° F) by giving up the heat in the air to the air passing over the fins of the intercooler. Now the cooler compressed air is fed into the engine. By the intake air being cooler, it is denser and even more fuel can be added. This results in a 20 - 30 HP increase just by adding the intercooler and making a few other changes. Also, the cooler air reduces detonation, so a higher constant boost level can be tolerated by the engine.

EGR SYSTEM

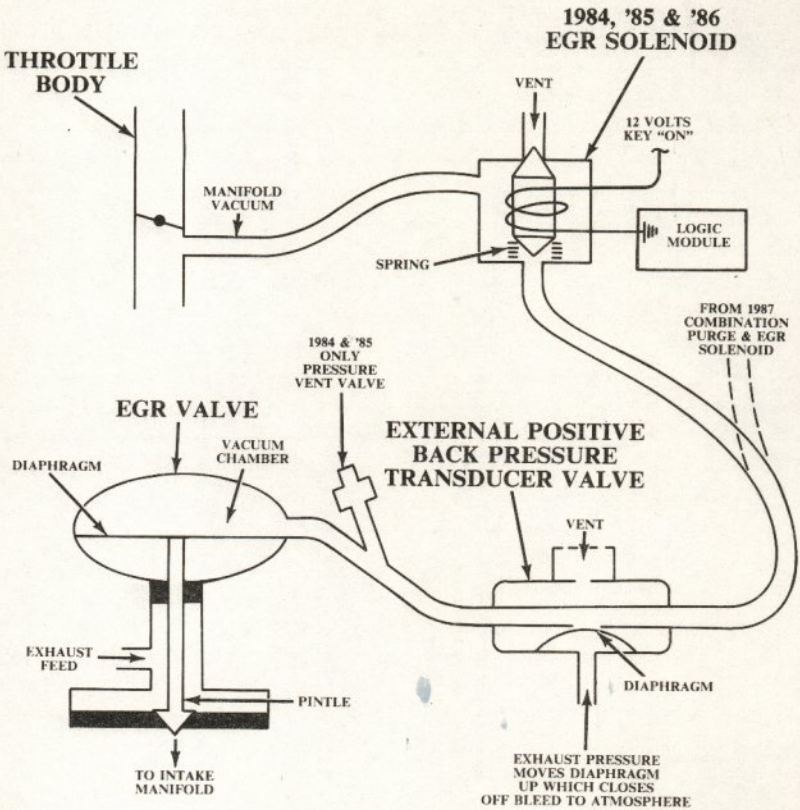


Fig. 74

EGR System

The EGR system is the same for 1984 - '87 with only one exception. Only 1984 & '85 have a pressure vent valve in parallel with the vacuum line to the EGR valve. Anytime the logic module ungrounds the EGR solenoid manifold vacuum is sent to the external positive backpressure valve. See Fig. 74. The backpressure valve has a diaphragm that moves with exhaust pressure. As it moves up it starts to block off a vacuum bleed. When the bleed is totally blocked, the two ports on the backpressure valve are connected to each other. When the diaphragm is partially up, some of the vacuum is bled off to atmosphere. See Fig. 74. So the manifold vacuum that is fed to the backpressure valve will be partially bled off under a low engine load condition. This happens because under a low load, the exhaust pressure is lower. As the load increases, the exhaust pressure increases which blocks off more of the bleed and sends more vacuum to the EGR valve. So the backpressure valve regulates the EGR according to engine load. As the EGR valve has vacuum applied to its diaphragm, the pintle moves up and allows exhaust gases into the intake manifold. These gases cool down the combustion chamber temperatures which helps reduce detonation and NO_X emissions.

1984 &'85 PURGE SYSTEM

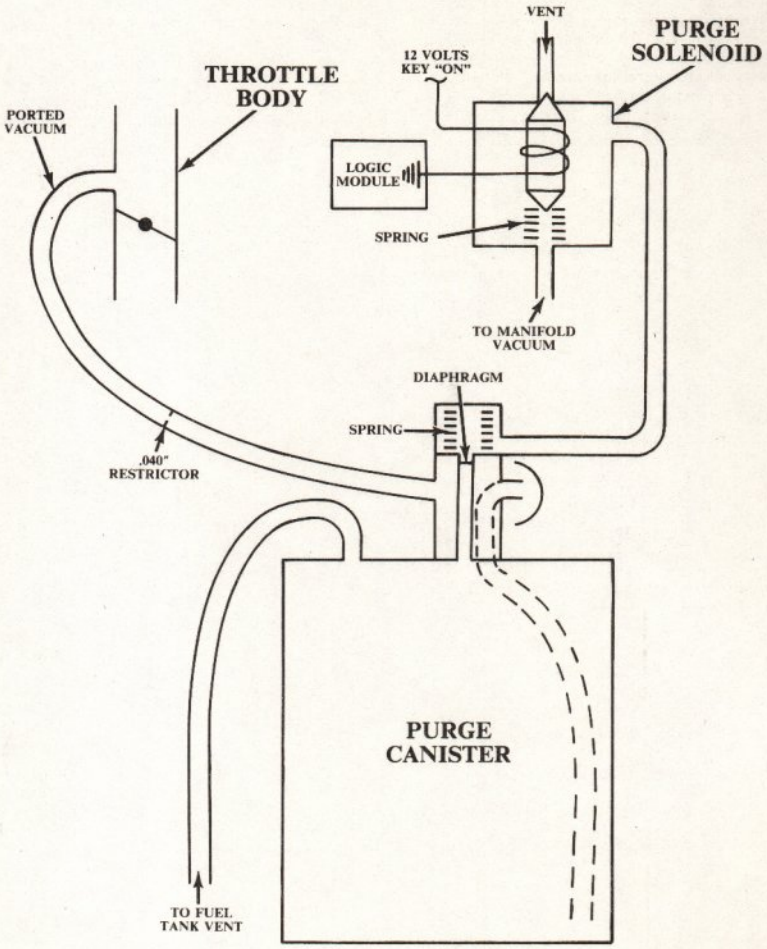


Fig. 75

Purge System

There are two types of purge systems used depending on the model year of the car. In 1984 & '85 the "normal" type of purge system is used. In this system, anytime the logic module has the purge solenoid ungrounded, when the throttle blade is opened, the canister is purged of fuel vapors with ported vacuum. See Fig. 75. In 1986 & '87, the Bilevel purge system is used. In this system, anytime the logic module has the purge solenoid grounded, the canister is purged of fuel vapors. At idle or closed throttle manifold vacuum through a .020" red restrictor purges the canister. See Fig. 76. When the throttle is opened, ported vacuum through a check valve purges the canister. The check valve is there to block air, from the ported vacuum fitting, that would go into the manifold vacuum fitting at closed throttle, if it weren't there.

1986 &'87 BILEVEL PURGE SYSTEM

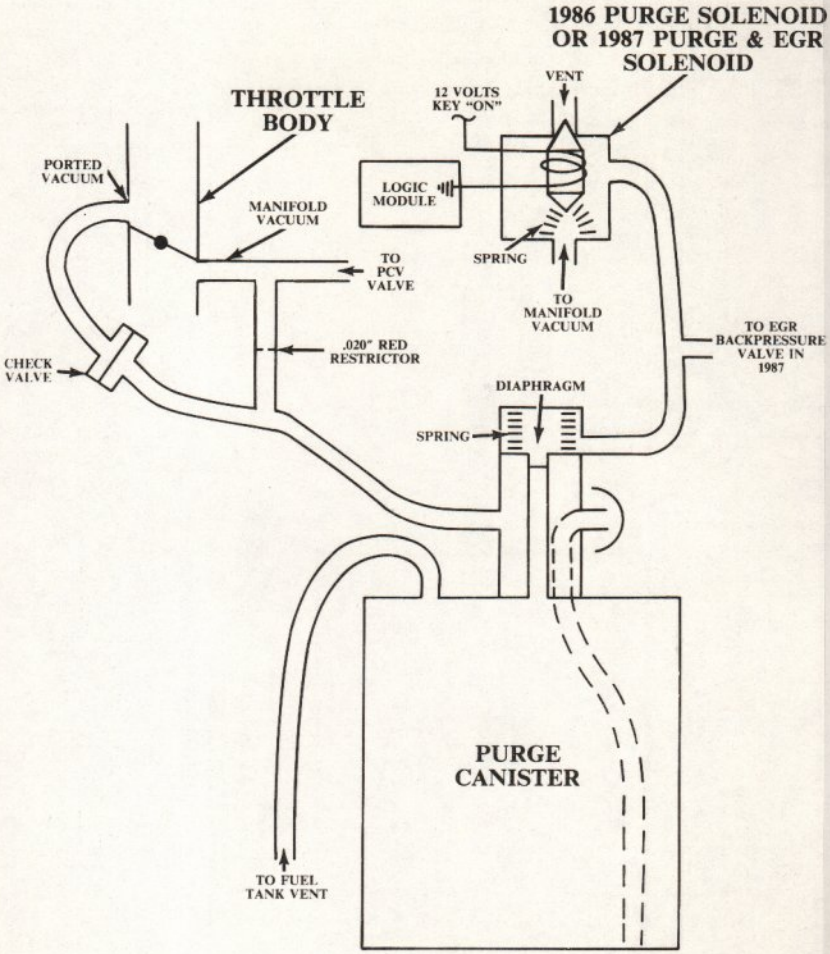


Fig. 76

Glossary

A

A/D Converter - Analog to digital converter. This IC chip takes any analog voltage that is fed into the logic module and converts it into a digital signal (binary number). The logic module cannot "understand" any analog voltage. The A/D converter can be thought of as the logic module's "voltmeter" allowing it to "see" the voltage being inputted to it.

C

Comparator - An IC chip with a digital output that indicates which of the two analog inputs are greater. In the case of the oxygen sensor circuit, it is hooked up as a unity gain amplifier. That is, its output is a digital average of its input.

CPI - Cemos Periphial Interface. This is an IC chip that is used to turn on a device (output function). Then at the command of the microprocessor it switches from driving the device to "reading" the device to "see" if its working OK (input function).

D

Darlington Amplifier - Two transistors hooked together in such a way that the emitter or collector of the control transistor feeds the base of the working transistor. This type of network offers high input resistance.

Differential Amplifier - An IC chip whose output is proportional to the difference between its inputs. This difference is then multiplied by a certain amount depending on the chip design.

F

Flip Flop - An IC chip is used as an, either or switch. When it receives voltage at its inputs, it outputs a voltage "blip" at one of its two outputs. This voltage "blip" then turns on the other type of Flip Flop. This type just needs the voltage "blip" to turn "on" and stay "on". Once "on" this type will output a constant voltage.

H

Hall Element - A thin sheet of conductive material that is passing current flow. When a magnetic field is exposed to the hall element perpendicular to current flow, a voltage is generated at the edges of the Hall element at right angles to the current flow. This voltage is proportional to the strength of the magnetic field.

H Drivers - (or called push pull) An IC chip that can output 12 volts or a ground when it receives a signal at its input. Its internal circuitry "decides" which output to provide. This used in the AIS motor circuit.

I

IC Chip - An integrated circuit that contains transistors, resistors and capacitors along with interconnecting conductors. This is all processed and contained on a single chip of silicon.

Injector Driver - An IC chip that acts as a transistor. What makes this different from a transistor is that is has current limiting capabilities once its turned "on".

Inverter - An IC chip whose output is always opposite from its input.

I/O - Input/Output. Part of an IC chip used in 1984 only, that could be used to turn on a device (output function). Then at the command of the microprocessor, it will switch from driving a device to "reading" a device (input function).

M

Microprocessor - An IC chip that is the "brains" of the logic module. It eventually receives all inputs and through other chips it commands the outputs.

N

NTC Thermistor - Negative Temperature Coefficient Thermistor. This is a resistor that will radically change its resistance with temperature. As the temperature increases, its resistance decreases. As the temperature decreases, its resistance increases.

P

Piezoelectric Crystal - A crystal found in nature that will output a voltage when squeezed or flexed.

Piezoresistive Chip - A silicon ship that has four resistors on it that are hooked up in a wheatstone bridge. When there is a change in pressure the resistance of the resistors changes.

Power Ground - A separate ground path for the logic and power module heavy current components (i.e., coil, injectors, etc.)

Pull-down Resistor - A resistor is hooked up to ground on one end and the other end is hooked up to the sensor or switch signal line. This serves to lower the voltage that is "seen" by the logic module chips.

Pull-up Resistor - A resistor that is hooked up to positive voltage on one end and the other end is hooked up to the sensor or switch signal line. If there is an open circuit (no current flow), the logic module will "see" all of the voltage. This "tells" the logic module that there is an open circuit by the signal line always being high.

PTC Thermistor - Positive Temperature Coefficient Thermistor. This is a resistor that will radically change its resistance with temperature. As the temperature increases, its resistance increases. As the temperature decreases, its resistance decreases.

R

RFI - Radio Frequency Interference. Classified as transient voltages that travel through the air an enter circuits unwantingly. Theses have to be suppressed (with a capacitor) for the circuit to work properly.

S

Schmitt Trigger - Used in the sync and reference pickups. It is fed a regulated voltage and provided a ground. It has one input and one output. If the voltage at its input goes above a certain internal set point, it turns "on" and outputs a voltage. If the voltage at its input falls below a certain point, the trigger turns "off" and outputs no voltage. So this changes a d.c. sine wave into a digital square wave.

Signal Ground - A separate path to ground for the logic and power modules. All the sensors (except the oxygen sensor) use this ground. Because of the low current flow through this circuit, there is little chance of voltage drift.

Solenoid Driver - An IC chip used to provide a ground for the solenoids and relays. When it receive a voltage at its input it turns "on" and provides a ground at its output.

U

Unity Gain Amplifier - An amplifier whose output is equal to its input. It is used when high resistance is needed in a circuit. Sometimes called a voltage follower.

W

Wheatstone Bridge - A series parallel circuit made up of four resistors in a "diamond" formation. The top is fed voltage and the bottom is provided a ground. The two sides are its outputs.

Z

Zener Diode - A diode that will pass current in the forward bias direction (anode to cathode) and if voltage goes above its breakdown voltage, it passes current in the reverse bias direction. This diode is doped so that damage won't happen when there is reverse current flow.